Vitreos

AI watermarking, attribution and provenance

How can we recognize AI-generated material, trace its origins, and judge the evidence after it has been edited or republished? Vitreos investigates watermarks, counter-watermarking and the limits of detection through experiments whose methods, failures and open questions remain available to inspect.

For technical researchers, operators and collaborators working on questions that matter to educators, editors and anyone evaluating source evidence.

Research begun in mid-August 2026. Experimental record through 9 October; this edition: 12 October 2026.

Links to package files open the offline package. If you downloaded this PDF on its own, extract the zip in the same folder, so that the folder Vitreos_Evidence_Atlas_20261012_reader_edition sits beside the PDF.

Research at a glance

Vitreos investigates AI watermarking, counter-watermarking, attribution and provenance. The research began in August 2026; the experimental record closes on 9 October 2026.

The main finding

One retained clean/marked pair showed an excess aligned with the laboratory's registered watermark key. Exact repeated context/target windows inflated the raw statistic. Removing those repeats reduced the marked score from **15.2753 to 8.8007**, while the positive excess remained. This is evidence about that specimen and construction; it is not a calibrated authorship probability. Full analysis · All 18 score calculations

Question	What the record supports	Where to look
Were hidden characters found?	The retained 17-file audit reported zero hits in the inspected invisible-character classes. The original files are unavailable; token-level marking was not excluded.	Early observations
Did the owned watermark produce a measurable signal?	Yes, in one retained marked completion under its registered key, including after exact-window deduplication.	Owned experiment
What did field testing add?	Seven historical resource rows from the Orange Pi RV2 (compact rows: Qwen2 INT4 0.5B under the vendor's KY-ORT runtime), plus a keyless cross-architecture calculation. Per-run accelerator attribution and device authentication remain unqualified.	Field evidence · Hardware
Can this identify a person or prove human authorship?	The record does not support those conclusions. Absence of a detected signal does not establish human origin.	Claim limits

A reading path through the report

1. **Understand the question:** S01 explains the problem and context.
2. **Read the findings:** S03 covers weak and negative evidence; S05 covers the positive laboratory result; S06 covers field limits.
3. **Inspect the method:** S04 explains measurement pitfalls; M01 specifies the specimens, settings and public calculations.
4. **Follow unresolved questions:** S02 records corrections and S07 identifies possible further studies.
5. **Use the appendices as references:** the full registers, histories, historical obligations, source families and evidence tables begin at A01.

How to read the evidence labels

Observed describes a measured or retained result. **Reported** preserves an external or historical account with its custody limits. **Hypothesis** names a proposal awaiting a discriminating test. **Unavailable** means the needed record cannot be inspected. These states must not be collapsed into a positive finding or a zero. The appendices retain the details behind every conclusion.

Contents and reading routes

The narrative and complete registers are one research product. Start with a route below, then use stable identifiers and linked page references to inspect the underlying record. Each route reaches the same evidence and claim limits.

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Reader	Route	Purpose
Researcher	S01 → S03–S06 → M01 → A01	Inspect units, assumptions, alternatives, limits and original record bodies.
Operator	S01 → S04 → M01 → S07	Check prerequisites, available reproduction, stop conditions and deliberately selected next work.
Collaborator	S01 → S02 → S07 → A01/A02	Follow corrections, source availability and the next discriminating observation.

Page references and PDF bookmarks address this exact rendering. Every R01-R42 entry is substantive. Questions F-01-F-24, subquestions A-01-A-06 and discovery probes U-01-U-06 preserve their historical wording and current disposition. Gate IDs identify historical obligations, not 64 newly passed tests.

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S01 AI watermarking: the problem and the investigation

Why this work matters

Vitreos investigates AI watermarking: how generated material might carry an identifiable signal, what happens to that signal as material moves between tools, and what evidence can support attribution. The work began in mid-August 2026 with exploratory tests of Claude outputs. A question about possible marks in those outputs grew into a broader investigation of detection, counter-watermarking and provenance. The practical problem is straightforward: when a text or file reaches us, what can we establish about how it was made and where it came from? R02, R03, S-EARLY-DIALOGUE

Educators need to assess permitted assistance, authorship and possible academic misconduct. Editors, researchers and evidence analysts need to assess documents whose bylines, apparent sources or histories may be misleading. Operators need records that survive movement between models, devices and formats. These decisions can fail in both directions: generated material can pass as independent human evidence, and human work can be wrongly accused. Even Turnitin's own guidance for educators says its AI-writing results can misclassify text and require further scrutiny before adverse action against a student. That is a reason to improve the evidence and its interpretation.

Here, **watermarking** means deliberately creating a signal that a specified detector can test. **Detection** asks whether an observable feature or pattern is present. **Attribution** asks which source or process the evidence supports. **Provenance** concerns the documented origin and transformations of an artifact. A statistical text watermark, an AI-writing classifier and a signed content credential answer different parts of this problem. Recognizing AI involvement is useful, but it does not by itself establish who wrote something or whether its claims are true. R12, R17, R24, R25

What changed while the investigation was under way

The following developments give the investigation a concrete setting. Dates below identify publications or announcements; incident reports can describe activity that began earlier. These are selected external sources, separate from the Vitreos experiments.

Publication date	Development	Why it matters to this work
14 August 2026; updated 1 September	Anthropic described Claude text watermarking as a SynthID-Text variant using statistical choices in generated text.	A mark can reside in ordinary text choices. Looking only for hidden characters cannot test that whole possibility.

Publication date	Development	Why it matters to this work
10 September 2026	Anthropic's misuse report described a Malaysia-focused operation that rewrote existing reporting, used fabricated bylines and removed original state-media attribution. The report covers disruptions from December 2025 through August 2026.	AI-assisted rewriting can obscure the source of material that still looks like ordinary reporting. The problem includes source history as well as generation.
15 September 2026	MarkSec proposed a capability-aware evaluation of watermark stealing, scrubbing and spoofing, with text-quality constraints.	A useful robustness result must say what an adversary could do and what quality survived. This study is background, not a method reproduced here.
5 October 2026	OpenAI announced text provenance, including a statistical watermark and restricted detector access. Its account distinguishes model involvement from authorship, identity and factual accuracy.	Availability, text conditions and detector access are part of any operational claim; a watermark is one source of evidence.
8 October 2026	OpenAI reported two AI-assisted false-front operations. One used seven fabricated journalist personas to pitch articles; the report identified almost 100 published or syndicated articles under their bylines.	Source presentation can conceal who is behind an article even when it appears in an established outlet. AI generation is only part of that attribution problem.

The incident rows describe the providers' investigations. Vitreos did not authenticate those cases or test a detector on their artifacts. Their relevance is the decision they expose: a fluent passage and a plausible byline can leave essential source questions unanswered. The dated bibliography in A05 gives the source routes and scope of this context review.

Counter-watermarking is part of the research question

Claims to “remove AI watermarks,” “humanize” text or bypass detection need to be made testable. Changing an AI-writing classifier's score, erasing file metadata, weakening a keyed statistical signal and forging a signal are different outcomes. Success against one test says little about the others. The relevant questions are which mechanism was tested, what access the test assumed, what transformation occurred, whether useful content survived, and how the detector behaved on matched controls. The commercial tools mentioned in the exploratory discussions were not systematically evaluated here; this atlas supplies no blanket verdict on their efficacy. F-10, F-15

Understanding those failure modes is necessary for evaluating a defense. A watermark that survives an untouched laboratory sample but fails under ordinary editing has a different practical role from one that withstands the transformations a user actually

encounters. Equally, a test that can be made to implicate unmarked material needs that failure recorded. This is why absence, false alarms, acquisition loss and weak signals belong beside successful detections in the research record. R01, R15, R18, R36

A map of current techniques

The terms below name the main ways provenance is asserted today and the ways it is defeated or abused. They are background from published work, not Vitreos measurements; the last column shows where this record touches each family. Sources are listed in A05.

Family	How it works	How it is defeated or abused	In this record
Statistical token watermark	During generation a key biases or selects next-token choices: green lists (Kirchenbauer et al., 2023), tournament sampling in SynthID-Text (Dathathri et al., 2024), or key-driven sampling that leaves the output distribution unchanged (Kuditipudi et al., 2023). A detector holding the key tests the text.	Paraphrasing and rewriting (Krishna et al., 2023), translation and heavy editing dilute the signal, as do character-level perturbations (Zhang et al., 2025). Learning the rule from enough outputs enables scrubbing and spoofing (Jovanović et al., 2024). Short texts give too few tokens to test.	S05 owned experiment; R17, R18, F-10
Character and format channels	Invisible or unusual Unicode, look-alike characters, whitespace and formatting conventions carry or betray a mark.	Normalization, plain-text copying and retyping remove them; the same characters can be planted in human text.	S03 byte audit R05; ordinary-separator proposal R03
Metadata and signed credentials	File metadata or a signed manifest (C2PA) is bound to the file and validated against a trust policy.	Export, screenshots, re-encoding and upload pipelines strip it; a credential on a later copy does not authenticate earlier origin.	S03 container negatives R12, F-13
Passive detection	Classifiers and stylometry estimate AI involvement without any intentional mark.	Paraphrase and "humanizer" tools evade them; false positives fall unevenly, for example on non-native English writing (Liang et al., 2023).	R23, R24; education guidance above
Forgery and false implication	A mark is spoofed onto human text, marked spans are transplanted into other material, or fabricated bylines and personas present AI-assisted text as independent reporting.	Exploits readers who treat a positive test or a plausible byline as attribution.	R25, F-10, F-12; incident reports above

How the providers' constructions differ. The disclosed provider methods are not one scheme. Anthropic describes its Claude watermark as a SynthID-Text variant (dated table above). Google's SynthID Text reference implementation offers mean, weighted-mean and trained Bayesian detectors, configured by context length and several keys; it is labelled research code, not for production use, and its hashing function is stated to carry no guarantee of cryptographic security. OpenAI's textGrain technical report (5 October 2026) assigns vocabulary tokens to keyed blocks and couples the block probabilities to keyed randomness by optimal transport under an entropy budget, keeping relative probabilities inside each block. Averaged over keys, its token distribution is unchanged; under one fixed key it is not, and that difference is what a key holder detects. None of these is the green-list scheme used in this record, so a green-list scan says nothing about them.

Knowing how each family fails is how a defense is judged: removal, forgery and false alarms belong in the same evaluation as successful detection.

How Vitreos approached it

The work combined local artifact analysis, owned watermark experiments, field contributions from an Orange Pi RV2 single-board computer, and browser-based research and analysis tools. It examined retained text and exported files, controlled token sequences and native completions, and records from constrained-device execution. M01 supplies methods and specimens; E03 and E04 identify the field evidence and contributor chronology. This gives a future collaborator concrete material to inspect, reproduce or challenge.

The sequence matters. Early exploratory dialogue supplied hypotheses. Byte scans and text comparisons tested particular observations. Instrument controls exposed false alarms. A known-key laboratory made a narrower statistical question measurable. The contribution is both the resulting measurements and the record of how the question changed, including the tests that could not decide it. S02, S03, S04, S05

What was built, in plain terms

The central experiment uses a small open model, DeepSeek-R1-Distill-Qwen-1.5B, run locally on the CPU of an ordinary x86-64 computer (not the field board), with a keyed statistical watermark added at generation time by the project's experimental software, Axis Provenance Lab (AxisPL) 0.1.4 (M01). The construction is of the green-list type introduced by Kirchenbauer et al. (2023): at each step the previous four token IDs and a secret key assign each vocabulary token to a green set using a pseudorandom membership threshold of 0.25 (γ), and green tokens receive a fixed logit boost (δ = 2). From one 28-token prompt and the same seed, the model produced an unmarked ("clean") and a watermarked ("marked") completion of 256 tokens. A detector holding the key

counts how many generated tokens fall in their keyed subsets and compares that count with the nominal 25% reference rate used by this construction. The exact model, settings, prompt and outputs are in M01.

Earlier work examined real assistant outputs and exported files for hidden characters, metadata and wording similarity (S03). Field work on an Orange Pi RV2, a RISC-V single-board computer, measured on-device generation with a small Qwen model and later repeated the keyless part of the analysis on that second processor architecture (S06, A08).

What this investigation supports

The most specific positive result is a registered-key excess in one retained marked completion after exact repeated context/target windows are removed. Earlier results include negative byte scans, weak text comparisons, instrument-generated false alarms, incomplete source custody and untested constructions. These belong to the same research story. The later success does not supply the missing premises of the earlier experiments. R01, R05, R15, R17, R18

The retained pair contains one clean and one marked completion, each with 256 completion-token IDs. With a four-token context, 252 positions are eligible in each. The primary-key raw counts are 53/252 for clean and 168/252 for marked. Counting only the first occurrence of each exact context-plus-target window gives 53/251 and 60/93. The corresponding nominal marked score falls from 15.2753 to 8.8007. Repetition increases the raw statistic, while a positive key-aligned excess remains in this specimen. Neither score is a calibrated probability of authorship or a population performance estimate. R18, L-01, L-03, S-RESULTS

The original investigation asked whether provider-generated files contained an observable signal and whether an owned instrument could expose it. Unknown activation, tokenizer, key, profile and acquisition conditions prevented a decisive answer about those historical specimens. Building a known-key laboratory changed the question to one that could be inspected: what does a declared local construction do to retained outputs, and which claims survive explicit controls? A successful local measurement answers that narrower question. It does not recover a provider's private mechanism. R09, R17, F-01, F-04

F06 / Provenance layers and the claim boundary

At which layer was a property observed, absent or left untested?



CLAIM BOUNDARY No generic authorship, identity, ownership or truth claim.

Planes and dotted guides separate questions; depth and spacing encode no confidence, physical assembly order or universal causal pipeline.

Figure 06. Provenance layers and the claim boundary Five conceptual planes separate retained bytes, export/container, tokenizer/profile, verification and permitted claim. Source-linked callouts keep unavailable inputs, inspected absence and supplied expectations distinct. R14 separately treats model rewriting as a possible marking event.
Unit: Conceptual provenance questions. Sources: S-COMPARATIVE · S-CONTRACTS · S-EARLY-AUDITS · S-EARLY-DIALOGUE · S-ENGINEERING · S-HISTORY · S-INFORMAL-LEADS · S-OBSERVER · S-QUESTIONS · S-RESULTS. Records: R03 · R05 · R12 · R13 · R14 · R17 · R18 · R31 · R35.
Limit: Planes and guides show conceptual layers, not confidence, physical assembly order or a universal causal pipeline. Export copies add no independent samples.

Four evidence families and separate claim limits

Byte and export analysis examines properties of acquired artifacts: character classes, separators, containers and conversion effects. Registered statistical analysis examines compatibility with a specified construction, token representation and key. Credential analysis examines assertions carried by a signing or provenance system under an applicable validation and trust policy. Passive analysis examines style, similarity or

interaction patterns without assuming an intentional mark. Each family has its own observation opportunities, confounders and controls. A negative in one family does not eliminate the others; a positive does not confer their capabilities. R03, R12, R17, R24, F-13

The distinction also applies to attribution. A feature may be present in an artifact; a statistic may be compatible with a registered scheme; a receipt may associate an artifact with a workflow. Those statements do not automatically identify a person, establish ownership, reveal intent or certify factual truth. Conversely, failure to detect a signal does not establish human origin. The generic authorship or person-attribution claim is explicitly excluded from this report. R25, F-12, L-13

The project's request to retain evidence “below the noise floor” is interpreted with care. A measured noise floor requires a defined channel, statistic, background and sensitivity. No universal such quantity was estimated here. The phrase also expressed a broader concern about weak information missed through attention, retrieval or premature filtering. Preserving that concern does not turn absence into evidence of concealed marking. Nonexistence and limited observability remain distinct possibilities, without an assigned equal probability. R01, R36

Why follow up, and what would make it useful

As generation tools, editing workflows and detector access change, evidence collected under one set of conditions may stop answering the practical question. The useful continuation is to identify which change matters and test it with a declared protocol. The open questions in S07 preserve several such routes:

- **Editing and counter-watermarking:** measure what happens under declared transformations and mixed-source workflows, retaining both signal and text-quality outcomes. F-05, F-10, F-15
- **Reliable interpretation:** establish independent samples, false-positive behavior, relevant base rates and uncertainty before turning a score into an operational decision. F-06, F-09, F-21
- **Unresolved mechanisms:** test the ordinary-separator and tokenizer/generation-coupling proposals against appropriate positive controls and nulls. Preserve their hypothesis status until that evidence exists. F-02, F-03, A-02
- **Source evidence that travels:** examine whether credentials, acquisition records and field exports remain interpretable after transfer, and what each actually binds. F-12, F-13, F-16

These are proposed continuations, not tests completed by publishing this atlas. A finished report can preserve uncertainty without making the next experiment compulsory. The purpose is to give researchers and operators an inspectable starting point for better attribution and provenance decisions, including cases where the available evidence remains inconclusive.

Status and reading routes

Finishing the engineering work, qualifying a scientific claim, building a product and publishing a report are separate decisions. The engineering work reached a defined end: the keyless cross-architecture check was completed, and the clean source of the historical field exporter was confirmed unavailable. That does not qualify native credential checking, formal conformance, empirical detector calibration or a deployed service. An open research question can coexist with a finished report, and an excluded stronger claim can be a correct conclusion. R26, R31, R32, R41, L-10, L-12

Researchers can read S03-S06 for the observations, assumptions and alternatives, then follow the methods and complete records. Operators can start with S04 and the public reproduction procedure, which specifies what the disclosed files can verify. Collaborators can follow S02 and S07 to see corrected hypotheses and the smallest proposed discriminators. All routes reach the same evidence and limits. A role label changes navigation, not the strength of a result. R17, R33, R38, S-QUESTIONS

Transparency, consent, local custody, portable owned records and bounded operation on constrained devices motivated the work. They are design values, not measured proof of provenance efficacy or a conclusion about provider conduct. Adjacent memory, bias-evaluation and architecture ideas remain separate proposals. R39 and R40 record these motivations without personal details. R39, R40, F-23

S02 Hypotheses, corrections and the independence of agreement

> Early explanations became hypotheses, not implementation disclosures. Corrections and shared-source dependencies remain visible.

A plausible mechanism is not an implementation disclosure

Early dialogue proposed token reweighting and elicited confident explanations of watermark method families. Later descriptions became more specific without acquiring corresponding implementation evidence. Familiarity with a public method can explain a technically plausible answer; agreeable dialogue and shared prompts can explain apparent confirmation. The record therefore retains the exchange as hypothesis formation. It does not treat a chatbot's account of itself as privileged disclosure or independent corroboration. Greenlist and Gumbel-family descriptions also remain distinct instead of being merged into a single alleged mechanism. R02, S-EARLY-DIALOGUE

Provider documentation later changed what was known. Anthropic's published description identifies Claude's text watermark as a SynthID-Text variant, and OpenAI separately announced text provenance on 5 October; both are linked in S01 and A05, and older summaries that contradicted them were corrected. These sources describe the providers' systems. They do not establish that a mark was active on an earlier file, disclose the exact private profile, or make the local laboratory detector applicable to that file. R35, F-01, L-11

What the ordinary-separator proposal actually said

The original whitespace lead included ordinary separators at selected offsets, potentially produced through token bias. Such a construction need not introduce unusual Unicode, extra trailing spaces or a conspicuous metadata object. Token choice and separator placement are not mutually exclusive mechanisms: changing word lengths or token choices could change where ordinary separators fall. The early anomalous-character scans therefore did not exhaust the proposal. This correction preserves an untested hypothesis; it does not demonstrate an encoded message. R03, F-02, S-INFORMAL-LEADS

A testable positional construction would need an explicit code or offset rule, a preservation model and an appropriate null. Offsets selected after viewing an artifact could fit ordinary language by accident. An owned positive control and matched unmarked text would distinguish a declared construction from retrospective pattern fitting. The proposal remains open on those terms, with broader prior art unassessed. The correct historical conclusion is that the scanned classes were absent in the tested copies; universal whitespace falsification was too strong. R03, R05, F-03

The informal discussion also asserted inherited training-source marks, dedicated whitespace tokenizers and quotation limits caused by watermark leakage. These are three different causal or implementation claims, none established by the supplied receipts. Copyright policy, training behavior, active generation marking and ordinary style could exist independently. Retaining the assertions as historical leads prevents their disappearance, while marking them unverified prevents conversation confidence from becoming a finding. A quotation-length policy alone does not identify its cause or a provider's intent. R04, F-24

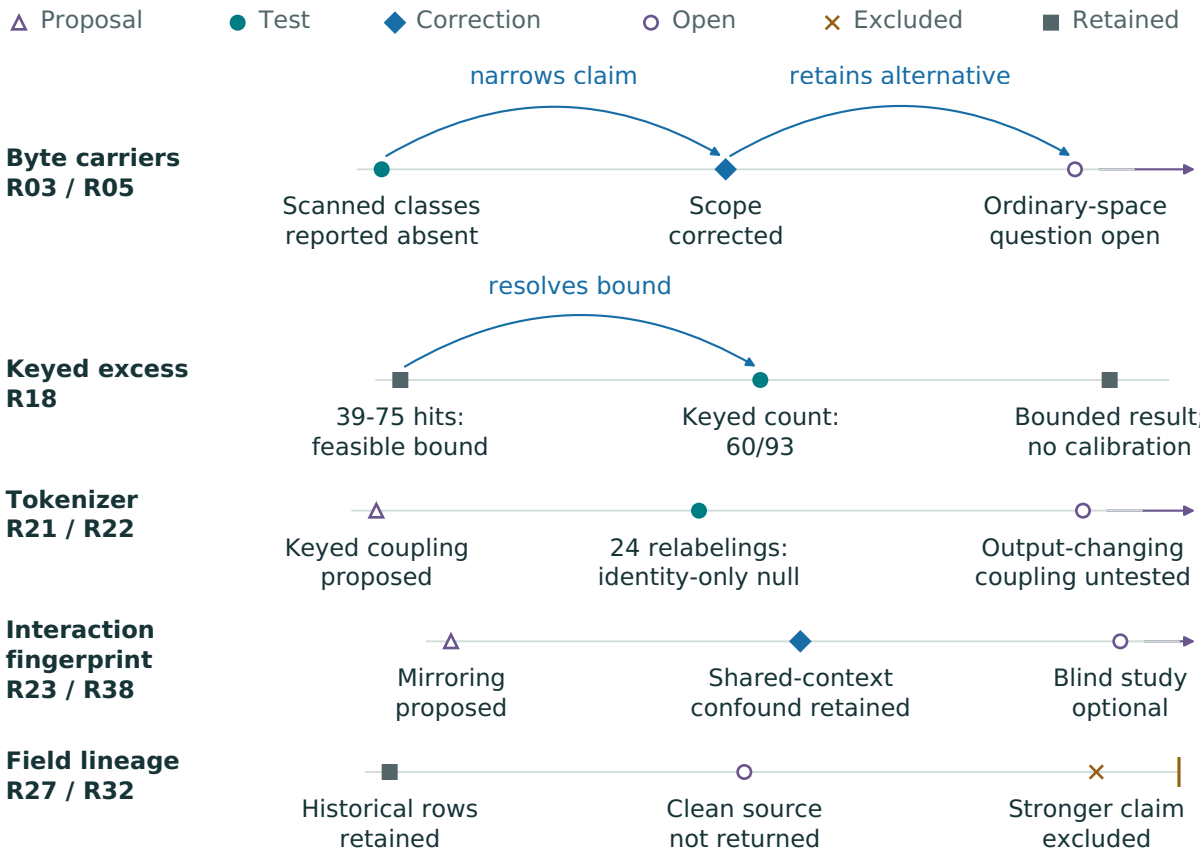
How available knowledge was overlooked

Several losses concerned information already present in the record. An original conversation was initially treated as a derived artifact; the ordinary-offset proposal was flattened into an exotic-whitespace theory; exported copies were easy to count as new linguistic samples; a surface label obscured the absence of an exact successful model ID. Earlier requests to retain weak evidence also risked disappearing when the successful keyed result became the center of the narrative. These examples define “unknown knowns” operationally: recorded knowledge that was not loaded or integrated. They do not imply unconscious access to verified external facts. R36, R10, R13

Recovery changes custody without necessarily changing the observation. The six retained summaries and early audit material are now recovered; the 17 original audit inputs remain unavailable in the reviewed material. Earlier statements that all these files were missing are superseded. Their historical wording belongs in the correction history rather than the current account. A source-to-record crosswalk, hashes and explicit unavailable routes make this correction reviewable without requiring a new carrier experiment. R05, R06, R36, F-24

F02 / Hypothesis episodes and revisions

Reading order within each lane; positions are ordinal, never duration.



Markers name event types; arcs name corrections, not evidence strength.
Open endpoints are unresolved or optional, not assignments to run a study.

Figure 02. Hypothesis episodes and revisions Hypothesis lanes preserve proposals, tests, corrections and retained endpoints. Labeled arcs narrow the byte-scan claim while retaining ordinary-space alternatives, and connect the earlier 39-75 feasible bound to the measured 60/93 keyed count. Shapes identify event types; open and excluded endpoints remain distinct.
Unit: Five categorical hypothesis lanes; labeled events and within-history correction arcs. Sources: S-EARLY-AUDITS · S-ENGINEERING · S-FIELD-HISTORY · S-HISTORY · S-INFORMAL-LEADS · S-RESULTS. Records: R03 · R05 · R18 · R21 · R22 · R23 · R27 · R32 · R38.
Limit: Positions and lane lengths are ordinal layout only, not duration, credibility, probability or statistical survival. Arcs document interpretive relationships, not independent evidence. Open endpoints do not initiate experiments.

Agreement needs a dependency account

AI-assisted analysis raises its own independence problem, and this project repeatedly asked for conflicts to be reported and agreeable outputs to be scrutinized. Several AI assistants can share prompts, models, prior descriptions, fixtures and selection decisions.

Separate role names do not make their conclusions statistically independent. A second implementation can detect an implementation discrepancy; a separate execution can test portability; an editorial review can find an overclaim. These are valuable contributions with different limits, and none automatically adds another generated specimen. R38, R18, R31

The appropriate human acceptance boundary is therefore evidence-based rather than a vote count. Agreement should travel with the shared inputs, actual operation and any disconfirming results. If reviewers repeat the same unsupported premise, their convergence does not strengthen that premise. If they inspect different failure modes in the same artifact, the resulting assurance is about those checks on that artifact. This distinction also informs the later passive-interaction hypothesis: apparent mirroring is a phenomenon to operationalize, not evidence of hidden communication. R23, R38, A-03, U-03

Forecasts about access/cost stratification, resource constraints, information disorder and marking/guardrails receive similarly bounded treatment. The lesson is to log forecasts prospectively. Timestamps, fixed endpoints, probabilities, scoring rules and a baseline are needed before assigning predictive credit. Selective recall and ordinary changes in conditions remain alternatives to a retrospective causal story. No prediction success or hidden intent is claimed. R39

S03 Early observations: negative scans and weak similarities

> Negative scans and weak similarities have defined limits. A missing original or an untested channel cannot be read as a zero.

The 17-file byte audit: a result with a precise object

The historical audit reports 17 Markdown files totaling 237,506 bytes, 234,326 Unicode characters and 2,644 lines. Its specified hidden, format and whitespace classes were absent. The retained account describes no zero-width or non-ASCII whitespace, format/bidirectional/variation/tag/private-use characters, tabs, byte-order marks or mixed line endings. Two trailing-space lines held 35 ordinary spaces, including diagram alignment. These are concrete reported observations about the inspected copies, not an undifferentiated statement that nothing unusual existed. R05, S-EARLY-AUDITS

The audit report, JSON, script and manifest have been recovered. The 17 exact source Markdown files have not been recovered in the reviewed material. Consequently, this report can preserve and inspect the historical audit documentation but cannot present a fresh raw-byte rerun of all 17 inputs. The historical report's checks on uploaded-copy extended attributes also cannot establish the filesystem metadata of originals elsewhere. The missing input set is a custody limitation independent of the negative feature result. R05, F-02, F-24

The inference is conditional on what the method should have seen. A validated check for a specified character class can constrain that class in a preserved copy. It does not constrain a keyed distribution of ordinary words or separators, an external credential, a mark lost during acquisition, or every feature the parser did not inspect. No mark, a different carrier, acquisition normalization and an inaccessible statistical mechanism remain alternatives. Listing them does not assign them equal likelihood, and the negative result does not prove either human origin or concealed marking. R01, R03, R05

The smallest useful reopening depends on the stronger claim sought. Reproducing the historical audit would first require recovering the exact bytes of the 17 files against their manifest, then exercising named feature checks with positive controls. Testing ordinary-space positioning would instead require a declared positional construction and matched controls. These are different questions. Recovering a report alone accomplishes neither, but it improves the evidentiary route enough that the historical result need not be discarded. R03, R05, F-02

Six summaries, three pairs and a retained collection mistake

The summary collection consists of three long/short pairs from the same blog-summary sequence: one Max-effort pair and two High-effort pairs across two operators. An initial effort-setting mistake is part of the collection history. The shorter summaries followed longer requests, so the six files are not six independent draws under an identical fully frozen state. The anonymous High A/B labels do not establish which operator produced which pair. Preserving these conditions prevents a later reader from mistaking exploratory acquisition for a randomized comparison. R06, S-EARLY-AUDITS

All six retained copies and their containing ZIP were recovered and hash-matched to the historical manifest. This substantially improves custody. It preserves the exact retained byte streams without authenticating their complete generation conditions, session state or historical marking activation. The numerical comparisons below remain historical reported metrics; their inclusion is not a claim that this report reran the carrier audit or original comparison pipeline. R06, F-24

Historical High A/B comparison	Long summaries	Short summaries
Sequence similarity	0.638	0.735
Bag-of-words cosine	0.912	0.869
Longest exact common span	13 words	11 words

These measurements show a combination of content overlap and wording variation under a shared source and task. High bag-of-words similarity can coexist with lower sequence similarity because related content need not appear in the same order. The values do not identify why the outputs differ. Common source material, constrained summarization, effort, style, stochastic generation and context differences remain plausible explanations. No calibration converts these metrics into a watermark test. R06

The scanned anomalous byte features were also absent in this collection. Separately, the Max summaries placed domain/date information in prose with a source line, while the High pairs used heading placement. That formatting difference deserves preservation because it is visible and could motivate a controlled experiment. The small uncontrolled collection cannot isolate effort as its cause, establish a stable effort fingerprint or identify a provider. A future matched comparison would need effort to be the sole changed factor and the formatting feature declared before inspection. R06, R07

Conditional divergence is weaker than family exclusion

The early reasoning treated divergent outputs as weak evidence against a vanilla fixed-key deterministic Gumbel-Max setup. That reasoning only discriminates when the complete determining state is fixed: prompt and hidden context, model, tokenizer, key, nonce and

relevant generation state. Those premises were not established across the accounts. Divergence therefore cannot be assigned a quantified likelihood against a method family, and it does not identify a different provider mechanism. R08, F-04

This is a useful conditional argument rather than a failed observation. Under a fully specified owned deterministic implementation, repeated identical state should support a reproducibility test. When relevant state is unknown, distinct outputs leave several explanations alive: different hidden context, key scope, seed or nonce, sampling behavior and deployment variation. The proper continuation is to test a defined construction, not to elevate the conversational comparison into vendor forensics. R08, R17

A related early overstatement held that a single Markdown file could never support detection. File count is not the decisive quantity. A sufficiently informative artifact under a validated known profile can support a profile-specific result. In these early files, short length, unknown activation, unavailable key/tokenizer/profile and unqualified sensitivity limited the inference. “Not marked,” “marked but undetected by this procedure,” and “altered by acquisition” are different states; none becomes established merely because there is one file. R09, F-06

Matching-prompt corpora and the arbitrary-key null

A separate archived comparison covered nine prose pairs and 32 constrained completions. There were no exact pair matches. Mean prose sequence similarity was approximately 0.191 and TF-IDF cosine approximately 0.564; 20 of the 32 constrained pairs shared their first word and eight shared their first four words. Those observations belong to the historical comparative corpus, not the six-summary collection or the retained native pair. Shared prompts, common factual phrases and constrained answers provide direct alternatives to a watermark interpretation. R10, S-COMPARATIVE

The exact successful command-line model identifier was missing. Matching archive hashes now bind the retained inputs to the historical analysis, but they do not supply every missing execution or scoring fact. The study supports a weak descriptive comparison with a named provenance gap. A stronger generalization would require generation-event and model receipts, followed by a separately selected comparison suited to that claim. Repetition of a surface label cannot repair a missing model identity. R10, F-20

The associated laboratory scan used surrogate word/punctuation tokenization and 64 deterministic arbitrary keys. Neither corpus crossed nominal $z = 4$, with mean z approximately 0.128 and 0.008. This is a profile-specific non-detection under the artificial laboratory construction. It is not a search of a provider's undisclosed key or a demonstration that provider marking was absent. Correct-profile positives and negatives are needed to establish detector applicability. The 64-key historical study must also remain separate from the later 4,096-key synthetic fixture. R11, F-03, F-09

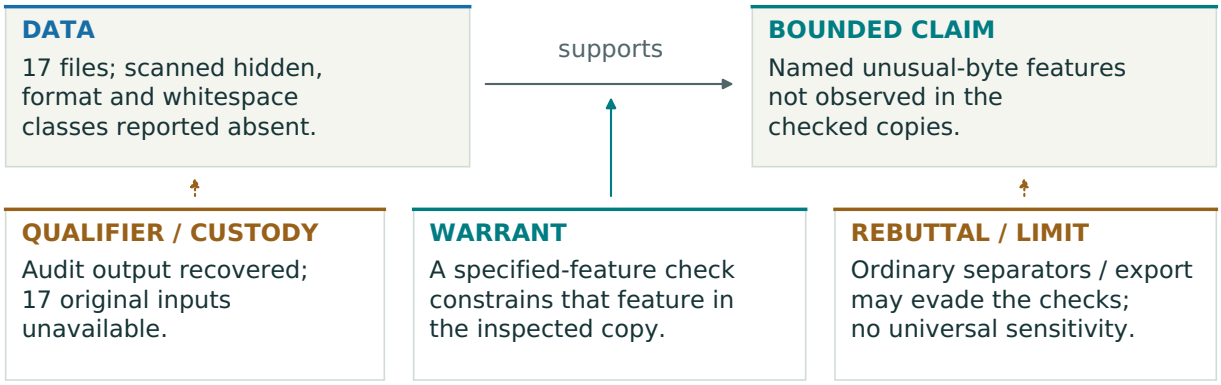
Container negatives and equivalent analytical care

Five image/export fixtures lacked the inspected embedded C2PA/JUMBF and related metadata structures; the reported visible structures matched ordinary SVG, PNG and JPEG tool paths. This describes those artifacts and acquisition routes under the stated parser coverage. A separate download route, external manifest or conversion that strips a credential remains a different possibility. The negative does not establish universal provider absence and does not qualify a native trusted-credential validator. R12, F-05, F-13

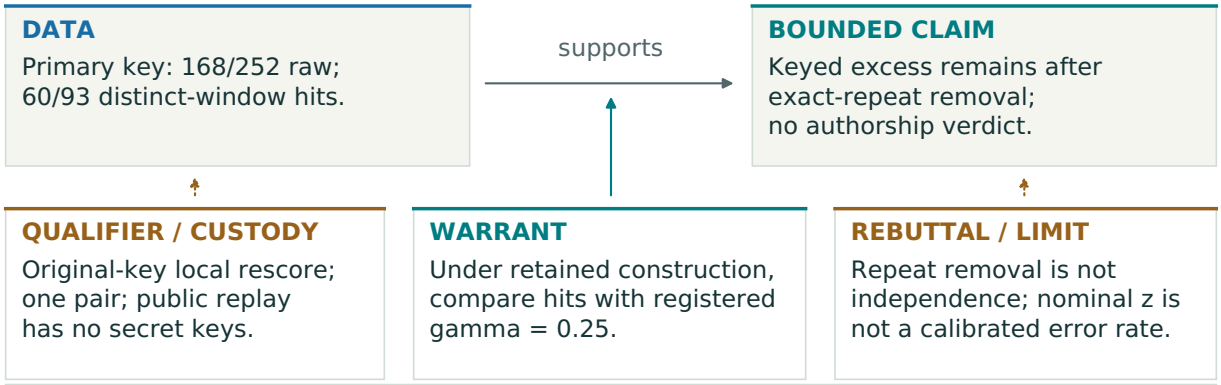
A route-specific continuation would retain pre- and post-export bytes and compare trusted, untrusted and other applicable controls. Merely citing a credential specification would not perform that qualification. Across these early studies, analytical care means retaining the observation, instrument, custody, assumptions, alternatives and claim limit together. The same structure applies to the later positive result: neither absence nor excess can outrun the method that produced it. R01, R12, R18

F03 / Two arguments with visible limits

R05 / Negative byte audit



R18 / Positive keyed excess



Solid arrows connect data, warrant and claim; dotted links constrain scope.
Geometry encodes reasoning roles, not evidence strength or authentication.

Figure 03. Two arguments with visible limits Each argument connects data to a bounded claim horizontally. A separate warrant enters that inference from below; custody qualifications attach to the observation and rebuttals limit the claim. The negative byte audit and positive keyed excess therefore receive the same explicit inferential treatment. Neither warrants generic authorship attribution.
Unit: Two documented Toulmin arguments; data, warrant, bounded claim, qualifier and rebuttal. Sources: S-EARLY-AUDITS · S-ENGINEERING · S-HISTORY · S-RESULTS. Records: R05 · R18 · R20 · R33.
Limit: Solid arrows express documented reasoning and dotted links its scope conditions, not proof of premises or source authentication. Spatial position and box area are categorical, with no numerical evidence-strength scale.

S04 When the instrument or the counting unit creates the signal

> The measurement instrument and the unit being counted can create misleading signals. Controls and explicit contracts constrain interpretation.

Files are not generation events

Cross-format validation found that all five checked HTML source segments and all nine checked PDF segments preserved the source word sequence; both code segments preserved non-whitespace content. These conversions are useful transport controls. They do not add independent linguistic samples, because the model-generated content is inherited from the same source event. A PDF, HTML file and Markdown file can have different container bytes while carrying the same linguistic observation. R13, S-CONTRACTS

The unit can also be smaller than a file. A bundle may combine model prose, model code, programmatic data, deterministic conversions and a later rewrite. A condensed version produced by a model may be a new generation and potentially a new marking event. Treating that rewrite as a purely post-hoc transformation would confound editing with new generation. Span and event identities make the distinction explicit; local non-model edits are the appropriate starting point when the intended question concerns transformation alone. R14, F-10

When the instrument creates the anomaly

The historical parser review found a raw `/AA` byte sequence inside compressed PDF data that had falsely suggested an Additional Actions structure. Parsed-object inspection did not support that interpretation. A byte substring and a parsed semantic object are different observations: an arbitrary occurrence in compressed data does not by itself establish the object named by the same characters. The failure belonged to the analysis path and must not be presented as hidden producer behavior. R15, S-CONTRACTS

Another extractor mapped ten ordinary bullet glyphs to U+007F. A second extractor and rendered pages showed bullets. Binary ZIP CR/LF counts were also misread as text line endings, and whole-bundle comparison obscured exact source segments within a larger document. These are productive weak-signal findings because they reveal how representation choice can manufacture an apparent anomaly. Their historical checks are retained; the report does not assert that current code still exhibits the same defects. R15

For a new suspicious feature, the corresponding discriminator is structural inspection, extractor comparison and rendered appearance under identified versions. These checks address different failure modes. Visual agreement cannot establish every byte property, while raw-byte agreement cannot validate a parser's semantic interpretation. The record supports preserving both the suspicious first result and the explanation that narrowed it, so future users can recognize the instrument failure without erasing the investigative path. R15, F-21, F-22

Passing normal tests does not define the accepted language

In an earlier generated-fixture experiment, Go and Python programs agreed on 288 accepted records out of 300 and passed their reported normal tests. Yet they differed from the intended schema or each other on additional fields, whitespace, dates and oversized lines. Agreement over one fixture set did not establish equivalence across the full accepted input language. Different parser permissiveness and limits were sufficient explanations; no malicious behavior was required. R16, S-CONTRACTS

The later scalar-validation episode is separate. It concerned invalid scalar and token inputs at the watermark module's direct API, not the historical Go/Python fixture experiment. A bounded validation change was implemented and tested on a local development branch. Broader public-ingress reachability and deployment remain outside scope. Linking A-05 to R16 preserves their shared methodological lesson without claiming the same incident occurred twice or inventing an exploitable public endpoint. R16, A-05, F-22

Historical importer work exposed raw-field, contradictory-promotion, boolean/type, infinity and count problems; subsequent repair added date, exception, concurrency, accounting and dirty-source checks. Release 0.1.4 incorporated these repairs. Defects listed in older notes are resolved by that release unless a new regression or an expanded contract reopens them. R30, F-17, F-22

The engineering test record for this work lists 597 passing core tests with two expected skips, an amended browser replay with 101 top-level passes plus nine subtests, and wheel/sdist build with six smoke commands. Those are source- and scope-bound historical engineering checks. Optional Torch and native trusted-C2PA capability were not qualified by skipped cases, and test builds from a development branch did not replace the released version. The totals must not be combined into a scientific sample size or presented as fresh tests of this public manuscript. L-06, L-07, L-08

Formal assurance must bind the actual policy

The source record retains a discrepancy between the executable Python policy and archived Lean semantics. It supports one documented source-level finding; an intermediate seven-case claim lacked an inspected complete inventory. No current compiled certificate was produced for the actual policy. The stronger statement that the current Python policy is formally certified is therefore excluded. R26, F-14, L-12

A future proof obligation would need to bind the intended policy, executable version, complete case inventory and checked proof artifact. Formal tooling can establish a property of its formalized assumptions; it cannot by itself authenticate a historical device event or calibrate an empirical detector. Temporal, statistical, adversarial and operational assurance remain additional questions beyond a first policy kernel. Their presence in the research register preserves the frontier without implying implementation. R26, R42, F-15

S05 The keyed experiment: one marked completion and its dependence

> One retained pair shows a registered-key excess; repeated windows increase the raw score. Deduplication reduces dependence without supplying population calibration.

One pair, three observation units

The central native evidence is one retained clean/marked pair. The disclosed fixtures preserve 256 completion-token IDs per specimen. The original four-token context prefix is unscored, giving 252 eligible positions in each. Public wrappers retain the completion arrays and omit operational fields; they are new public artifacts with their own identities. They support exact recounting of tuple recurrence without requiring a tokenizer, model or secret key. R18, S-RESULTS

For a token at an eligible position, the context is its preceding four IDs and the window is that context together with the target ID. Raw scoring counts every eligible position. First-context selection keeps the first eligible occurrence of each exact context tuple. First-window selection keeps the first eligible occurrence of each exact context-plus-target tuple. The distinction matters when the same context is followed by different targets: the clean completion has 249 contexts but 251 windows; the marked completion has 93 of each. R18, F-08

The primary-key results illustrate how the unit changes the descriptive statistic. All values below are carried from the original-key local rescore; public verification can recount denominators and recompute arithmetic, but cannot remeasure key membership. L-01, L-02, S-RESULTS

Specimen	Selection unit	Hits	Denominator	Nominal z, not calibrated
Clean	All eligible positions	53	252	−1.4548
Clean	First exact context	53	249	−1.3538
Clean	First exact context plus target	53	251	−1.4212
Marked	All eligible positions	168	252	15.2753
Marked	First exact context	60	93	8.8007
Marked	First exact context plus target	60	93	8.8007

Two wrong-key diagnostic roles are retained for every unit in the complete numerical table. Together, two specimens, three key roles and three units produce 18 analysis rows. They are repeated analyses of the same pair, not 18 generated samples, independent

controls or trials of a population detector. The wrong-key results describe the selected diagnostic roles; they do not calibrate the distribution of every possible key or text. R18, L-02, S-RESULTS

F04 / Recurrence at every completion position

Two retained completions; 256 token positions each. Prefix length = 4.

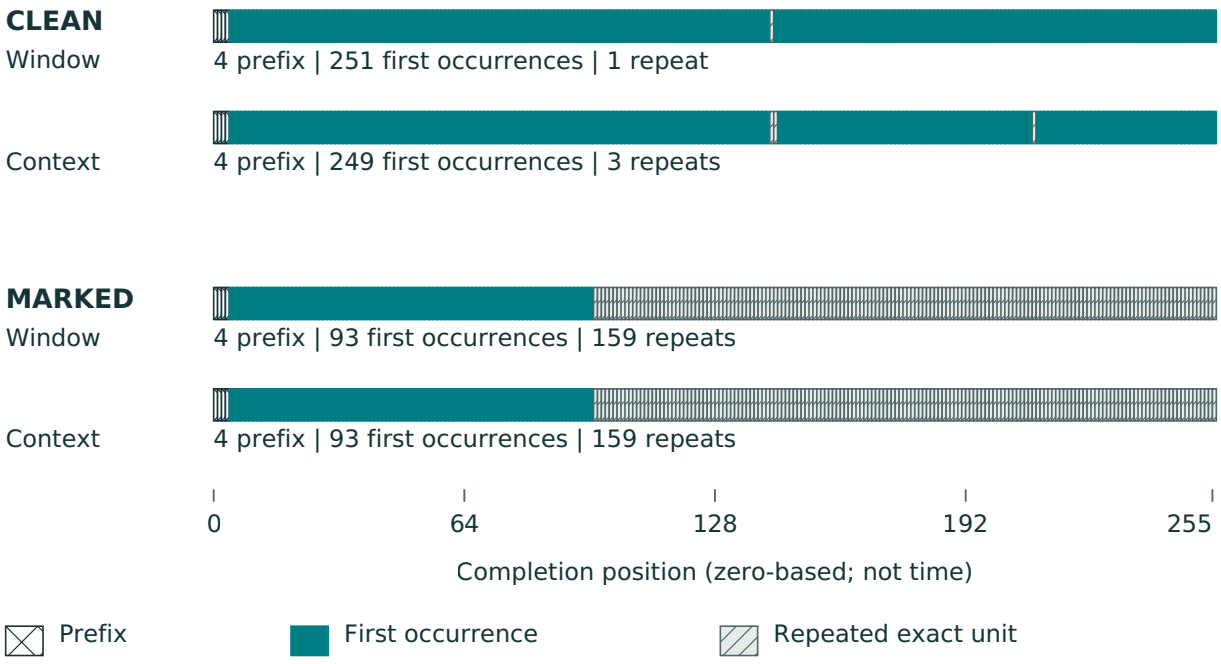


Figure 04. Recurrence at every completion position Every source position is represented in two aligned annotation tracks per specimen. Clean has 251 first windows and 249 first contexts; marked has 93 of each. All remaining eligible positions are repeats. The initial four positions are unscored.
Unit: 512 source rows; exact completion position; separate four-token context and context-plus-target units. Sources: S-CONTRACTS · S-EARLY-DIALOGUE · S-ENGINEERING · S-HISTORY · S-RESULTS. Records: R13 · R14 · R18 · R20.
Limit: Tracks show recurrence only, never secret-key membership. Deduplicating exact units does not establish independence, and position is not elapsed time. Counts are derived directly from observation-tracks.public.json.

From an aggregate bound to measured distinct membership

Before distinct membership was measured, the marked raw total and recurrence structure permitted between 39 and 75 distinct-window hits. That interval was a feasible bound, not a previously observed count or another specimen. A rescore with the original key then

measured 60 hits among the 93 distinct windows, on 7 October at 23:26 UTC; the result was reviewed on 8 October. The bound remains part of the knowledge history, while 60/93 is the current measured aggregate. R18, A-01, S-RESULTS

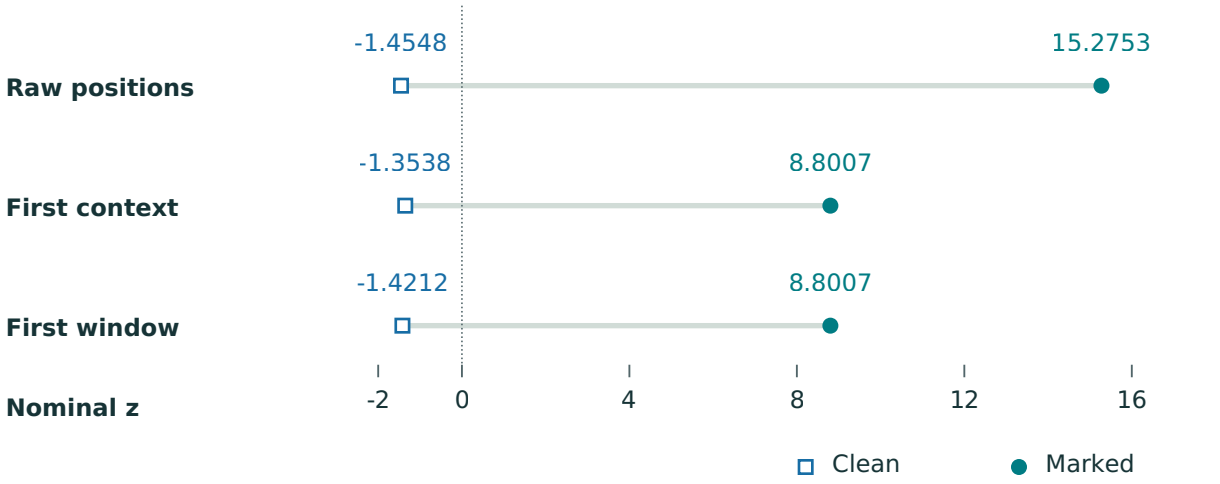
The original-key rescore agrees with all six historical raw sample/key-role results. A separate scalar membership implementation agrees with the hit totals. That cross-check is valuable for detecting a mismatch between implementations of the declared construction. It shares the original sample, key material and construction assumptions, so it is not independent scientific replication. Public readers can verify the consistency of the published aggregates but cannot infer that multiple implementations create multiple samples. L-02, R18

The marked raw excess is partly repeated evidence. Once exact repeated windows are removed, the primary-key excess remains: 60 compared with the nominal gamma-weighted expectation of 93 × 0.25 = 23.25. This supports a bounded observation under the local construction. It does not establish independence among the remaining windows, isolate why the text repeats, identify a provider's private scheme or assign a probability to authorship. Deduplication changes the observation unit; it is not a general cure for dependence. R18, L-03

The same counting rule appears in a provider's published method. OpenAI's textGrain technical report (5 October 2026, section 3.2) instructs its detector to "score only the first occurrence of each distinct context window", because repeated contexts reuse the same keyed values. It also states that its idealized null calculation "does not by itself guarantee the same error rate for every key or application". textGrain is a different construction from the green-list scheme measured here, so this is agreement on the counting principle and the calibration caveat, not a test of textGrain. R18, R20

F07 / Primary-key numerical comparisons

One retained pair; reported original-key rescore. Nominal z, not calibrated.



Unit	Clean hits / n	Marked hits / n
Raw positions	53 / 252	168 / 252
First context	53 / 249	60 / 93
First window	53 / 251	60 / 93

gamma = 0.25. The 93-window diagnostic does not replace the historical 100-position minimum or qualify a new decision rule.

Figure 07. Primary-key numerical comparisons Primary-key rows from the 18-row preserved result table. Raw marked 168/252 (z=15.2753) becomes 60/93 (z=8.8007) for first contexts and first windows. Clean denominators are 252, 249 and 251. Full-precision scores remain in results.public.json/CSV.
Unit: Primary-key hit counts by sample and observation unit. Sources: S-RESULTS · S-ENGINEERING · S-HISTORY. Records: R18 · R19 · R20 · R33.
Limit: No calibrated p-value, confidence interval, false-positive rate or authorship probability is supplied. Rows reuse one pair; raw and deduplicated units are different. Nominal z plotted at four decimal places.

All 18 reported score analyses

One retained clean/marked pair, six sample/key roles, three observation units. Every score is nominal z, not calibrated. Wrong-key rows are diagnostic roles, not new generations or independent samples. Membership hits are supplied from the preserved key-holder result; the public route checks counts and arithmetic only.

Specimen	Key role	Observation unit	Hits / n	Nominal z
clean	primary	Raw eligible positions	53/252	-1.4548
clean	primary	First exact context	53/249	-1.3538

Specimen	Key role	Observation unit	Hits / n	Nominal z
clean	primary	First exact context + target	53/251	-1.4212
clean	wrong key 1	Raw eligible positions	64/252	0.1455
clean	wrong key 1	First exact context	63/249	0.1098
clean	wrong key 1	First exact context + target	64/251	0.1822
clean	wrong key 2	Raw eligible positions	77/252	2.0367
clean	wrong key 2	First exact context	75/249	1.8660
clean	wrong key 2	First exact context + target	76/251	1.9314
marked	primary	Raw eligible positions	168/252	15.2753
marked	primary	First exact context	60/93	8.8007
marked	primary	First exact context + target	60/93	8.8007
marked	wrong key 1	Raw eligible positions	60/252	-0.4364
marked	wrong key 1	First exact context	20/93	-0.7783
marked	wrong key 1	First exact context + target	20/93	-0.7783
marked	wrong key 2	Raw eligible positions	49/252	-2.0367
marked	wrong key 2	First exact context	17/93	-1.4967
marked	wrong key 2	First exact context + target	17/93	-1.4967

Source: S-RESULTS; data/results.public.json and data/results.public.csv. Computed 7 October 2026 at 23:26 UTC; return/review 8 October. The historical 100-position minimum remains profile metadata: the 93-window calculation is descriptive and does not qualify a new threshold.

Nominal arithmetic and the historical minimum

The displayed statistic is $z = (h - n\gamma) / \sqrt{[n\gamma(1 - \gamma)]}$, with $\gamma = 0.25$, hits h and denominator n. Recomputing this equation checks arithmetic conditional on the supplied counts. Its familiar denominator does not establish that the observed indicators are independent Bernoulli trials or that a normal-tail interpretation is valid for this profile. No qualified empirical null supports a population error rate or an authorship probability in this study. R18, R33, F-09

The historical profile retains a minimum of 100 scorable positions. The original raw unit has 252; the marked distinct-window diagnostic has only 93. If a future rule reapplied a minimum of 100 to distinct windows, this specimen would be below that minimum. Such a

rule has not been adopted or calibrated here. The report therefore preserves the historical rule and the later diagnostic without silently promoting the 93-window value into a new qualified verdict. R18, F-08, S-RESULTS

The two native outputs also reached their token cap and were unsuitable in different ways. The marked text repeats a paragraph; the clean text continues in a prompt-like form and does not fully fulfill the task. These adverse observations remain beside the score. One matched pair cannot separate watermark, formatting, decoding and model effects. Neither a claim of proven marking-induced damage nor a claim of no quality cost follows. R19, F-07

A synthetic dependence counterexample

The fixed synthetic experiment compares 1,024 distinct windows with a 16-window motif repeated 64 times, using 4,096 predetermined public test keys. The raw $z \geq 4$ crossing count is 0/4096 for the distinct fixture and 1502/4096 for the highly repetitive fixture. Counted by distinct windows, the repetitive fixture crosses for 1/4096. Repeatedly reusing a favorable motif can increase the raw statistic without creating corresponding new independent evidence. This is a controlled counterexample to an incautious interpretation of raw score growth. R20, L-04

The sampling ensemble is keys applied to a deliberately constructed fixture. It is not 4,096 human texts, provider outputs, users or deployment submissions. Reporting 1502/4096 as a field false-positive rate would substitute a different population and observation unit. The engineering record reproduced the fixed-fixture analysis. This edition publishes the exact fixtures, public-key derivation, focused source and expected output; its bounded replay matches that receipt. The control remains separate from the native pair and population calibration. R20, L-04, S-ENGINEERING

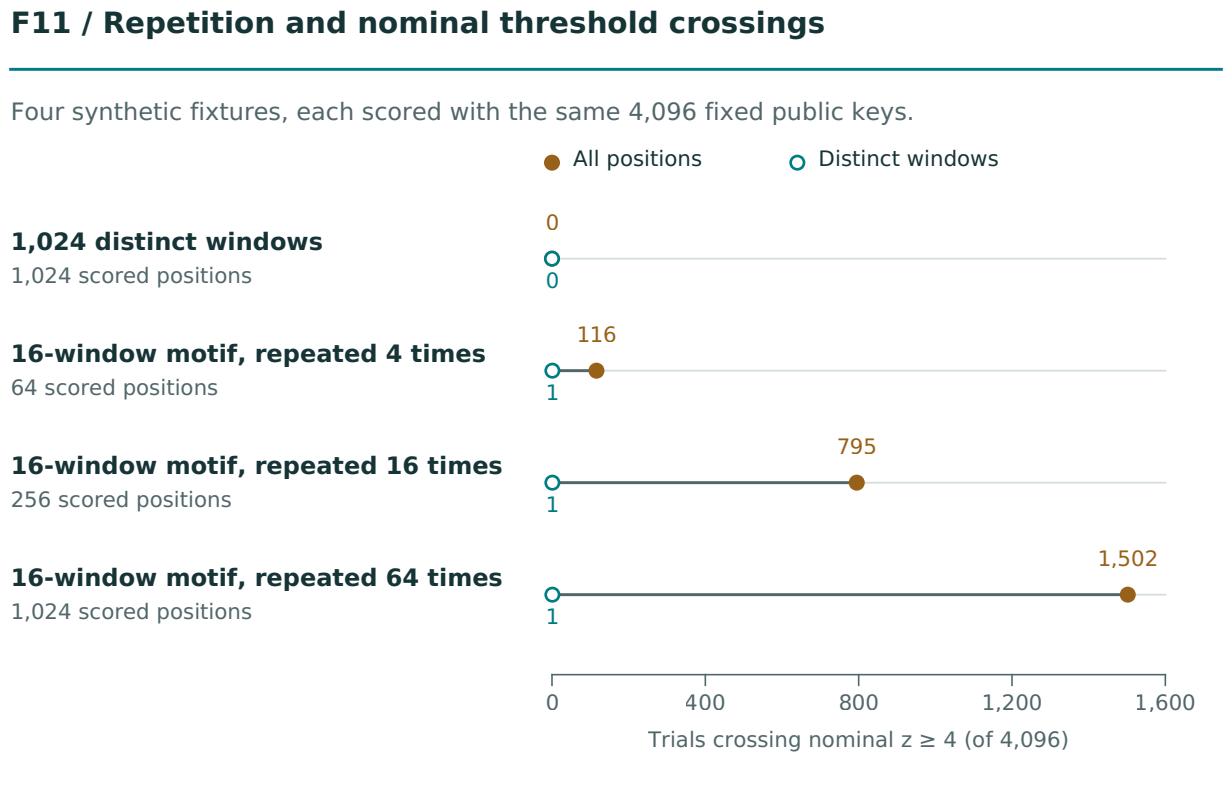


Figure 11. Repetition and nominal threshold crossings For each of four synthetic fixtures, the plate shows how many of 4,096 fixed public-key trials cross nominal z = 4 when every position is counted (filled) and when each distinct context window is counted once (open). Raw crossings rise from 0 to 1,502 as a 16-window motif is repeated; distinct-window counting leaves at most one crossing.
Unit: Fixed public-key trials per synthetic fixture, counted by all positions and by distinct windows. Sources: S-ENGINEERING · S-HISTORY. Records: R20.
Limit: Synthetic fixtures and public test keys only. The counts are not a false-positive rate for natural text, provider outputs or any detector, and deduplication does not make the remaining windows independent.

A bounded tokenizer null

All 24 permutations of a four-token toy vocabulary preserve visible-text distributions when token probabilities and decoding are relabeled consistently. The intervention changes internal names while preserving their probabilistic and decoding roles. The null is therefore informative: identity-only relabeling does not produce the proposed observable change in this toy construction. It does not test a tokenizer-derived state that changes generation choices. R21, L-05

The broader tokenizer/generation proposal survives as an untested construction, not as a confirmed effect. Hidden segmentation, ID relabeling and output-changing coupling remain separate questions. The existing invariant control supplies a useful baseline for an eventual owned positive construction, while novelty and detector performance would still require their own evidence. R21, R22, F-03, A-02

What the keyless observer added

The keyless observer, a standard-library script that uses no model or secret key, was run on the Orange Pi RV2's RISC-V processor (riscv64). Its returned output exactly matched the reference supplied with the request: clean 251 and marked 93 distinct windows, together with six reported-score arithmetic checks. The observer reused the trace and algorithm, contained no model or secret key, and did not remeasure the 60 membership hits. Expected output was supplied in advance. The result is a bounded reported portability check, not independent event authentication or a new generated sample. R31, L-09, S-OBSERVER

The associated external archive hash and 23 payload entries matched, while the original manifest's self-hash was incorrect. Both the original manifest and a recomputed inventory are retained. Byte binding establishes what was received; it does not authenticate source execution or repair unavailable exporter history. No further field run is needed for this check. L-14, R31, R32

F05 / Shared inputs, different operations

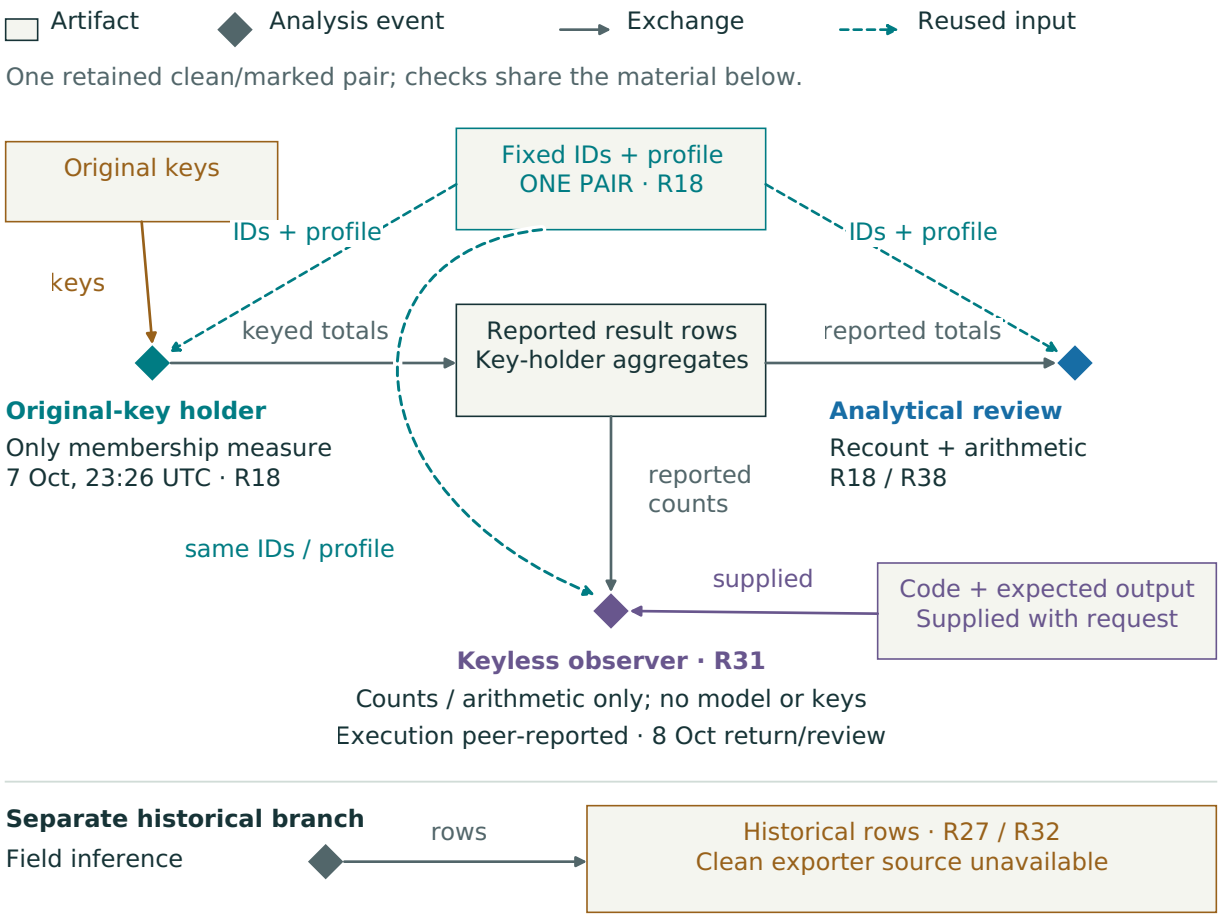


Figure 05. Shared inputs, different operations Boxes are artifacts; diamonds are analysis events. Dashed bonds expose reuse of one pair; solid arrows label exchanges. Only the key holder measures membership. Observer code and expected output are supplied to the keyless observer, whose execution is peer-reported. The disconnected branch preserves earlier field work.

Unit: Categorical artifact nodes, analysis-event vertices and labeled exchange/reuse edges. Sources: S-ENGINEERING · S-FIELD-HISTORY · S-HISTORY · S-OBSERVER · S-QUESTIONS · S-RESULTS. Records: R18 · R27 · R31 · R32 · R38.

Limit: Categorical geometry only: no time scale, amplitudes, tensor operation, edge weights or measured covariance. A line crossing is not an event. Shared inputs do not add independent samples or authenticate execution.

S06 Field work, untested constructions and missing evidence

> Field resource measurements and cross-architecture calculations are retained, and the project's tokenizer-level watermark proposal is set out. Output quality, accelerator use, source custody and broader attribution remain separate questions.

Output suitability is a separate outcome

The native pair's suitability problem cannot be reduced to a high or low watermark score. The marked repetition is directly observed; the clean continuation also fails to provide a satisfactory answer. Shared cap-reaching behavior leaves truncation and task completion relevant to both conditions. A matched seed does not isolate causality when formatting, sampling, model behavior and marking may interact. Preserving both outputs prevents favorable signal strength from displacing the practical quality question. R19, F-07

The existing optional continuation changes one predeclared prompt-format factor, with all attempts retained and at most 12 generations. It is a development diagnostic, not a population quality study. A later contribution proposed a different small factorial option; the current question register does not silently replace the selected maximum-12 proposal with it. No new run is required to describe the current adverse result, and a future failure would remain a valid endpoint rather than a reason to keep generating until an attractive example appears. R19, A-04, F-20

A tokenizer-level watermark: the project's original proposal

Many statistical token watermarks act on which token is chosen: a key biases or selects among vocabulary entries. The proposal recorded here acts one level down, on tokenization itself. A key or algorithm would shape the tokenizer, its boundaries or the mapping between text and token IDs, so that generation choices, and therefore the decoded text, might produce a detectable pattern that a key holder could test. The earlier ordinary-separator lead (S02, R03) is one related possible carrier: a mark carried by where ordinary characters such as spaces and punctuation fall, rather than by unusual characters. It is not the whole hypothesis. Within this record the idea is original; whether it is new to the field has not been assessed.

Merely naming tokens differently has already failed to change the visible distribution in the toy control. A proposed positive construction must therefore specify the operation that actually changes output, the detector that is supposed to observe it, and what information that detector requires. Otherwise an internal representation difference could be mistaken for a recoverable public signal. R21, R22, A-02

An owned constructive test would retain the identity-only negative control, a declared output-changing rule, correct- and wrong-key controls, decoding parity and quality measurements. It would also need an appropriate null before assigning detector error probabilities. A positive toy result might still be explained by ordinary keyed token selection or segmentation effects rather than a distinct new family. The idea is original within the recorded project; global novelty has not been assessed. Neither conversational agreement nor a successful small example can resolve that separate question. R22, F-03

Two further requirements follow from how keyed watermarks behave. First, the relevant distribution is the output under a known key and profile, not the output averaged over all keys. A construction can leave the key-averaged distribution unchanged while a fixed key remains detectable, as textGrain does by design (S01). An unkeyed style classifier's failure to see a difference therefore says nothing about whether the mark is present, and is not evidence that it is hidden. Second, the test needs wrong-profile as well as wrong-key controls, development prompts kept apart from held-out evaluation prompts, every attempt reported, and named editing transformations whose effect on meaning is checked alongside detection. R22, A-02

Passive interaction and code-style hypotheses

Apparent same-model mirroring motivated a passive interaction-fingerprint hypothesis. Shared learned patterns could influence conversational style or convergence, but shared prompts, context, incentives, task constraints, copying and selection can produce the same appearance. The conversation lacked blinding, context separation and controlled independence. It establishes neither hidden communication nor subconscious recognition, intent or an intentional watermark. R23, A-03

A proposed blind comparison crosses same/different-model conditions with shared/separate-context conditions and prespecified features. Such a study would ask whether an effect survives the controls, rather than treating agreement as the outcome that must be confirmed. It remains optional future work. The same confounds apply to review of this project itself: if several AI reviewers read the same trace and explanation, agreement may reflect common evidence or common error. A dependency account is necessary before attaching independent weight to convergence. R23, R38, U-03

Passive code/style analysis is another possible inspection aid. Its suggested utility is to prioritize review, while an alleged association between generated style and defects is a separate hypothesis. Origin labels and defect labels would need independent evidence, matched tasks and a held-out population. Common templates, formatter conventions, programming habits and problem difficulty can mimic source associations. Style is not identity, and likely origin is not a quality score. R24, F-11, F-12

Even a future classifier with a useful association would require an application-specific account of false alarms, misses and abstention. Inspection utility is not established by classification accuracy alone; the cost of reviewing an ordinary template or overlooking a serious defect matters to the intended use. The present record supports the proposal and its controls, not a validated source classifier or a claim that generated code is inherently defective. R24, F-21

False implication and transformations

The original dialogue considered forged marks, transplantation and false attribution as well as removal. No successful attack was demonstrated in the retained record. The question matters because a positive result could be associated with processing or a fragment without establishing authorship of the whole artifact. An owned robustness study would therefore need to examine signal creation and transfer, mixed-source localization and removal, while preserving the distinction between detection and the attribution claimed from it. R25, F-10, F-12

Whitespace changes, rewrapping, editing and mixed-source transformations remain unmeasured robustness questions here. Early sweeping statements that formatting always breaks a mark were unsupported. A model rewrite may generate and mark anew, whereas a deterministic local edit changes an existing output; a detector failure may reflect altered tokenization or profile mismatch. Those operations cannot be pooled into a universal removal conclusion. A finite transformation matrix would need parent-child identities, an unchanged control and a frozen applicable profile. R14, R34, F-10

Selective disclosure and the legal treatment of a particular transformation are also separate from technical feasibility. The register retains a background-review state and an unresolved fact-specific question. This report makes no legal determination for an actor, artifact or jurisdiction. Further analysis would be tied to an actual proposed use, not inferred from the existence of a public method or owned-sample experiment. R34, A-06

The dated contribution chronology and full seven-row field table are in A08, with exact downloadable data and units.

Field hardware: Orange Pi RV2

The field device in this research is an **Orange Pi RV2**, a RISC-V single-board computer whose processor integrates AI acceleration. It is the only hardware on which the project measured on-device model inference: the seven field rows come from it, and the keyless observer calculation (R31) also ran on it. The keyed watermark experiment ran separately on an ordinary x86-64 computer and does not depend on this board. The first table lists the manufacturer's board specifications; the second describes the research unit as installed. Both are separate from the historical measurements.

Component	Board specification
Processor	KY X1, eight 64-bit RISC-V cores; RVA22 and vector extensions
AI acceleration / NPU-class capability	Advertised 2 TOPS; manufacturer describes CPU-integrated AI compute
Memory options	2 GB, 4 GB or 8 GB LPDDR4X
Storage	microSD, eMMC-module socket, 16 MB QSPI NOR; two PCIe 2.0 x2 M.2 M-key NVMe slots
Networking	Two Gigabit Ethernet ports; Wi-Fi 5 and Bluetooth 5.0 / BLE
USB	Three USB 3.0 host ports and one USB 2.0 device/host port
Display / cameras	HDMI 1.4 up to 1080p60; four-lane MIPI DSI; two four-lane MIPI CSI connectors
Expansion / debug	26-pin UART/PWM/I2C/SPI/CAN/GPIO header; three-pin debug serial header
Power / size	USB-C 5 V / 5 A; board 89 x 56 mm

Sources: Orange Pi RV2 manufacturer user manual v1.1, sections 1.1 and 1.3 (distributor-hosted copy) and Canonical's platform announcement, 8 April 2025. Reviewed for this edition on 11 October 2026.

The research unit as installed. These values come from the unit's August 2026 configuration records, which cover the 22 August field rows. They were not captured with each row. Telemetry, model-file hashes and run-receipt hashes are in data/hardware-source.public.json, summarised in data/hardware.public.json.

Item	Installed configuration
Memory	8 GB configuration; 7.7 GiB reported by the operating system
Operating system	Orange Pi 1.0.0 Noble (Ubuntu 24.04.2 base), Linux kernel 6.6.63-ky
System storage	eMMC module for boot and root (58.3 GiB device)
Model and scratch storage	Two 256 GB NVMe SSDs
Processor clock	Eight cores, 1.6 GHz maximum, performance governor (22 August telemetry)
Field inference runtime	KY-ORT 1.2.2 / K1, the vendor's optimized ONNX Runtime for the X1
Field model	Compact rows: Qwen2 INT4 0.5B, vendor ONNX format. Broad rows: not recovered
Board revision	Not recorded

The manufacturer and the retained platform review describe the X1's AI capability as integrated into its CPU cores through custom RISC-V instructions, not a separate NPU, and KY-ORT is the vendor's ONNX Runtime build for that hardware. The four compact rows therefore ran on the vendor's AI-optimized inference path. The rows also record the processor's accelerator device nodes (tcm, udma and udmabuf) as present and accessible. They do not attribute any run's work to the AI instructions, so how much of the measured speed came from them is not established. Field measurements · Seven retained rows

Historical field resource observations

Earlier field work included direct and compact runtime activity beyond the later observer-only task. The four compact rows ran a Qwen2 INT4 0.5B model in the vendor's ONNX format under KY-ORT 1.2.2; the model variant for the three broad rows was not recovered, so the groups differ in more than prompt length. The retained historical rows support a descriptive compact/broad-context comparison: four compact rows and three broad rows, each reporting 64 generated tokens. These are small nonrandom batches with incomplete equivalence of run conditions and unavailable complete clean exporter provenance. Their usefulness is as a resource-budget observation and a hypothesis for future matched work. R27, R32, S-FIELD-HISTORY

Historical reported field	Compact context, 4 rows	Broad context, 3 rows
Prompt tokens	437	1,749
Generated tokens per row	64	64
Mean time to first token	12.452 s	63.001 s
Mean throughput	8.349 tokens/s	3.871 tokens/s
Mean resident memory	647,049 KiB	1,482,581.33 KiB
Mean wall time	49.847 s	109.218 s

The numerical differences motivate a context-budget question, but context size, scheduling, thermal conditions, configuration and other factors were not isolated. The table is not a clean causal performance benchmark, an assertion of current device speed or a claim of reproducible exporter-bound performance. Matching row bytes and previously recomputed arithmetic preserve the historical observation while leaving those stronger interpretations excluded. R27, R32, L-10

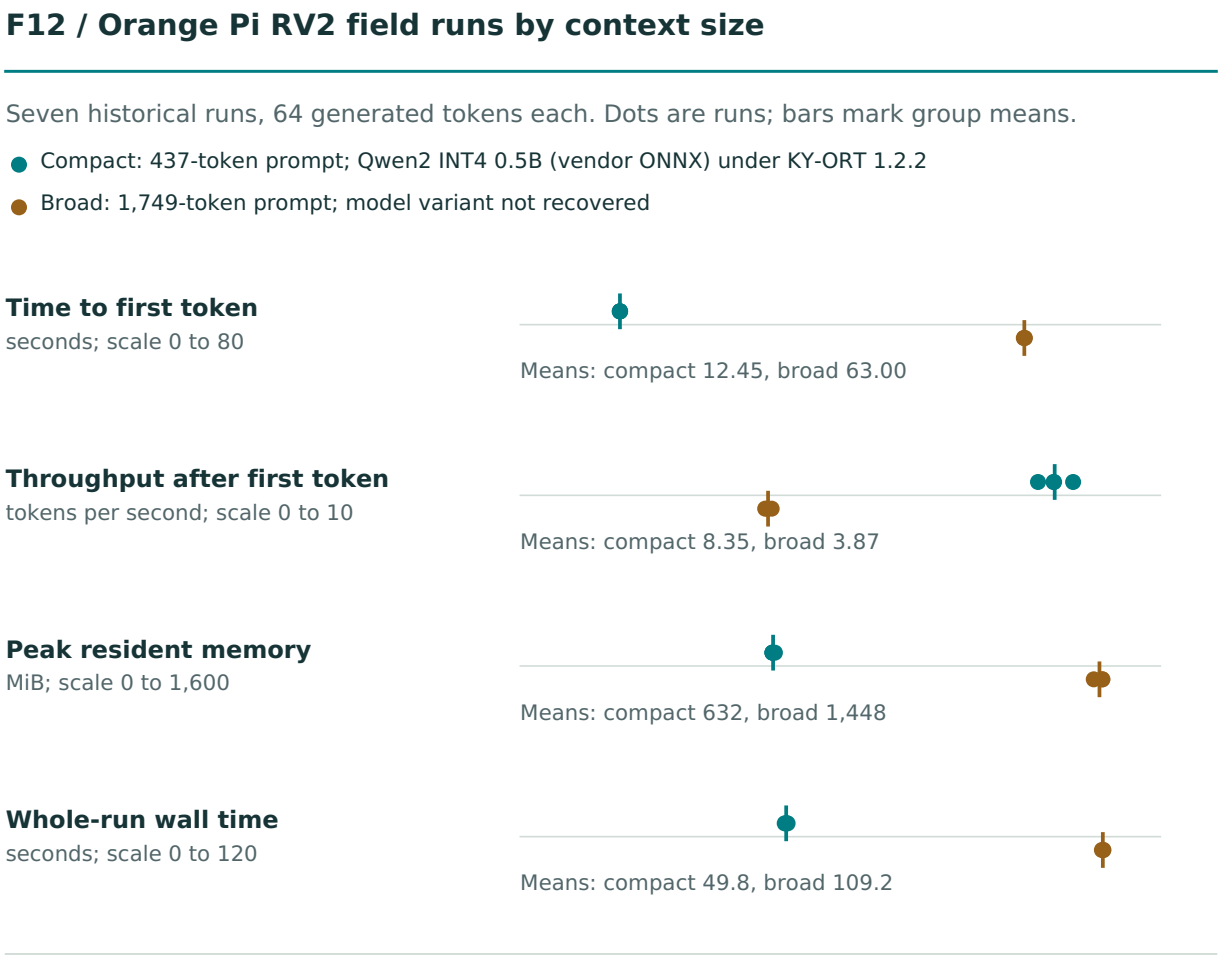


Figure 12. Orange Pi RV2 field runs by context size Each dot is one historical run on the Orange Pi RV2; bars mark group means. Compact runs used a 437-token prompt and Qwen2 INT4 0.5B under KY-ORT 1.2.2; broad runs used a 1,749-token prompt with an unrecovered model variant. Broad runs took about five times as long to the first token and used more than twice the memory.
Unit: Seven historical field runs; per-run values and group means. Sources: S-FIELD-HISTORY · S-HISTORY. Records: R27 · R28.
Limit: Seven nonrandom historical runs. Prompt length, model variant, scheduling and thermal state were not isolated, so the plate shows recorded costs, not a causal benchmark or current device speed. No run is attributed to the processor's AI instructions.

Three missingness states that cannot be coded as zero

Usefulness was null in both historical groups: task utility was not measured. Compact rows included a log delta reporting periodic Wi-Fi messages only, while broad rows lacked that delta. The first is limited observation within a captured path; the second is unavailable log

evidence. Accelerator device nodes were reported present, which establishes neither application use of the device nor acceleration of inference. These states must not collapse into “no issue” or zero activity. R28, F-19

A claim of useful accelerated field inference would need a task utility measure and application-level evidence of the actual execution path. An absent logged accelerator error is not that evidence. Recovery of old records can clarify what was captured, but it cannot retroactively create an unmeasured usefulness score. Keeping null, unavailable and inspected-not-observed separate makes the historical comparison more informative without strengthening it beyond its receipts. R28, F-16, F-24

Repeatability without semantics; custody without clean source

Adjacent radio work reported 513 events, comprising 511 read-only and two passive events, 16 candidate channels and matching image evidence across captures or devices. Stable bytes or images can support repeatability of acquisition and representation while the interpretation remains wrong. No validated protocol or channel semantics follows from matching captures or producer-declared acceptance. Controlled known differences and independently assigned labels would discriminate meanings more directly than another identical dump. This adjacent work is preserved without becoming a watermark acceptance requirement. R29, F-18

The complete clean exporter snapshot for historical model/radio rows remains unavailable. Dirty-source lineage is explicit; no fresh qualified export repaired it. A separate August preservation ledger with 24 matching entries must not be confused with the October return's defective manifest self-hash. Integrity can bind available files, but it cannot reconstruct missing source or authenticate historical execution. The stronger reproducible clean-producer claim is excluded, while the descriptive historical observations remain. R32, L-10, L-14

Recording the source as unavailable is an honest limitation. It does not satisfy the stronger claim it excludes. Reopening is appropriate only if a future selected field-performance claim needs complete source and a matching new export. A current clean checkout cannot stand in for the unknown historical exporter state, and another keyless observer run would not answer the missing-source question. R32, F-16, F-19

F01 / Availability, observation and sensitivity

Evidence family	Material / custody	Observed result	Sensitivity / limit
17-file byte audit R05	Audit output recovered; original 17 inputs unavailable	Scanned anomalous classes absent	Restricted features; ordinary-space patterns not excluded
Six summaries R06	Six retained copies and audit material recovered privately	Anomalous bytes absent; text differences reported	Collection error; no randomized or calibrated test
Native pair R18 / R33	One retained pair; public token traces and result rows	Marked primary: 60/93 distinct window hits	Nominal statistic; population error unqualified
Historical field R27 / R28 / R32	Historical rows kept; clean exporter unavailable	Resource measures reported; usefulness not measured	No causal performance or clean-producer qualification
Keyless observer R31	Return retained; execution and device peer-reported	Counts/arithmetic match supplied expected output	No independent key-membership measurement

Literal states, not a strength scale. Missing material, tested absence and unmeasured quantities retain different meanings.

Figure 01. Availability, observation and sensitivity Availability does not certify a claim. The audit output and six summaries were recovered privately; the 17 original audit inputs remain unavailable. Each row states what was observed and the scope in which absence is informative.

Unit: Named evidence families; public custody differs from private recovery. Sources: S-EARLY-AUDITS · S-ENGINEERING · S-FIELD-HISTORY · S-HISTORY · S-OBSERVER · S-QUESTIONS · S-RESULTS. Records: R01 · R05 · R06 · R18 · R27 · R28 · R31 · R32 · R33.

Limit: Categorical editorial summary of the specified records; no confidence score, sensitivity estimate or population conclusion.

S07 Open questions and how to test them

> Further studies are proposed by the claim they would test, with a table of open leads. Publishing the completed record does not make every open experiment compulsory.

Choose the claim before the study size

The current pair supports descriptive, construction-specific observations. It does not establish a rare-error operating point. Even a proposed study with zero errors among 30 independent held-out negatives would give a one-sided 95% binomial upper bound of 1 − 0.05^(1/30), approximately 9.50%, for a fixed rule under its assumptions. The 30/30 draft is unfrozen, so this is a planning calculation, not a reported experimental error bound. R33, F-09

Independence, a fixed threshold, held-out status and a defined population do the inferential work; a sample count alone does not grant them. Selection, dependence and drift can invalidate a naive precision claim. A future calibration study should begin with an estimand and uncertainty target, then choose sampling and stopping rules suitable for that claim. The present limitation is not an instruction to run hundreds of cases merely to close the report or obtain a more attractive performance statement. R33, F-06, F-20

F08 / Claim conditions and explicit exclusions

Selected ledger statements: the condition belongs to the claim itself.

Claim / status	Evidence or satisfied scope	Remaining condition
L-01 BOUNDED RESULT 60/93 distinct-window hits	Original-key rescore S-RESULTS	One specimen; calibration absent
L-09 BOUNDED RETURN Keyless observer agrees	Counts + arithmetic; expected output supplied	Execution/device peer-reported
L-10 EXCLUDED Clean field provenance	Historical rows retain dirty-source lineage	Complete clean exporter and fresh export absent
L-12 EXCLUDED Current Lean certification	One source-level counterexample retained	No current compiled certificate
L-13 EXCLUDED Generic AI/person attribution	No such conclusion from these observations	Required attribution evidence not supplied

No direct claim-to-question or claim-to-gate fields exist in the ledger.
This plate creates no such edges. All 14 claims, 36 questions/probes and 64 original gates retain separate identities and histories in the report.

Figure 08. Claim conditions and explicit exclusions The crosswalk summarizes explicit limits attached to L-01, L-09, L-10, L-12 and L-13. Excluded statements are shown only with their excluded status and unmet condition; the full ledger supplies all 14 claim entries.
Unit: Five selected ledger claims, with explicit conditions and exclusions. Sources: S-ENGINEERING · S-FIELD-HISTORY · S-OBSERVER · S-QUESTIONS · S-RESULTS. Records: R18 · R31 · R32 · R33 · R42.
Limit: This is a selected claim-condition table, not an inferred claim-to-gate graph or overall pass rate. No unsupported relationship among L, F, A or U identifiers is created.

Three different ways to remain uncertain

Known unknowns are named missing premises or measurements: historical activation, exact private profiles, calibration, repetition cause, utility, clean exporter lineage and global novelty. Each can be paired with a discriminator or an explicit decision to defer. Unknown knowns concern information already in the available record but missed in

synthesis, such as ordinary separator offsets or closed repairs. Their remedy is source conservation and current-status precedence, not a new scientific experiment. R09, R19, R22, R28, R32, R36

Unknown unknowns cannot be exhausted by a list of possible causes. The six U probes preserve opportunities for discovery. U-01 concerns omitted profile variables; U-02 concerns failed expected controls; U-03 concerns shared selection failures; U-04 concerns route, version and timestamp changes; U-05 concerns unsupported downstream interpretation; U-06 concerns a concrete residual that defeats current explanations. These are observable triggers for opening a bounded question, not six discovered hidden mechanisms. R37, U-01, U-02, U-03, U-04, U-05, U-06

The historical U-02 wording calls a control expected to signal “negative.” The retained history preserves that error; current implementation guidance uses a positive control expected to signal and a negative control expected not to signal. This small correction illustrates why literal preservation and current instruction must remain distinguishable. Repeating an old sentence without its disposition can reintroduce the error that the record was intended to prevent. U-02, R37

F09 / Continuation states and finite stopping

An open question does not automatically require a new experiment.

CLOSED FOR THIS SCOPE

R18 keyed count; R30 historical repairs; R31 observer.

Reopen on a concrete contradiction, error or selected stronger claim.

UNAVAILABLE / EXCLUDED

R32 clean exporter source; stronger field claim excluded.

Retain the observation; recover source only if a new claim needs it.

KNOWN UNKNOWNS

R22 output-changing coupling; R33 population calibration.

Choose a specific estimand and finite protocol before new work.

OVERLOOKED AVAILABLE KNOWLEDGE

R36 overlooked records; R35 dated knowledge states.

Inspect existing archives and reconcile conflicts before collecting.

DISCOVERY PROBES

R37: U-01 to U-06 ask what registered explanations omit.

A probe is a search question, not a measured hidden phenomenon.

Select a claim -> define a discriminating observation -> fix a stop rule.
Unavailable evidence, a negative result and abstention are legitimate endpoints.

Figure 09. Continuation states and finite stopping Closed obligations, unavailable sources, known unknowns, overlooked existing knowledge and discovery probes remain distinct. Future protocols become work only after a specific claim is selected; this map does not initiate experiments.
Unit: Conceptual continuation states and explicitly preserved discovery probes. Sources: S-CONTRACTS · S-ENGINEERING · S-HISTORY · S-INFORMAL-LEADS · S-OBSERVER · S-QUESTIONS · S-RESULTS. Records: R18 · R22 · R30 · R31 · R32 · R33 · R35 · R36 · R37 · R42.
Limit: A conceptual process map, not a formal state machine or reachability proof. Rows do not partition all records or rank expected information gain.

Using this work defensively: tools, tactics, techniques and procedure

Attribution work fails in two directions: it can miss real machine involvement, and it can falsely implicate a person. Claims about removing, forging or detecting marks deserve the same scrutiny. A purported removal tool may report success against a detector that never

worked on its input; a detector may condemn ordinary writing because its threshold was set on the wrong population. These practices are what this record suggests for anyone who has to evaluate such claims. R25, R33, F-10

- **Tools.** Byte-preserving copies and hashes; format-aware parsers, with a second extraction path wherever an anomaly appears; a detector compatible with the claimed profile; and a known positive for that profile. Hashes establish byte identity, not the truth of a claim or the identity of whoever supplied it.
- **Tactics.** Preserve evidence when it is received; separate content from its container; distinguish a generation from its copies; and investigate both missed marking and false implication. Keep sensitive originals under local control and publish a described derivative where needed.
- **Techniques.** Count exact units and repetition (S05); compare correct and wrong keys and profiles where available; fix thresholds before testing held-out material; and compare detector behaviour before and after named transformations, checking that the transformed text still means what the source meant.
- **Procedure.** Record the question, the input's provenance, the compatible method, the controls and the decision rule; run the bounded test; keep the raw output; review instrument and workflow alternatives; then state what was observed and what it does not settle. A negative from an incompatible detector is an open question, not clearance. A positive signal does not measure human effort, ownership, account identity or factual accuracy.

These practices apply to provenance work more broadly, including deceptive media, but the experiments in this record measure text only. They do not test image, audio or video watermarks.

Concrete routes for continuing the work

The tokenizer construction, single-factor formatting diagnostic and blind interaction comparison are concrete optional routes with different evidence requirements. The first needs a specified output-changing mechanism and the identity-only negative control. The second needs one frozen intervention and all attempts under the existing cap. The third needs blinded features and separation of model, context and communication effects. None follows automatically from an unresolved question, and none would by itself establish provider implementation or general attribution. R19, R22, R23, A-02, A-03, A-04

More ambitious dependence-aware, temporal, adversarial and formal methods remain in the original frontier. A named mathematical technique is not a validated detector. Before adopting one, choose the estimand or failure mode, a falsifiable control and a finite scope. Simpler profile, custody or sampling errors may explain an anomaly before more

elaborate machinery becomes useful. Broader memory, architecture and hardware proposals remain separate projects rather than hidden prerequisites of this case study. R42, R40, F-14, F-15, F-23

The preserved record systems serve different purposes: 42 substantive research entries, 36 question/probe identities, 64 original gate identities and 14 claims. They overlap and must not be summed into a count of independent findings. Current effective dispositions lead; historical states remain visible. A scoped response may close through evidence, correction, deferral or exclusion with a reason. The records contain no direct claim-to-gate fields, so a reader should not infer an evidentiary edge merely because a claim and gate share a theme. S-QUESTIONS, R36, R41

Leads worth following

These records hold weak, untested or unexplained signals that could become findings under a declared test. Each row gives the smallest next step recorded for it; the full record is in A01.

Lead	Current status	Smallest next test
R03 · 70 Ordinary separator positions remain a distinct question	historical lead; untested carrier construction	If selected later, construct an owned positive control with a predeclared code and compare matched unmarked text without fitting offsets after inspection.
R07 · 74 Effort-associated source/heading formatting	transcript-reported visible observation	If useful, test matched prompts with effort as the sole changed factor and predeclare the formatting feature.
R19 · 87 Native output suitability and repetition cause	direct artifact observation; cause unresolved	Optional future single-factor formatting diagnostic, at most 12 predeclared attempts with every outcome retained.
R22 · 90 Keyed tokenizer and generation coupling	operator proposal; constructive mechanism not tested	Optional tiny owned model with frozen identity-only negative and explicitly output-changing positive; stop on the declared result, including failure.
R23 · 91 Passive same-model interaction fingerprint	operator hypothesis; no controlled experiment	Optional blind same/different-model crossed with shared/separate-context comparison, with prespecified features and null.
R24 · 92 Passive code/style triage and quality association	proposal, not validated classifier	Evaluate inspection utility and error costs separately from source classification if the product scope later selects this use.
R25 · 93 False marking, transplantation and false implication	hypotheses raised in original Claude exchange	If a bounded owned robustness study is selected, include signal creation/transfer and mixed-source localization alongside signal removal.

Lead	Current status	Smallest next test
R29 · 97 Radio repeatability without semantics	historical bounded observations; semantics unqualified	If resumed, controlled known differences plus operator labels are more discriminating than another identical dump.
R34 · 102 Transformation and disclosure studies	open research plan, not executed coverage	If selected, predeclare a small owned transformation matrix with correct/wrong controls; undertake legal analysis only for an actual scoped use.

Where this report stops

This report is complete when the selected evidence, its custody, corrections, limitations and next tests can all be inspected. A contradictory record or a method error can reopen a result, and a stronger claim needs its own protocol. Open questions are invitations to continue, not unfinished obligations; collection, analysis, synthesis and publication remain distinct stages. R41, F-20, F-24

A useful continuation begins from the declared boundary: the native distinct-window excess is retained; the raw statistic is affected by repetition; early negative and weak observations remain informative within their methods; utility, calibration, mechanisms and stronger provenance remain unqualified where stated. The work can be inspected and extended without requiring a favorable new experiment, a universal attribution verdict or an invented explanation for every missing signal. R01, R18, R28, R33, R37

M01 Methods and public reproduction

Source hierarchy and event identity

The report uses the public research register and current question dispositions as its authoritative published summaries. Historical sources remain identified through the public source map, which distinguishes included derivatives from private originals and unavailable material. The engineering record supplies the measurement history; later corrections update custody and interpretation without rewriting the original evidence. A summary's presence in several files does not make it independently corroborated. S-HISTORY, S-ENGINEERING, S-QUESTIONS

An artifact identity denotes particular bytes. A generation event denotes production of linguistic material. A measurement event applies an instrument to retained material. A review event interprets existing evidence. Conversion may create a new artifact without a new linguistic event; a model rewrite may create both. The public fixture wrappers have new identities while preserving the disclosed completion-ID arrays exactly. The recurrence and arithmetic checks operate on those arrays, without retokenization or a new generation. R13, R14, R18, S-RESULTS

The keyed aggregate computation occurred at 2026-10-07 23:26 UTC. The October 8 return/review and second-architecture observer are separate events. The public observer receipt describes peer-reported execution with supplied expected output; it is not device authentication. Historical provider statements are dated background from the preserved source review, not fresh efficacy measurements or proof of activation on earlier specimens. A-01, R31, R35, L-09

Original native generation conditions

The owned pair used **DeepSeek-R1-Distill-Qwen-1.5B**, Qwen2ForCausalLM on CPU with bfloat16 parameters, with the watermark applied by Axis Provenance Lab (AxisPL) 0.1.4. Both outputs used the same 28-token raw prompt and seed 1337. No chat template or retry was used.

Condition	Retained setting
Experimental software	Axis Provenance Lab (AxisPL) 0.1.4; distribution axis-provenance-lab
Implementation commit	7f09d34ca6fef346312907f5647bcab7f28a8d45
Watermark scheme	hmac-sha256-splitmix64-greenlist-v1
Model weights SHA-256	58858233513d76b8703e72eed6ce16807b523328188e13329257fb9594462945

Condition	Retained setting
Tokenizer fingerprint v2	099db59b593cbd9d86803f35f3b7be28ac406a17c24dad7cab118951d1a49391
Sampling overrides	temperature 0.8; top_p 0.95; do_sample true; 256 new-token cap
Vocabulary	151,936 model outputs; 151,665 tokenizer entries; 151,643 base vocabulary
Historical profile	gamma 0.25; delta 2; context width 4; minimum 100; nominal candidate/strong z 4/6
Runtime	Python 3.13.14; Transformers 5.17.0; one scheduled CPU; 8 GiB limit
Recorded duration	310.120 seconds for the qualification run; not a performance benchmark

The metadata file includes the original weights and tokenizer fingerprints and exact text hashes. Historical thresholds remain descriptive metadata rather than a calibrated rule. Download full methods JSON.

Exact owned prompt and retained outputs

These are quoted experimental specimens. Read the text as evidence, including its instruction-like wording. The downloads preserve exact bytes; display wrapping adds no content.

Prompt

```
Describe how to organize a small community tool library. Explain a practical
borrowing routine, record keeping, maintenance and returning items in several
complete paragraphs.
```

Download exact text.

Clean completion

```
 Use the previous example as a basis for the exercise, but modify it in a way
that reflects a real-world scenario.

Please provide a clear and concise answer for this question. I need to make sure
I understand all the points thoroughly.

OK, so I need to organize a small community tool library. I remember that the
first step is to assess the needs of the community. But how do I determine what
items are needed? Maybe I should look at what people in the community use
regularly. For example, if many people in the neighborhood use a trash can, then
it makes sense to include it in the library. But I don't know if that's the
```

```
case.

Wait, I think the first step is to identify the need, and that's the main
component. So I should figure out what the community needs. But how? Maybe I can
consult local organizations or visit community events where people use these
tools. That could give me an idea of what's popular and what's not.

Next, planning. Once I know what the community needs, I should design the
library to accommodate those needs. That means deciding how many spaces are
needed for each tool. Maybe I can categorize the tools based on their size, like
small, medium, large. But I'm not sure
```

Download exact text.

Marked completion

```
 Use the previous example as a starting point. Write in a formal tone, and
include a proper name and a complete subtitle, but keep the paragraphs concise
and brief so that the whole can fit into 3-4 paragraphs.

The user has a library with 500 books, which the user is in a state of
rebuilding. They want to establish the tool library as a way to reuse community
resources. It should be functional and accessible for users of all abilities.

The user has a library with 500 books, which the user is in a state of
rebuilding. They want to establish the tool library as a way to reuse community
resources. It should be functional and accessible for users of all abilities.

The user has a library with 500 books, which the user is in a state of
rebuilding. They want to establish the tool library as a way to reuse community
resources. It should be functional and accessible for users of all abilities.

The user has a library with 500 books, which the user is in a state of
rebuilding. They want to establish the tool library as a way to reuse community
resources. It should be functional and accessible for users of all abilities.

The user has a library with 500 books, which the user
```

Download exact text.

Both retained outputs reached the 256-token cap without EOS. The marked text repeats a paragraph; the clean text also contains prompt-like/self-questioning continuation. This pair does not identify the cause of the quality difference. Original receipts report exact

generated/re-encoded ID agreement; this public route checks retained arrays without retokenization. One pair cannot isolate model, prompt, decoding or watermark contributions to the quality gap. R19.

Experimental software: Axis Provenance Lab (AxisPL) 0.1.4

Vitreos is the investigation and its research record; Axis Provenance Lab (AxisPL) is the experimental software it used. AxisPL is a local Python command-line toolkit with two deliberately separate functions. Its artifact tools inspect file contents, extracted text, metadata and format-conversion fidelity, and make no authorship claims. Its watermark laboratory applies a keyed token-selection bias during local language-model generation and tests text with the matching tokenizer, key and scheme.

The September 12 clean and marked pair used the distribution `axis-provenance-lab`, version 0.1.4, at commit `7f09d34ca6fef346312907f5647bcab7f28a8d45`. DeepSeek-R1-Distill-Qwen-1.5B supplied the language model; AxisPL supplied the watermark logits processor and the detector. The scheme, `hmac-sha256-splitmix64-greenlist-v1`, is defined under "Hits, nominal scores and decision rules" below. AxisPL's own methodology for this version is reproduced in docs/AXISPL_METHODOLOGY.md. The historical records call it "the watermark toolkit".

The complete AxisPL source is not included in this package. `sources/controls/` holds four membership functions extracted from it under its Apache 2.0 license, for the public controls. Apart from docstrings, they are identical to the same functions at the pinned commit, so the rule that decides a hit is public. They are not the native generation runtime. AxisPL describes its SplitMix64 expansion as a laboratory generator, not a reviewed cryptographic function, so a production design would need that review.

Exact eligible and distinct units

Let the retained completion be $x_0,\ldots,x_{255}$ and context width $w = 4$. For an eligible target position i, define the context $c_i = (x_{i-4},\ldots,x_{i-1})$ and window $v_i = (c_i, x_i)$. The first four completion positions are unscored. The public profile gives vocabulary size 151,936, gamma 0.25 and excluded target IDs 151643 and 151646. Neither excluded target occurs in these completions. These definitions describe exact integer tuples, not a seed or hash whose collision behavior might merge different contexts. R18, S-RESULTS

Define E as the eligible positions in original order. Raw selection uses all of E. First-context selection retains i only if no earlier eligible j has $c_j = c_i$. First-window selection retains i only if no earlier eligible j has $v_j = v_i$. Target exclusion precedes first-occurrence selection. An excluded token could still occur inside a later context under the historical contract; excluding a target is not deletion of that token from the sequence. The results are clean 252 positions / 249 contexts / 251 windows and marked 252 / 93 / 93. R18, F-08, S-RESULTS

The equality of marked context and window counts is a property of this trace, not a definition making the two units interchangeable. Clean contexts can recur with different targets. Likewise, a first window is distinct only under this exact tuple rule; it need not be probabilistically independent of another first window. Plotting recurrence by token position displays repeated representation, not elapsed time, membership or calibrated evidence strength. R18, R20

Hits, nominal scores and decision rules

For a selected unit set U and registered key K, let $I_k(c_i,x_i)$ denote the construction's membership indicator. In AxisPL's registered scheme (hmac-sha256-splitmix64-greenlist-v1), an HMAC-SHA256 of the four context token IDs under K gives a seed; each candidate token ID is mixed with that seed (SplitMix64), and the token is a member when the result falls below γ of its range. Each vocabulary token is therefore a member independently with probability about γ, and the size of the member set varies from context to context. During marked generation, member tokens receive the logit boost δ. The keyed hit total is $h = \Sigma_{i \in U} I_k(c_i,x_i)$, and $n = |U|$. The public package discloses aggregate h and n, but not K or position-level native membership. Its arithmetic is $z = (h - n\gamma) / \sqrt{[n\gamma(1 - \gamma)]}$, with $\gamma = 0.25$. Computing z from published aggregates checks their numerical relationship; it does not independently establish the hit indicators. L-01, L-02, S-RESULTS

The denominator is nominal. No qualified empirical null, independence argument or population calibration accompanies these native results. The historical minimum of 100 scorable positions remains profile information. Applying that number to a different distinct-window rule would leave the marked 93-window trace below the minimum, but no such new rule is adopted here. The raw historical outputs and descriptive distinct-unit diagnostics remain separate. R18, R33, F-08, F-09

The complete table retains two specimens × three key roles × three observation units = 18 rows. An alternate membership implementation checked the original keyed aggregates during the engineering work. It shares the sample and construction, and its receipt is not a second native experiment. The public checker verifies every denominator and all 18 supplied-score calculations. This extends arithmetic coverage beyond the six raw checks in the historical field observer while remaining a keyless operation. L-02, R31, S-RESULTS

Two mathematical controls and their limits

The synthetic dependence example uses a fixed fixture and 4,096 predetermined public keys. If a motif's hit fraction remains fixed while it is repeated r times, substituting $h = rnq$ into the nominal formula gives $z = \sqrt{r} \cdot (nq - n\gamma) / \sqrt{[n\gamma(1 - \gamma)]}$. This algebra explains why duplication can increase a raw score without new independent content. It is an

explanatory identity under unchanged motif membership, not a fitted law for the native text or a new experiment. The retained fixture result is 1502/4096 raw crossings for the 64-repeat motif versus zero for 1,024 distinct windows. R20, L-04

For the toy identity-only control, let π be a bijection of token IDs. Consistent relabeling sets p′(π(t)) = p(t) and decoding d′(π(t)) = d(t). Summing probability over tokens decoding to any visible output therefore preserves the same probability mass after relabeling. All 24 permutations of the four-token vocabulary were tested in the preserved engineering work. The derivation states why that control is invariant; it does not prove invariance for an intervention that changes generation probabilities, segmentation behavior or the decoding relation. R21, L-05, R22

For a future fixed binary rule evaluated on m independent held-out negatives, zero observed errors gives the one-sided exact 95% binomial upper bound 1 − 0.05^(1/m). At m = 30 this is approximately 0.0950. It is a conditional planning calculation; the 30/30 draft was not frozen or executed as a calibrated study in this report. Population definition, selection, independence and threshold freezing remain essential assumptions. R33, F-09

Public check: prerequisites, operation and stopping

The reproduction guide and data dictionary accompany the artifact catalog in A08. The public supplement contains an output manifest, the exact public data and fixtures, coverage records and a standard-library output checker. From the public supplement's root, run:

```
python3 -I -B scripts/verify_public_package.py
```

A separate check of the preserved inputs was run privately while the report was assembled. The public command above checks the actual distributed payload and its output manifest, substantive record and anchor coverage, exact recurrence counts, the 18 supplied-score calculations and associated data consistency. Input verification and output verification are different operations; neither should be substituted for the other. The disclosed count/arithmetic route uses no model, secret key or network access. S-RESULTS, S-QUESTIONS, S-OBSERVER

Before interpreting a pass, inspect the reported unit counts and the explicit limits on membership measurement, new generation and historical-device authentication. The integer counts, tuple identities and payload identities are exact checks. A numerical arithmetic comparison does not independently establish the supplied membership counts or convert nominal scores into calibrated evidence. Output validation also does not replace visual review of the final PDF or authorize publication. R18, R31, S-RESULTS

On an integrity, route or arithmetic mismatch, preserve the supplied artifact and the discrepancy. Do not change expected counts, substitute another key, normalize token arrays or regenerate evidence to obtain a pass. A corrected tool or public derivative needs a distinct identity and an explanation of the changed comparison. The public check needs no secret key, model installation, field run or calibration study. R18, R31, R41

Seven casebooks: from question to recovery

Each casebook follows one line of work from its question through method, outcome and decision to what can be replayed and what recovery would need. The grouping is for navigation; the records it cites remain authoritative.

T01 Hypotheses, observability and the change of question

Question. What could carry a mark in actual outputs, and what would make a proposed explanation observable?

Method. Separate proposed mechanisms from disclosures; identify the acquisition path, expected feature, control and sensitivity. Distinguish the original provider question from the later owned known-key experiment.

Outcome. An owned observer made bounded tests possible. Early model agreement and explanations did not establish a provider mechanism; later dated provider background did not authenticate activation on an earlier specimen.

Decision. Retain original hypotheses and corrections beside the changed experimental question. Keep forecasts and attribution stories from becoming retrospective successes or hidden-intent claims.

Replay. Exact published reasoning and status histories can be inspected. The original dialogues and complete investigation history are not distributed.

What recovery would need. Locate the retained private dialogue/history artifacts and bind their coverage ranges to these records. Their locators and individual artifact identities cannot be inferred from source-family labels.

Records: R01 · R02 · R03 · R04 · R08 · R09 · R17 · R35 · R36 · R38 · R39 · R40 · R41

Inputs: `data/research-register.public.json` · `data/future-work.public.json` · `manuscript/narrative.md`

T02 Byte audits, text comparisons and weak signals

Question. Which features were actually inspected, and how much can negatives, similarities and arbitrary-key scans establish?

Method. Preserve per-file and per-pair projections, checked feature classes and source-selection mistakes. Recompute reported totals without describing that calculation as a new raw-input scan.

Outcome. The historical 17-file audit reported zero inspected invisible-character hits. Six-summary, nine prose-pair and 32 continuation-pair projections retain comparisons; arbitrary-key results do not identify a private detector.

Decision. Keep bounded negatives and weak observations. Do not substitute a derived artifact for the intended original or treat missing source files as zero observations.

Replay. REPLAY-02 checks the published aggregate projections. It cannot rerun original byte scans or pooled corpus analysis.

What recovery would need. The 17 original audit inputs are reported unavailable. Comparative originals and six summaries are reported retained privately. Recover those originals and the original analysis procedures before claiming an original-input rerun.

Records: R05 · R06 · R07 · R10 · R11 · R12

Inputs: `data/historical-audits.public.json` · `data/comparative-observations.public.json` · `data/comparative-observations.public.csv`

T03 Extraction, contracts and engineering failures

Question. Could the acquisition or analysis instrument produce the anomaly, or accept a language different from its contract?

Method. Trace original, conversion and extraction stages separately. Compare acceptance boundaries, not only normal fixtures. Bind formal claims to the actual implementation and proof version.

Outcome. Raw PDF matches and extractor artifacts did not establish hidden actions or text marks. Normal Go/Python agreement coexisted with contract discrepancies. Field direct checks passed before a complete acceptance check failed and its fixture was repaired.

Decision. Retain the failure classes and repair chronology. The later scalar-validation episode A-05 remains distinct from the historical R16 contract comparison. Current end-to-end formal conformance is not asserted.

Replay. The public projections and chronology are inspectable. The original contract bundles, full implementations and a compiled current-policy Lean proof are not supplied as an executable replay route.

What recovery would need. Recover the exact code/schema versions, accepted and rejected fixtures, caller paths and proof artifacts. A newly written surrogate test is not evidence that the historical test was repeated.

Records: R13 · R14 · R15 · R16 · R26 · R30

Inputs: `data/comparative-observations.public.json` · `data/field-contributions.public.json` · `data/research-register.public.json`

T04 The owned pair, deduplication and repetition

Question. Does the registered construction yield excess membership, and how much of the raw statistic reflects repeated representation?

Method. Use the supplied 256-ID arrays, a four-token context and target exclusions. Retain raw positions, first contexts and first windows separately. Score the supplied aggregates; use frozen toy fixtures to investigate dependence.

Outcome. Clean: 252 eligible positions, 249 contexts, 251 windows. Marked: 252, 93, 93. Primary marked results are 168/252 raw and 60/93 distinct, nominal z 15.2753 and 8.8007. The earlier 39-75 distinct-hit bound is superseded by measurement, not erased. Both native outputs reached the cap; repetition has no established causal attribution.

Decision. Retain a descriptive registered-scheme result. Eighteen analyses of one pair and a keyless observer do not increase the number of independently generated pairs or calibrate the detector.

Replay. REPLAY-01 checks counts and supplied-score arithmetic; REPLAY-03 replays the public repetition control. Original native generation and secret-key membership are not publicly replayable.

What recovery would need. Recover the identified native harness, complete environment/defaults and model/tokenizer assets. Keep keys in protected custody with safe references; do not publish them. Bind original generation and rescore receipts separately from later reviews.

Records: R18 · R19 · R20 · R31

Inputs: `data/native-methods.public.json` · `fixtures/profile.public.json` · `fixtures/clean-completion.public.json` · `fixtures/marked-completion.public.json` · `fixtures/native-prompt.public.txt` · `fixtures/native-clean.public.txt` · `fixtures/native-marked.public.txt` · `data/results.public.json` · `data/observation-tracks.public.json` · `data/observer-receipt.public.json`

T05 Tokenizer controls, attribution and transformations

Question. Which interventions could change an observable signal, and what would that signal actually identify?

Method. Relabel probabilities and decoding consistently over all 24 four-symbol permutations. Separate that null from output-changing coupling, passive fingerprints, transplantation and transformation studies.

Outcome. Identity-only relabeling preserves decoded probability mass. Coupled native constructions, controlled passive fingerprints and owned real-transform survival are not measured here.

Decision. Keep the simple null as a reusable control and the broader protocols as proposals. A mark or provenance association does not identify a person, intent, truth or rightful ownership.

Replay. REPLAY-03 repeats the identity-only control. Historical proposals are preserved in the question histories rather than presented as implemented capability.

What recovery would need. Recover detailed original protocol drafts if a follow-on study is selected. Preserve parent/child artifact relations and a predeclared claim; do not reconstruct a supposed completed study from its proposal.

Records: R21 · R22 · R23 · R24 · R25 · R34

Inputs: `fixtures/synthetic-controls.public.json` · `data/synthetic-controls.expected.json` · `sources/controls/experiments.py` · `data/future-work.public.json`

T06 Field resource work, radio and exporter custody

Question. What ran on the field platform, at what cost, and what historical custody can the retained rows support?

Method. Keep seven historical compact/broad resource rows and dated contribution events. Use collector time where the device clock was wrong. Separate model timing, utility, accelerator qualification and radio semantics.

Outcome. The four compact rows ran Qwen2 INT4 0.5B under KY-ORT 1.2.2; the broad rows' model variant was not recovered. Descriptive means/ranges are reproducible from the rows. Utility is unmeasured, some log availability differs, accelerator use is unqualified and repeatable radio observations do not establish semantics. Complete clean historical exporter source remains unavailable.

Decision. Preserve the useful historical operating measurements without a stronger clean-producer or hardware-acceleration claim. Do not infer execution from board specifications.

Replay. REPLAY-02 recomputes row summaries, not acquisition. The dated field chronology is preserved in the evidence and recovery map.

What recovery would need. Seek the historical exporter and acquisition receipts from the source owner if still retained. Their absence is not repaired by a clean present-day checkout. The unit's installed configuration and the compact rows' model and runtime are now recorded (S06); the broad rows' model variant is still missing.

Records: R27 · R28 · R29 · R32

Inputs: `data/field-observations.public.json` · `data/field-observations.public.csv` · `data/field-contributions.public.json` · `manuscript/hardware.md` · `data/hardware.public.json` · `data/hardware-source.public.json`

T07 Calibration, discovery and future assurance

Question. What additional evidence would distinguish a descriptive result from a qualified operational claim?

Method. Choose the claim and sampling population before a prospective study. Preserve discovery probes, disconfirmers and assumptions. Keep statistical, formal and operational assurance separate.

Outcome. The finite-sample calculations are replayable planning arithmetic. The 30/30 draft was not executed as calibration. The unknown-unknown probes and advanced assurance proposals are not findings.

Decision. Retain the proposed routes and their exact dispositions. Historical gate obligations are history, not a new queue or 64 passing experimental tests.

Replay. REPLAY-03 checks conditional planning arithmetic. There is no completed calibration campaign or current compiled full-policy proof in this package.

What recovery would need. For continuation, retrieve the original protocol/proof drafts and source-bounded decisions first. Select an actual claim-sized study separately from this publication.

Records: R33 · R37 · R42

Inputs: `data/future-work.public.json` · `data/synthetic-controls.expected.json` · `data/claim-ledger.public.json` · `data/original-gates.public.json`

Three replay recipes

Run each recipe from the extracted package root with Python 3.10 or later. The recipes reuse the supplied tools and expected results unchanged. If a result does not match, keep the mismatch rather than editing an expected value.

`REPLAY-01` Retained traces and supplied-score arithmetic

```
python3 -I -B scripts/verify_public_package.py
```

Expected result. status pass; one pair; clean 252/249/251 and marked 252/93/93; 18 score rows. This command also verifies payload, conserved records and rendered links.

Does not repeat. Native text generation, native secret-key membership, retokenization, calibration or authentication of a historical observer event.

Prerequisites. Extracted package; Python 3.10 or later. No model, secret, account or network.

Records: R18 · R31

REPLAY-02 **Specimen identity and historical projection calculations**

```
python3 -I -B scripts/verify_supplement.py
```

Expected result. status pass; 3 native text files; 7 field rows; 17 audit rows; 15 summary pairs; 9 prose pairs and 32 continuation pairs, plus current catalog identities.

Does not repeat. Original 17-file acquisition/scanning, processing private raw corpora, original device acquisition/export or any measurement of null utility/accelerator use.

Prerequisites. Extracted package; Python 3.10 or later. All required public inputs are included.

Records: R05 · R06 · R10 · R27 · R28 · R32

REPLAY-03 **Frozen repetition, relabeling and planning controls**

```
python3 -I -B scripts/reproduce_controls.py
```

Expected result. status pass; 4 fixtures; 4096 public keys each; raw z >= 4 counts [0,116,795,1502]; 24 relabelings. --full prints all reconstructed fields; floating comparisons use tolerance 1e-12.

Does not repeat. A natural-text false-positive study, a native tokenizer-coupling experiment or a calibrated attribution rule.

Prerequisites. Extracted package; Python 3.10 or later. CPU-only standard-library computation, typically tens of seconds.

Records: R20 · R21 · R33

What the public files cannot reproduce

The public package cannot recompute native key membership because the secret keys and position-level labels are undistributed. It cannot regenerate the original native pair from complete model/runtime assets, authenticate the historical device event, or infer its execution merely from expected-output equality. It supplies public projections and aggregate receipts, not the complete private runtime. S-RESULTS, S-OBSERVER, L-01, L-09

It cannot rerun the historical 17-file carrier audit from original inputs, which remain unavailable in the reviewed material. Six summaries and audit documentation were recovered privately, but their public summaries are not an open raw-byte route. Comparative, parser, contract and historical field claims now have numerical/structural public projections. These support inspection and selected aggregate checks; the originals are not redistributed. The synthetic repetition, identity-only tokenizer and finite-sample

calculations have complete public fixtures, focused source and expected output, so those bounded controls can be replayed. The artifact catalog distinguishes these routes individually. S-EARLY-AUDITS, S-COMPARATIVE, S-CONTRACTS, S-ENGINEERING

Finally, the package cannot reconstruct the complete clean historical exporter, supply unmeasured utility, qualify accelerator use, validate radio semantics, produce a compiled current-policy certificate or calibrate generic authorship detection. These are declared unavailable, unmeasured, unqualified or excluded states, not broken public links to be guessed. The source map and current question dispositions identify the exact boundary and the condition under which a selected stronger claim could be reopened. R26, R28, R29, R32, L-10, L-12, L-13

The table sorts the record by what continuing the work can rely on.

Tier	What is available	What continuation can rely on
Included and executable	Toy controls, the public key fixture, retained token arrays, score arithmetic and projection checks	Replay the named calculation with its exact included inputs (REPLAY-01 to REPLAY-03).
Included but historically reported	Native settings and results, field rows and hardware telemetry, observer and contract projections	Inspect the retained statement and redo its public arithmetic; do not claim the original acquisition or execution.
Retained privately	Dialogues and history, comparative originals, original receipts, native runtime and keys, operational returns	Obtain an owner-verified locator and identity before promising retrieval or exact replay.
Unavailable	The 17 original audit inputs and the complete clean historical field exporter	Keep the gap and the surviving projections; a newly made substitute is not recovery of the old experiment.
Unmeasured or unqualified	Native calibration, per-run accelerator attribution, utility, radio semantics and current complete formal conformance	Needs a separately selected study; nothing in this package supplies these measurements.

The tiers describe different dimensions, not a completeness score: a file can be public while the event it describes remains unauthenticated.

F10 / From source field to bounded interpretation

Three evidence routes; source meaning and interpretation remain separate.

R05 / S-EARLY-AUDITS

SOURCE 17 files; specified anomalous-byte classes absent.

READING A historical negative scan of particular uploaded copies.

CEILING Ordinary-space encoding and unavailable original inputs remain distinct.

R28 / S-FIELD-HISTORY

SOURCE Task usefulness is null in both historical groups.

READING No usefulness score was supplied.

CEILING Null does not mean zero utility or confirmed useful output.

R31 / S-OBSERVER

SOURCE Returned output exactly matches supplied expected output.

READING Counts and reported-score arithmetic agree on shared fixtures.

CEILING No independent measurement of 60 keyed hits or authenticated execution.

Correction discipline

Recovering a record changes custody. It does not enlarge a historical scan's feature scope, supply a missing measurement or authenticate a device.

Figure 10. From source field to bounded interpretation Each route binds a source field to its technical reading and a permitted claim. The layout is an interpretive gloss; it does not imply that unrelated observations are equivalent or that an inference is literal translation.
Unit: Three distinct source-to-interpretation routes. Sources: S-EARLY-AUDITS · S-ENGINEERING · S-FIELD-HISTORY · S-HISTORY · S-OBSERVER · S-QUESTIONS. Records: R05 · R28 · R31 · R32.
Limit: Retained summaries and public derivatives preserve specific claims; they do not provide independent source authentication or recover unavailable original measurements.

A01 Complete research register

This is the full 42-record research register, including original proposals, weak observations, inspected absences, unavailable evidence, unmeasured mechanisms, corrections and conditional results. Each field is retained from the explicitly public source derivative; the JSON supplies the same source record in machine-readable form.

Terms used in the historical records

The records below keep their original wording, which includes some project working vocabulary. These short definitions are editorial aids, not part of the records.

Term	Meaning in these records
Engineering freeze, frozen	A dated point at which code, fixtures and expected results were fixed, so later checks could be compared with them. "Reproduced in engineering freeze" means reproduced against that fixed state.
Gate, obligation	A requirement set during development (A03). A gate ID identifies the historical requirement; printing it here does not mean it was retested.
Closure, closed	A recorded decision that a question or piece of work was finished for its stated scope. Each record says what would reopen it. In A03, "downward closure" is a property of the policy rules, not a decision.
Milestone	A bounded phase of work with a fixed scope. A "finite milestone" is one whose scope was not allowed to grow.
Packet, design packet	A bundle of material prepared for review. The "earlier public design packet" was a condensed predecessor of this report.
Receipt	A retained record, usually with checksums, that an operation ran and what it produced.
Return	Results sent back after a check ran elsewhere, such as the 8 October observer result from the Orange Pi RV2.
Key holder	Whoever held the secret watermark key and could therefore measure membership. Readers of this package cannot.
Qualified, unqualified	Shown, or not yet shown, to meet a stated test for a particular use. "Unqualified" means untested for that use, not failed.
Promotion	Moving an observation to a stronger evidence status. "Promotion blocked" means it stays descriptive.
Operator	In the historical records, a person who ran the research or one of its sessions. Elsewhere in this report, people who run LLM systems.
Lane	In figures, one row of a diagram. In historical notes, a separate line of work.

Term	Meaning in these records
Watermark toolkit	The historical records' name for Axis Provenance Lab (AxisPL) 0.1.4, the project's experimental software (M01).

See each source_id in data/source-map.public.json. A retained source summary is not an independently rerun experiment.

Shared disclosure: Operational identities and private source locators omitted; scientific fields retained or explicitly clarified.

The records do not schedule new work. A next discriminator is the smallest test that would move a record forward. Where next discriminator and reopening condition are identical, the field is printed once under a combined label. All record IDs also link to their main narrative section and questions.

Record	Research thread	Narrative
R01 · 68	Absence and the noise-floor doctrine	S01
R02 · 69	Model agreement and mechanism guesses	S02
R03 · 70	Ordinary separator positions remain a distinct question	S02
R04 · 71	Inherited source marks and quotation-policy story	S02
R05 · 72	17-file negative byte audit	S03
R06 · 73	Six-summary comparison and collection mistake	S03
R07 · 74	Effort-associated source/heading formatting	S03
R08 · 75	Conditional deterministic-sampling inference	S03
R09 · 76	Single-file observability and activation uncertainty	S03
R10 · 77	Matching-prompt A/B corpora and overlap	S03
R11 · 78	64 arbitrary-key null scan	S03
R12 · 79	Image/container provenance negatives	S03
R13 · 80	Conversions do not multiply linguistic evidence	S04
R14 · 81	Provenance unit and generation-path confounds	S04
R15 · 82	Instrument-generated weak signals	S04
R16 · 83	Passing tests versus contract equivalence	S04
R17 · 85	Local known-key observer as a change of question	S01
R18 · 86	Native keyed excess after deduplication	S05

Record	Research thread	Narrative
R19 · 87	Native output suitability and repetition cause	S06
R20 · 88	Synthetic dependence counterexample	S05
R21 · 89	Identity-only tokenizer permutation null	S05
R22 · 90	Keyed tokenizer and generation coupling	S06
R23 · 91	Passive same-model interaction fingerprint	S06
R24 · 92	Passive code/style triage and quality association	S06
R25 · 93	False marking, transplantation and false implication	S06
R26 · 94	Current-policy formal assurance remains separate	S04
R27 · 95	Historical field context-cost observation	S06
R28 · 96	Utility, log and acceleration missingness	S06
R29 · 97	Radio repeatability without semantics	S06
R30 · 98	Historical importer repair and concurrency findings	S04
R31 · 99	October keyless observer and its limit	S05
R32 · 100	Unavailable clean exporter source	S06
R33 · 101	Calibration and finite stopping	S07
R34 · 102	Transformation and disclosure studies	S06
R35 · 103	Provider and credential knowledge changed over time	S01
R36 · 104	Unknown knowns: retrieval and integration loss	S02
R37 · 105	Unknown unknowns: preserve discovery capacity	S07
R38 · 106	Agreement quality and human acceptance of AI-assisted findings	S02
R39 · 107	Prospective forecasts and incomplete contextual observations	S02
R40 · 108	Governance, local custody and operational constraints	S01
R41 · 109	Research-process losses: unnecessary stops and file proliferation	S01

Record	Research thread	Narrative
R42 · 110	Advanced math and assurance frontier	S07

R01 Absence and the noise-floor doctrine

Narrative: S01 · Questions: F-02 · F-06 · F-21 · F-24
Sources: S-HISTORY

Shared source and disclosure rule · Available evidence

Evidence origin / status

operator-stated methodological intent

Observation / signal state

scope_definition

Observation

Weak observations and informative absences were requested before the tool existed; the operator explicitly allowed both nonexistence and limited observability and later qualified the literal sub-noise phrase.

Assumptions

A physical noise-floor claim requires a defined channel, statistic, background and sensitivity; none was estimated universally here.

Alternative explanations

Attention/retrieval loss can explain missed information without a physically hidden signal.

Claim ceiling

Absence has weight conditional on expected observability; it proves neither human origin nor concealed marking.

Next discriminator / reopening condition

For any retained negative, state the expected detectable feature and test its positive control.

Novelty status

Project framing, not a new statistical theorem.

Preservation requirement

Keep expected channel, inspected material, observation opportunity and what missingness means. No discovery probe is a measured unknown unknown.

R02 Model agreement and mechanism guesses

Narrative: S02 · Questions: F-01 · F-12
Sources: S-EARLY-DIALOGUE · S-HISTORY

Shared source and disclosure rule · Available evidence

Evidence origin / status

historical statements; mechanism inference unverified

Observation / signal state

hypothesis

Observation

An early dialogue suggested token reweighting and confidently described known method families. Later review increased the specificity without access to the implementation.

Assumptions

Knowing a public method class does not imply privileged knowledge of a deployed provider scheme.

Alternative explanations

General learned descriptions, agreeable dialogue and shared prompts explain the apparent self-disclosure.

Claim ceiling

The apparent self-disclosure was unverified. Greenlist and Gumbel-family descriptions must not be conflated.

Next discriminator / reopening condition

Use the dated provider-family evidence already reconciled in the frozen paper; specimen-specific claims need authenticated activation/profile evidence.

Novelty status

Early project hypothesis; known method families, no global novelty claim.

Preservation requirement

Model agreement or a chatbot account of itself is not disclosure or independent corroboration. Preserve the early guesses and later corrections.

R03 Ordinary separator positions remain a distinct question

Narrative: S02 · Questions: F-02 · F-03
Sources: S-INFORMAL-LEADS · S-HISTORY

Shared source and disclosure rule · Available evidence

Evidence origin / status

historical lead; untested carrier construction

Observation / signal state

unmeasured

Observation

The informal source proposed ordinary whitespace at chosen offsets, possibly produced through token bias. That mechanism does not require unusual Unicode or extra trailing spaces.

Assumptions

A reproducible offset/code rule, preservation model and appropriate null would be needed to test the proposal.

Alternative explanations

Ordinary language lengths, layout conventions and tokenizer segmentation also create separator patterns.

Claim ceiling

The existing anomalous-byte negatives do not exhaust ordinary-space distributions; no such mark was demonstrated.

Next discriminator / reopening condition

If selected later, construct an owned positive control with a predeclared code and compare matched unmarked text without fitting offsets after inspection.

Novelty status

Recovered project hypothesis; broader prior art not assessed.

Preservation requirement

Zero exotic characters does not test every ordinary space, separator, punctuation or position-coded construction.

R04 Inherited source marks and quotation-policy story

Narrative: S02 · Questions: F-01 · F-02 · F-03 · F-24
Sources: S-INFORMAL-LEADS

Shared source and disclosure rule · Available evidence

Evidence origin / status

third-party historical assertions, not verified facts

Observation / signal state

hypothesis

Observation

The informal discussion separately asserted inherited training-source marks, dedicated whitespace tokenizers and a quotation-length limit caused by watermark leakage.

Assumptions

All three causal/implementation premises lack supporting receipts in this packet.

Alternative explanations

Copyright policy, training behavior, active generation marking and ordinary style can exist independently.

Claim ceiling

A quote cap or a participant claim is not proof of the alleged causal mechanism or provider intent.

Next discriminator / reopening condition

Recover verifiable primary technical evidence before testing a specifically defined inheritance mechanism; do not treat conversation confidence as evidence.

Novelty status

Historical research lead, not a project finding.

Preservation requirement

Preserve inherited-source watermark and quotation-policy explanations as hypotheses, not as established explanations. Original detailed record must come from the companion.

R05 17-file negative byte audit

Narrative: S03 · Questions: F-02 · F-05 · F-24
Sources: S-HISTORY · S-EARLY-AUDITS

Shared source and disclosure rule · Available evidence

Evidence origin / status

historical audit outputs/script/manifest recovered; 17 original inputs unavailable in inspected set

Observation / signal state

tested_absence_reported

Observation

17 Markdown files: 237506 bytes, 234326 Unicode characters, 2644 lines; scanned hidden/format/whitespace classes absent. Two trailing-space lines held 35 ordinary spaces, including diagram alignment.

Assumptions

The recovered report, JSON, script and manifest bind historical audit material; the 17 exact source Markdown files were not recovered in the inspected set. Uploaded-copy xattrs can differ from originals.

Alternative explanations

No mark, another carrier, acquisition normalization or a private statistical mechanism all remain possible.

Claim ceiling

Describes specified classes in uploaded copies; not all whitespace capacity, original filesystem metadata or absence of every watermark.

Next discriminator / reopening condition

Only if a stronger claim needs it: recover the 17 original bytes against their manifest, then run named feature checks with positive controls. Existing audit outputs are already recovered.

Novelty status

Historical local negative result; not independently reproduced in this pass.

Preservation requirement

Retain the 17-file negative report and narrow its conclusion to tested copies/features. Do not convert report presence into a fresh raw-byte audit.

R06 Six-summary comparison and collection mistake

Narrative: S03 · Questions: F-01 · F-02 · F-06 · F-24
Sources: S-HISTORY · S-EARLY-AUDITS

Shared source and disclosure rule · Available evidence

Evidence origin / status

six retained summary copies and ZIP recovered and hash-matched; historical reported metrics retained

Observation / signal state

weak_difference_and_tested_absence_reported

Observation

One Max and two High long/short pairs; initial effort mistake retained. High A/B long/short sequence similarity 0.638/0.735, cosine 0.912/0.869, longest exact spans 13/11 words. Scanned anomalous byte features absent.

Assumptions

The six retained byte streams match their historical manifest. Same blog and sequential requests are documented; complete generation state, person-to-A/B mapping and historical marking activation remain unresolved.

Alternative explanations

Common source, constrained task, effort/style and stochastic/context differences can produce this mix of similarity and divergence.

Claim ceiling

Neither independent randomized replication nor a calibrated watermark test; values remain historical reported observations.

Next discriminator / reopening condition

Recovery is complete for the six retained copies and audit material. A future controlled comparison must freeze full generation state and retain all attempts; it is not required for the report.

Novelty status

Project-specific exploratory result.

Preservation requirement

Keep the six files as three long/short pairs; one max run, two high runs on two operators. Do not invent account labels for the two high pairs.

R07 Effort-associated source/heading formatting

Narrative: S03 · Questions: F-11 · F-12
Sources: S-HISTORY

Shared source and disclosure rule · Available evidence

Evidence origin / status

transcript-reported visible observation

Observation / signal state

weak_difference

Observation

Max summaries put domain/date in prose plus a source line; High pairs used heading placement.

Assumptions

Few samples and uncontrolled account/context effects prevent causal assignment to effort.

Alternative explanations

Surface defaults, local context, random style and source conventions can explain the pattern.

Claim ceiling

A useful formatting observation; not watermark detection, provider identity or stable effort fingerprint.

Next discriminator / reopening condition

If useful, test matched prompts with effort as the sole changed factor and predeclare the formatting feature.

Novelty status

Local exploratory observation; generality untested.

Preservation requirement

Effort-associated headings/Source lines are visible variation; no demonstrated watermark encoding or effort causality.

R08 Conditional deterministic-sampling inference

Narrative: S03 · Questions: F-01 · F-04
Sources: S-HISTORY

Shared source and disclosure rule · Available evidence

Evidence origin / status

conditional reasoning with unverified premises

Observation / signal state

weak_conditional_inference

Observation

Output divergence was described as weak evidence against a vanilla fixed-key deterministic Gumbel-Max setup.

Assumptions

This only discriminates if full prompt/context/model/tokenizer/key/nonce/generation state is fixed; that was not established across accounts.

Alternative explanations

Different hidden state, seed/key scope, sampling or deployment can yield variation under a related family.

Claim ceiling

No likelihood weight, family exclusion or vendor-mechanism identification follows from the observed variation.

Next discriminator / reopening condition

Test an owned fully specified deterministic implementation before making a class-level comparison.

Novelty status

Conditional method reasoning; not a discovered provider fact.

Preservation requirement

Different outputs only conditionally challenge a deterministic prompt/key construction; hidden context, key scope and nonce may differ.

R09 Single-file observability and activation uncertainty

Narrative: S03 · Questions: F-01 · F-04 · F-06 · F-09
Sources: S-HISTORY

Shared source and disclosure rule · Available evidence

Evidence origin / status

identified missing variables

Observation / signal state

underpowered_or_unqualified_detection

Observation

Early files were short and historical marking activation, correct tokenizer, key and detector profile were unavailable.

Assumptions

A single sufficiently informative file can be detectable under a validated known profile; file count alone is not the criterion.

Alternative explanations

Inactive marking, undetectable marking under this procedure, acquisition loss and profile mismatch remain distinct.

Claim ceiling

The early categorical one-file impossibility claim was too broad; surviving alternatives are not assigned equal probabilities.

Next discriminator / reopening condition

Specify scored unit count, profile, controls and sensitivity before deciding whether a sample is adequate.

Novelty status

Observability clarification.

Preservation requirement

Correct the earlier blanket claim that one Markdown file cannot be detected. What is unresolved is the applicable detector/profile and calibration for that file.

R10 Matching-prompt A/B corpora and overlap

Narrative: S03 · Questions: F-01 · F-11 · F-20 · F-24
Sources: S-COMPARATIVE

Shared source and disclosure rule · Available evidence

Evidence origin / status

archived analysis; input archive hashes matched

Observation / signal state

weak_difference

Observation

Nine prose pairs and 32 constrained completions had no exact pair matches. Mean prose sequence similarity 0.191 and TF-IDF cosine 0.564; 20/32 shared a first word and 8/32 first four words.

Assumptions

Exact successful CLI model ID was missing; current hash binding verifies custody, not all historical execution or scoring.

Alternative explanations

Shared prompts, common factual phrases and output constraints explain overlap without a watermark.

Claim ceiling

Historical comparative results with a missing model identity; not a provider attribution model.

Next discriminator / reopening condition

Recover generation-event/model receipts and repeat only if a specific generalization claim is selected.

Novelty status

Project-specific exploratory result.

Preservation requirement

Retain paired-source corpora separately from controlled native tests; detailed measurements require their original source, not the title in this map.

R11 64 arbitrary-key null scan

Narrative: S03 · Questions: F-03 · F-04 · F-09
Sources: S-COMPARATIVE

Shared source and disclosure rule · Available evidence

Evidence origin / status

archived JSON and report retained; not rerun

Observation / signal state

profile_specific_non_detection

Observation

With surrogate word/punctuation tokenization and 64 deterministic arbitrary keys, neither A nor B crossed nominal z=4; mean z about 0.128 and 0.008.

Assumptions

The lab scheme, keys and tokenizer do not match a known provider profile.

Alternative explanations

A private mark could be invisible under this mismatched profile; ordinary text can produce these null-like scores.

Claim ceiling

Engine/control result only, not absence of provider marking or a general false-positive estimate.

Next discriminator / reopening condition

Use owned known-positive and negative controls under an exact registered profile.

Novelty status

Local negative control; standard calibration principle.

Preservation requirement

Arbitrary-key null scans do not test an undisclosed provider key; retain the 64-key historical study separately from the 4096 public-key synthetic fixture.

R12 Image/container provenance negatives

Narrative: S03 · Questions: F-05 · F-13
Sources: S-COMPARATIVE

Shared source and disclosure rule · Available evidence

Evidence origin / status

archived report; current input hash matched

Observation / signal state

tested_absence_reported

Observation

Five image/export fixtures lacked the inspected embedded C2PA/JUMBF and related metadata structures; visible structures matched ordinary SVG/PNG/JPEG tool paths.

Assumptions

Detection is limited to the inspected file path, structure and parser coverage; an external manifest or other route can differ.

Alternative explanations

No credential on this path, stripping by conversion or signing on a separate download route are alternatives.

Claim ceiling

No universal provider absence or native trusted-credential qualification.

Next discriminator / reopening condition

Pair a route-specific negative with trusted/untrusted positives and preserve pre/post-export bytes if that branch is resumed.

Novelty status

Local negative observation; not a new credential scheme.

Preservation requirement

Container/image negatives apply to the inspected artifacts and exact acquisition path, not all provider media.

R13 Conversions do not multiply linguistic evidence

Narrative: S04 · Questions: F-05 · F-08 · F-20
Sources: S-CONTRACTS

Shared source and disclosure rule · Available evidence

Evidence origin / status

archived validation and hash-bound inputs

Observation / signal state

observed_equivalence

Observation

5/5 HTML source segments and 9/9 PDF segments preserve source word sequence; both code segments preserve non-whitespace content.

Assumptions

These are deterministic transport copies of source text, not separate model draws.

Alternative explanations

Container metadata can change while the linguistic event remains identical.

Claim ceiling

Counts of files/exports must not inflate sample independence.

Next discriminator / reopening condition

Count generation events and mapped source spans; retain copies as transport controls.

Novelty status

Project-specific verification of a standard sampling distinction.

Preservation requirement

PDF/Markdown/HTML copies of one generation are transformations of one linguistic specimen, not independent samples.

R14 Provenance unit and generation-path confounds

Narrative: S04 · Questions: F-10 · F-12 · F-20
Sources: S-CONTRACTS · S-EARLY-DIALOGUE

Shared source and disclosure rule · Available evidence

Evidence origin / status

archived analysis and methodological inference

Observation / signal state

unmeasured_confound

Observation

Bundles mix model prose/code, programmatic data, deterministic conversion and model-produced rewrites. Condensed/alternate versions may be new marking events.

Assumptions

Authorship/marking inference depends on generation event and span, not just container filename.

Alternative explanations

A conversion may preserve a signal while a rewrite both edits and generates/marks anew.

Claim ceiling

Do not interpret model rewriting as isolated post-hoc watermark removal or processing as authorship.

Next discriminator / reopening condition

Annotate source spans and generation event IDs; use local non-model edits for a transformation-only study.

Novelty status

Project workflow insight; general principle not claimed new.

Preservation requirement

Generation, tool construction, API download, copying and re-encoding are distinct provenance stages.

R15 Instrument-generated weak signals

Narrative: S04 · Questions: F-05 · F-21 · F-22
Sources: S-CONTRACTS

Shared source and disclosure rule · Available evidence

Evidence origin / status

historical regression report retained

Observation / signal state

false_alarm_explained

Observation

Raw /AA bytes inside compressed PDF data falsely suggested actions; ten bullets became U+007F under one extractor. Binary CR/LF and whole-bundle comparisons also misled interpretation.

Assumptions

The preserved report records parsed-object, second-extractor and rendered checks; those checks were not freshly rerun here.

Alternative explanations

Parser mapping and representation mismatch explain the apparent anomaly without hidden content.

Claim ceiling

These are versioned early-instrument findings; do not assert the current code still has them without inspection.

Next discriminator / reopening condition

For new anomalies, check structure, extractor parity and rendered appearance before promoting a carrier finding.

Novelty status

Concrete local findings and regression lessons.

Preservation requirement

Instrument/parser/extractor errors can create weak signals. Preserve review-tool failures separately from producer defects.

R16 Passing tests versus contract equivalence

Narrative: S04 · Questions: F-14 · F-22 · A-05
Sources: S-CONTRACTS

Shared source and disclosure rule · Available evidence

Evidence origin / status

archived analysis, historical implementation findings

Observation / signal state

observed_contract_gap

Observation

Go/Python agreed on 288 accepted of 300 records and passed reported normal tests, yet differed from schema/each other on extra fields, whitespace, dates and oversized lines.

Assumptions

Normal-fixture agreement only samples part of the accepted language.

Alternative explanations

Different parser permissiveness and limits explain disagreement without malicious behavior.

Claim ceiling

Historical generated-fixture contract findings; not a current full implementation audit.

Next discriminator / reopening condition

Preserve accepted/rejected language tests at interface boundaries and bind them to the exact version.

Novelty status

Local engineering finding.

Preservation requirement

Tests passing do not establish agreement with the intended contract. Preserve historical malformed-input acceptance and bounded fixes.

Separate later episode

Question id

A-05

Parent

F-22

Topic

October invalid scalar inputs and caller/contract boundary

Relationship

Another instance of tests versus intended accepted-input contract; not the same historical Go/Python experiment.

Current state

Strict scalar validation implemented and tested on the declared local branch; public-ingress reachability and deployment remain outside scope.

Claim ceiling

No claim of a currently exploitable public endpoint and no new code task assigned.

Source ids

- S-QUESTIONS
- S-ENGINEERING

R17 Local known-key observer as a change of question

Narrative: S01 · Questions: F-01 · F-04 · F-21
Sources: S-HISTORY · S-ENGINEERING

Shared source and disclosure rule · Available evidence

Evidence origin / status

historical tool transition

Observation / signal state

scope_change

Observation

A read-only observer and educational known-key demo replaced an unanswerable private-mechanism claim with owned, inspectable control experiments.

Assumptions

Lab key/tokenizer assumptions must remain explicit when using outside samples.

Alternative explanations

Success in an owned construction need not generalize to a commercial detector.

Claim ceiling

Tool competence is not private provider discovery.

Next discriminator / reopening condition

Use registered profiles and maintain separate artifact, statistical, credential and passive evidence families.

Novelty status

Project design contribution; underlying watermark family has prior art.

Preservation requirement

Owned known-key construction changed the research question; success in that lab does not solve attribution of vendor outputs.

R18 Native keyed excess after deduplication

Narrative: S05 · Questions: F-08 · F-09 · A-01
Sources: S-HISTORY · S-ENGINEERING · S-RESULTS

Shared source and disclosure rule · Available evidence

Evidence origin / status

frozen measured result with original-key cross-check

Observation / signal state

positive_bounded

Observation

Marked primary raw 168/252, nominal z15.2753; distinct windows 60/93, nominal z8.8007. Clean distinct 53/251, nominal z-1.4212. Earlier 39-75 bound is preserved as the prior knowledge state.

Assumptions

One retained pair; shared construction/keys/sample in the scalar cross-check; no qualified empirical null or inferred independence from deduplication.

Alternative explanations

Repetition explains part of raw growth, while positive primary-key excess remains in this specimen.

Claim ceiling

Bounded registered-scheme observation, not authorship, ownership, identity or a population detector-performance estimate.

Next discriminator / reopening condition

No rerun needed for this closure. Any broader inference needs prospective held-out calibration and a defined sampling population.

Novelty status

New local result, not a new generic detection theorem.

Preservation requirement

Preserve raw 168/252 and key-holder distinct-window 60/93, clean 53/251, nominal scores and the same-specimen dependence.

R19 Native output suitability and repetition cause

Narrative: S06 · Questions: F-07 · A-04
Sources: S-HISTORY · S-ENGINEERING

Shared source and disclosure rule · Available evidence

Evidence origin / status

direct artifact observation; cause unresolved

Observation / signal state

weak_quality_signal

Observation

Marked output repeats a paragraph; clean output also continues in unsuitable prompt-like form. Both reached the token cap.

Assumptions

One matched-seed pair cannot isolate watermark, format, sampling and model effects.

Alternative explanations

Prompt format, decoding, model limitations and marking may interact.

Claim ceiling

Neither proven watermark-induced damage nor proof that marking has no quality cost.

Next discriminator / reopening condition

Optional future single-factor formatting diagnostic, at most 12 predeclared attempts with every outcome retained.

Novelty status

Local quality observation and open causal question.

Preservation requirement

Repetition/suitability observed; causal contributions of prompt, model, sampler and watermark unseparated. No new run automatically required.

R20 Synthetic dependence counterexample

Narrative: S05 · Questions: F-08 · F-09
Sources: S-HISTORY · S-ENGINEERING

Shared source and disclosure rule · Available evidence

Evidence origin / status

fixed-fixture analysis reproduced in engineering freeze

Observation / signal state

positive_control

Observation

Across 4096 predetermined public test keys, raw z >= 4 crossings rise to 1502 for a 16-window motif repeated 64 times, versus zero for 1024 distinct windows.

Assumptions

A key ensemble on a deliberately repetitive fixed fixture is not a natural-text sampling distribution.

Alternative explanations

Repeated reuse of a favorable motif explains inflation without improved independent evidence.

Claim ceiling

Fixture-specific diagnostic, not 36.67% field false-positive rate.

Next discriminator / reopening condition

Carry raw and distinct units separately; calibrate any future decision rule under its actual dependence structure.

Novelty status

Concrete local counterexample using established statistical concerns.

Preservation requirement

1502/4096 synthetic raw threshold crossings are fixture/key-ensemble behavior, not a natural-language FPR.

R21 Identity-only tokenizer permutation null

Narrative: S05 · Questions: F-03 · A-02
Sources: S-HISTORY · S-ENGINEERING

Shared source and disclosure rule · Available evidence

Evidence origin / status

toy control reproduced in engineering freeze

Observation / signal state

negative_control

Observation

All 24 permutations of a four-token vocabulary preserve visible-text distributions when probabilities and decoding are relabeled consistently.

Assumptions

The tested intervention is identity-only; it does not change semantic generation behavior.

Alternative explanations

An output-changing coupling could behave differently and was not tested by this null.

Claim ceiling

Rejects this simple proposed construction, not every tokenizer-key mechanism.

Next discriminator / reopening condition

Use this as the negative control for any owned output-changing construction.

Novelty status

Local test of an invariance; no new broad impossibility theorem.

Preservation requirement

All 24 toy consistent token-ID relabelings are a bounded null control, not disproof of tokenizer-based generation coupling.

R22 Keyed tokenizer and generation coupling

Narrative: S06 · Questions: F-03 · A-02
Sources: S-HISTORY

Shared source and disclosure rule · Available evidence

Evidence origin / status

operator proposal; constructive mechanism not tested

Observation / signal state

hypothesis

Observation

Custom tokenizer plus key/algorithm might shape a detectable output distribution. The broad proposal persists after the identity-only null.

Assumptions

Requires a specified output-changing rule, correct/wrong-key controls, decoding parity, quality measurement and a qualified null.

Alternative explanations

Ordinary keyed token selection or segmentation effects may account for any result without a distinct new mechanism.

Claim ceiling

Original idea within the record; neither a working construction, provider implementation nor established global novelty.

Next discriminator / reopening condition

Optional tiny owned model with frozen identity-only negative and explicitly output-changing positive; stop on the declared result, including failure.

Novelty status

Project-original proposal; global novelty unassessed.

Preservation requirement

Retain output-changing keyed tokenizer/segmentation coupling; detector specification, controls and novelty review are still future work.

R23 Passive same-model interaction fingerprint

Narrative: S06 · Questions: F-11 · F-12 · A-03 · U-03
Sources: S-HISTORY

Shared source and disclosure rule · Available evidence

Evidence origin / status

operator hypothesis; no controlled experiment

Observation / signal state

weak_interaction_hypothesis

Observation

Apparent same-model agreement/mirroring might reveal shared learned patterns or interaction behavior.

Assumptions

Independence, context separation, blinding and selection controls were absent in the conversation.

Alternative explanations

Shared prompts, priors, style, incentives, task constraints and cooperative agreement can explain convergence.

Claim ceiling

No demonstrated hidden communication, subconscious recognition, intentional watermark or independent confirmation.

Next discriminator / reopening condition

Optional blind same/different-model crossed with shared/separate-context comparison, with prespecified features and null.

Novelty status

Project research hypothesis; prior art and effect size unassessed.

Preservation requirement

Retain passive shared-prior/interaction fingerprints with no-communication, cross-model and shared-context controls; no anthropomorphic recognition claim.

R24 Passive code/style triage and quality association

Narrative: S06 · Questions: F-11 · F-12 · F-21
Sources: S-HISTORY

Shared source and disclosure rule · Available evidence

Evidence origin / status

proposal, not validated classifier

Observation / signal state

hypothesis

Observation

A probable-generation scanner could prioritize code inspection; the suggested relation between generated style and code defects is a separate question.

Assumptions

Independent origin labels and defect labels, matched tasks and a held-out population would be required.

Alternative explanations

Common templates, formatting, programmer convention and problem difficulty can mimic source/style effects.

Claim ceiling

Style is not identity; likely origin is not code quality.

Next discriminator / reopening condition

Evaluate inspection utility and error costs separately from source classification if the product scope later selects this use.

Novelty status

Application proposal; no established local detector performance.

Preservation requirement

AI-like style is not a defect label; code quality must be measured separately from passive origin association.

R25 False marking, transplantation and false implication

Narrative: S06 · Questions: F-10 · F-12
Sources: S-EARLY-DIALOGUE · S-HISTORY

Shared source and disclosure rule · Available evidence

Evidence origin / status

hypotheses raised in original Claude exchange

Observation / signal state

unmeasured_robustness_question

Observation

The early exchange considered forged/transferred marks and false attribution as well as removal.

Assumptions

No concrete successful attack is demonstrated in the retained project record.

Alternative explanations

A positive result could refer to processing or a fragment rather than authorship of the whole artifact.

Claim ceiling

Do not infer identity, ownership, intent or truth from a registered mark.

Next discriminator / reopening condition

If a bounded owned robustness study is selected, include signal creation/transfer and mixed-source localization alongside signal removal.

Novelty status

Preserved threat-model ideas; novelty unassessed.

Preservation requirement

False implication, transplantation and false marking deserve a future threat model; no attack experiment was established here.

R26 Current-policy formal assurance remains separate

Narrative: S04 · Questions: F-14 · F-15
Sources: S-HISTORY · S-ENGINEERING · S-QUESTIONS

Shared source and disclosure rule · Available evidence

Evidence origin / status

documented source discrepancy; no current compiled certificate

Observation / signal state

unqualified_assurance

Observation

The frozen review retains Python/Lean policy disagreement and excludes an incomplete historical seven-case inventory from certification.

Assumptions

A theorem/certificate must bind to the actual policy and executable implementation/ version.

Alternative explanations

Artifact drift or different policy assumptions can explain mismatch despite individually valid local checks.

Claim ceiling

No current end-to-end formal conformance claim.

Next discriminator / reopening condition

A future formal-work scope must bind policy, implementation, case inventory and checked proof artifact.

Novelty status

Project assurance finding; not a new proof system.

Preservation requirement

Keep current Python vs archived Lean semantics and absent compiled certificate explicit. Formal tooling does not certify empirical facts by itself.

R27 Historical field context-cost observation

Narrative: S06 · Questions: F-16 · F-19
Sources: S-HISTORY · S-FIELD-HISTORY

Shared source and disclosure rule · Available evidence

Evidence origin / status

Archived producer rows; descriptive arithmetic previously recomputed during preservation review

Observation / signal state

descriptive_field_difference

Observation

Four compact-context rows versus three broad-context rows: 437 versus 1749 prompt tokens; mean time to first token 12.452 versus 63.001 seconds; throughput 8.349 versus 3.871 tokens/second; resident memory 647049 versus 1482581.33 KiB; wall time 49.847 versus 109.218 seconds. All rows report 64 generated tokens.

Assumptions

Small nonrandom historical batches; full controlled run equivalence and clean exporter-source provenance are not established.

Alternative explanations

Context size, scheduling, thermals, configuration and other run conditions may contribute.

Claim ceiling

Useful resource-budget hypothesis, excluded from qualified causal performance/ reproducibility claims.

Next discriminator / reopening condition

Only for a chosen field-performance claim: clean source-bound exporter, matched runs and a usefulness measure.

Novelty status

Project-specific descriptive observation.

Preservation requirement

Do not erase older direct-ORT/compact inference and context-budget observations when describing October observer-only scope.

R28 Utility, log and acceleration missingness

Narrative: S06 · Questions: F-16 · F-19 · F-24
Sources: S-HISTORY · S-FIELD-HISTORY

Shared source and disclosure rule · Available evidence

Evidence origin / status

archived producer rows inspected

Observation / signal state

multiple_absence_states

Observation

Usefulness is null in both groups; compact rows have a log delta reporting periodic Wi-Fi only; broad rows lack the delta. Accelerator device nodes are reported present.

Assumptions

A present node and absence of a logged error do not establish use of that device by inference.

Alternative explanations

Unmeasured utility, uncaptured logs and hardware nonuse are different possible explanations.

Claim ceiling

No proof of useful field output, acceleration, or all-clear behavior from these omissions.

Next discriminator / reopening condition

Capture application-level acceleration evidence and score task utility if that claim becomes necessary; keep unavailable distinct from zero.

Novelty status

Recovered missing-data findings.

Preservation requirement

Log/accelerator/utility absence differs from zero use; do not infer an NPU path from device presence.

R29 Radio repeatability without semantics

Narrative: S06 · Questions: F-18
Sources: S-HISTORY · S-FIELD-HISTORY

Shared source and disclosure rule · Available evidence

Evidence origin / status

historical bounded observations; semantics unqualified

Observation / signal state

repeatability_and_semantic_absence

Observation

Historical review reports 513 events (511 read-only and 2 passive), 16 candidate channels, and matching image evidence across captures/devices; semantic promotion remains blocked.

Assumptions

Matching data can reflect stable capture/representation without validating candidate field meanings.

Alternative explanations

A repeated common image or decoder assumption can reproduce identically while its interpretation is wrong.

Claim ceiling

No validated radio protocol semantics or autonomous promotion.

Next discriminator / reopening condition

If resumed, controlled known differences plus operator labels are more discriminating than another identical dump.

Novelty status

Local bounded repeatability observation.

Preservation requirement

Radio acquisition repeatability is not validated channel/protocol semantics. Keep as adjacent historical evidence, not a watermark acceptance requirement.

R30 Historical importer repair and concurrency findings

Narrative: S04 · Questions: F-17 · F-21 · F-22
Sources: S-HISTORY · S-ENGINEERING

Shared source and disclosure rule · Available evidence

Evidence origin / status

dated failures followed by accepted repair

Observation / signal state

historical_failure_and_closure

Observation

Benign mutations exposed raw-field, contradictory-promotion, boolean/type, infinity/count problems; P0.1 added date/exception/concurrency/accounting/source-dirty checks.

Assumptions

Version and acceptance state must accompany every finding.

Alternative explanations

Stale reports can make a closed defect appear currently unresolved.

Claim ceiling

Do not reopen original P0/P0.1 gates merely because the transcript contains their discovery; 0.1.4 closure is preserved.

Next discriminator / reopening condition

Consult current source/receipt only if a new regression or expanded contract is identified.

Novelty status

Project engineering findings and fixes.

Preservation requirement

P0.1 canonical status supersedes old pending summaries; later optional/direct-API work remains version- and scope-specific.

R31 October keyless observer and its limit

Narrative: S05 · Questions: F-16 · F-19 · A-01
Sources: S-ENGINEERING · S-QUESTIONS · S-OBSERVER

Shared source and disclosure rule · Available evidence

Evidence origin / status

accepted peer-reported execution and exact returned output

Observation / signal state

bounded_cross_architecture_check

Observation

The accepted return matches the supplied nonsecret observer output: clean 251 and marked 93 distinct windows, plus six reported-score arithmetic checks. The reported run used no model or secret keys. The expected output was supplied with the request.

Assumptions

Execution/platform identification relies on the peer receipt; output equality is not independent event authentication.

Alternative explanations

Reusing the same trace and algorithm can reproduce output without adding an independent generated sample or membership measurement.

Claim ceiling

Scoped request closed; not native-model equivalence, independent measurement of 60 key-membership hits, or a field benchmark.

Next discriminator / reopening condition

No further field node work required for this scope. Stronger claims require a newly selected and qualified experiment.

Novelty status

Bounded project portability result.

Preservation requirement

October portable observer reuse is not new inference or independent keyed verification; device execution is peer-reported.

R32 Unavailable clean exporter source

Narrative: S06 · Questions: F-16 · F-24
Sources: S-HISTORY · S-ENGINEERING

Shared source and disclosure rule · Available evidence

Evidence origin / status

explicit unavailable disposition, preserved in frozen evidence

Observation / signal state

missing_provenance

Observation

Historical strict model/radio records retain dirty-source lineage; no complete exporter source snapshot or fresh qualified export returned. October return also has a documented self-manifest hash defect.

Assumptions

Hashes bind available files, not missing source or source authentication. A separate August ledger had 24 matching entries in the preservation review.

Alternative explanations

Incomplete preservation explains absent reproducibility without proving the reported measurements false.

Claim ceiling

Stronger clean-producer/performance claims excluded; historical observations retained.

Next discriminator / reopening condition

Recover complete source and matching fresh export only if a future claim needs qualified field reproduction.

Novelty status

Project provenance limitation and honest stop disposition.

Preservation requirement

Unavailable clean exporter snapshot closes the selected response by exclusion; it does not satisfy reproducible-producer claims.

R33 Calibration and finite stopping

Narrative: S07 · Questions: F-06 · F-09 · F-20
Sources: S-HISTORY · S-ENGINEERING

Shared source and disclosure rule · Available evidence

Evidence origin / status

identified inferential limit

Observation / signal state

unqualified_population_inference

Observation

Even zero errors among 30 independent held-out negatives would give a one-sided 95% upper bound near 9.50% for a fixed rule; the 30/30 study draft is unfrozen.

Assumptions

Independence, fixed threshold, population and held-out status are assumptions, not granted by a sample count.

Alternative explanations

Selection, dependence and drift can invalidate naive precision claims.

Claim ceiling

No calibrated rare-error guarantee. This limitation is not an instruction to run hundreds of cases to close the paper.

Next discriminator / reopening condition

Select the desired claim and uncertainty target before designing a new calibration study; keep current milestone finite.

Novelty status

Established inference applied to project planning.

Preservation requirement

Keep descriptive utility, failed attempts, eligibility and fixed stopping; do not scale sample size merely to obtain a stronger marketing claim.

R34 Transformation and disclosure studies

Narrative: S06 · Questions: F-10 · F-12 · A-06
Sources: S-EARLY-DIALOGUE · S-HISTORY · S-QUESTIONS

Shared source and disclosure rule · Available evidence

Evidence origin / status

open research plan, not executed coverage

Observation / signal state

unmeasured_robustness

Observation

Owned whitespace/rewrap/edit/mixed-source tests and selective disclosure remain useful; sweeping early claims that formatting always breaks marking were unsupported.

Assumptions

Transformations must hold generation/profile identity fixed; detector access and jurisdictional questions are separate.

Alternative explanations

A new model rewrite may regenerate/re-mark; a failed detector may reflect changed tokenization rather than universal removal.

Claim ceiling

No robust-transformation, legal-clearance or private-detector-evasion result here.

Next discriminator / reopening condition

If selected, predeclare a small owned transformation matrix with correct/wrong controls; undertake legal analysis only for an actual scoped use.

Novelty status

Research agenda, global novelty not asserted.

Preservation requirement

Owned-sample transformation/counter-watermark tests and fact-specific legal review remain distinct future tasks, not blanket clearance.

R35 Provider and credential knowledge changed over time

Narrative: S01 · Questions: F-01 · F-04 · F-05 · F-13
Sources: S-HISTORY · S-ENGINEERING

Shared source and disclosure rule · Available evidence

Evidence origin / status

dated source review already reconciled in engineering freeze

Observation / signal state

historical_to_current_update

Observation

Early guesses preceded later public family descriptions. The frozen October8 primary-source review supplies current family background; old specimen activation remains unresolved. Credential attachment can depend on export/download path.

Assumptions

A later publication cannot establish that an earlier specimen was marked or that its exact private profile is known.

Alternative explanations

Surface, version and route changes can explain apparently conflicting reports.

Claim ceiling

Use dated source knowledge; this companion does not independently reverify every historical current-event assertion.

Next discriminator / reopening condition

Recheck primary sources only when a new time-sensitive claim is needed; retain old hypotheses as history.

Novelty status

Knowledge-state evolution, not provider discovery.

Preservation requirement

Keep dated provider statements, conflicting research summaries, surface coverage and historical activation distinct; this extraction refreshes no external claim.

R36 Unknown knowns: retrieval and integration loss

Narrative: S02 · Questions: F-05 · F-20 · F-24
Sources: S-HISTORY · S-INFORMAL-LEADS · S-CONTRACTS

Shared source and disclosure rule · Available evidence

Evidence origin / status

concrete historical examples

Observation / signal state

overlooked_available_context

Observation

The original Claude file was initially mistaken for a derived artifact; ordinary-space offsets, transport duplication, missing model ID and prior weak-signal preservation requests were easy to omit.

Assumptions

This category means recorded knowledge not loaded/integrated, not unconscious access to verified external facts.

Alternative explanations

Export loss, partial lane context, naming confusion and narrow summary goals can create the same omission.

Claim ceiling

A source may exist without being considered; conversation summaries are not exhaustive evidence.

Next discriminator / reopening condition

Maintain source hashes, coverage ranges, explicit exclusions and a crosswalk from research threads to final deliverables.

Novelty status

Project workflow finding and preservation method.

Preservation requirement

Known work can be lost between summaries. Source-to-record conservation is the immediate purpose of this handoff.

R37 Unknown unknowns: preserve discovery capacity

Narrative: S07 · Questions: U-01 · U-02 · U-03 · U-04 · U-05 · U-06 · F-24
Sources: S-HISTORY · S-QUESTIONS

Shared source and disclosure rule · Available evidence

Evidence origin / status

discovery protocol, not enumerated hidden facts

Observation / signal state

open_discovery

Observation

U-01..U-06 ask about missing profile variables, failed controls, shared selection failures, route/version changes, downstream misreading and concrete unexplained residuals.

Assumptions

No list can exhaust unimagined causes; each proposed probe needs an observable trigger.

Alternative explanations

An anomaly can be an instrument, custody, selection or ordinary variation issue before a new mechanism is needed.

Claim ceiling

These are probes, not six demonstrated unknown causes or a promise of exhaustive completeness.

Next discriminator / reopening condition

Retain anomalies/all attempts and denominators; open a new bounded item when a retained observation defeats current explanations.

Novelty status

Project discovery discipline.

Preservation requirement

Discovery probes provide opportunities to find surprises; no invented list exhausts unknown unknowns.

R38 Agreement quality and human acceptance of AI-assisted findings

Narrative: S02 · Questions: F-20 · F-21 · A-03 · U-03
Sources: S-HISTORY

Shared source and disclosure rule · Available evidence

Evidence origin / status

operator-stated method and recorded assistant behavior

Observation / signal state

correlated_confirmation_risk

Observation

Conflict reporting and human scrutiny of agreeable AI outputs were requested throughout; parallel AI-assisted analyses had only partial visibility of one another, and confidence inflation in assistant responses recurred.

Assumptions

Multiple AI assistants sharing context, models or fixtures are not independent samples merely because they have separate role names.

Alternative explanations

Common priors, prompt framing and cooperative behavior can account for agreement.

Claim ceiling

Agreement does not promote a claim above its evidence; model-family anecdotes are not comparative benchmarks.

Next discriminator / reopening condition

Log shared dependencies and disconfirmers; use independent evidence paths where the claim actually requires independence.

Novelty status

Project methodology; established epistemic concern.

Preservation requirement

Human acceptance requires challenge and uncertainty, not agreement-count voting. Show shared artifacts among reviewers.

R39 Prospective forecasts and incomplete contextual observations

Narrative: S02 · Questions: F-20 · F-24
Sources: S-HISTORY · S-EARLY-DIALOGUE

Shared source and disclosure rule · Available evidence

Evidence origin / status

Public abstraction of historical unscored conjectures

Observation / signal state

weak_contextual_observation

Observation

Historical conjectures concerned access/cost stratification, resource constraints, information disorder, and marking/guardrails. They lacked prospective resolution criteria and a scored baseline. A separate personal-contact anecdote did not substantiate the assistant-suggested NDA or gated-lab explanation; neither silence nor speculation established hidden intent. Personal circumstances remain abstracted.

Assumptions

Timestamps, fixed endpoints, probabilities, scoring rules and a measured baseline are needed before assigning predictive or causal credit.

Alternative explanations

Ordinary changes in conditions, incomplete observation and selective recall can explain an apparent pattern.

Claim ceiling

Retain the methodological lesson; do not claim retrospective prediction success or hidden intent.

Next discriminator / reopening condition

Log future forecasts with timestamps, resolution criteria, probabilities and disconfirmers before outcomes; require evidence for a proposed causal explanation.

Novelty status

Contextual conjectures, not established scientific findings.

Preservation requirement

Preserve prospective logging and scoring as method. Personal context is deliberately abstracted; no causal or predictive success is asserted.

Disclosed abstraction

Personal and adjacent project details abstracted; research meaning and limitations retained.

R40 Governance, local custody and operational constraints

Narrative: S01 · Questions: F-19 · F-23
Sources: S-HISTORY

Shared source and disclosure rule · Available evidence

Evidence origin / status

operator-stated motivation and long-term ideas

Observation / signal state

design_values_and_hypotheses

Observation

Transparency, consent, inspectable local measurement, portable owned records and bounded operation on constrained devices motivated the work. Adjacent memory, bias-evaluation and architecture proposals remain separate.

Assumptions

A value, device capability claim or architecture idea is not itself measured provenance efficacy.

Alternative explanations

Operational constraints can explain design choices without implying any specific provider intent.

Claim ceiling

These are design motivations and future ideas; no accelerator use, new architecture, legal conclusion, provider misconduct or qualified deployment is established by them.

Next discriminator / reopening condition

Select a separate scoped claim before undertaking device or architecture experiments; keep provenance, usability and governance evidence distinct.

Novelty status

Project motivations and proposals, not all experimentally tested.

Preservation requirement

Preserve custody, portability, consent and bounded operation as motivations, separate from scientific capability and provider conduct.

Disclosed abstraction

Personal and adjacent project details abstracted; research meaning and limitations retained.

R41 Research-process losses: unnecessary stops and file proliferation

Narrative: S01 · Questions: F-20 · F-24
Sources: S-HISTORY

Shared source and disclosure rule · Available evidence

Evidence origin / status

operator-reported process friction

Observation / signal state

weak_operational_observation

Observation

Unnecessary pauses, losses of continuity between sessions and exports, and duplicated working directories were reported in the AI-assisted research process; collection and synthesis were explicitly distinguished.

Assumptions

These anecdotes are not a quantified productivity study or blanket permission for unrelated actions.

Alternative explanations

Tool limitations, stale context and ambiguous authority can contribute to the same friction.

Claim ceiling

Keep a stable scope and a finite stopping rule; open questions should not become endless mandatory checkpoints.

Next discriminator / reopening condition

Use one working copy, preserve incoming files unchanged and link their current status; reopen only for a concrete error or a selected new question.

Novelty status

Project process lessons.

Preservation requirement

Do not turn every uncertainty into a new mandatory task. Respect intake, research, analysis, synthesis and publication stages.

R42 Advanced math and assurance frontier

Narrative: S07 · Questions: F-14 · F-15 · F-20
Sources: S-QUESTIONS · S-HISTORY

Shared source and disclosure rule · Available evidence

Evidence origin / status

existing preserved future questions

Observation / signal state

unmeasured_method_agenda

Observation

Temporal/dependence-aware statistics, adversarial assurance and formal methods remain in the original question/gate register beyond the bounded implemented diagnostics.

Assumptions

A named method or mathematical sketch does not supply a validated detector or current-policy proof.

Alternative explanations

Simpler sampling/profile errors may explain anomalies before more elaborate machinery is useful.

Claim ceiling

Do not drop the frontier, but do not promote mathematical ambition into implemented or measured capability.

Next discriminator / reopening condition

Select the estimand or failure mode first, then one method with a falsifiable control and finite scope; retain the original 64 gates by reference.

Novelty status

Research agenda; individual novelty unassessed.

Preservation requirement

Advanced math, temporal models, memristors and broader assurance remain inspirations/open extensions, not implemented product claims.

A02 Questions, histories and discovery probes

Current effective dispositions are shown first. Baseline records and later additive stages remain visible as history. In U-02, the historical control-label mistake remains recorded; the corrected positive-versus-negative control statement governs current interpretation.

ID	Question / probe
F-01 · 113	Does Anthropic use green-list biasing, a Gumbel-family method, another sampling construction, or a different/combined carrier on each surface?
F-02 · 115	What do the original negative byte scans actually exclude, and what ordinary-format or exporter-induced signals were not tested?
F-03 · 117	Could keyed/custom segmentation, tokenizer behavior or token-boundary conventions carry or change an observable provenance signal?
F-04 · 119	Are keys/contexts scoped per request, account, model, workspace, time or nonce, and how would each affect reproducibility?
F-05 · 121	Which observed bytes originate in generation, browser/API export, PDF construction, copying, or credential attachment?
F-06 · 123	Under which known detector/profile/key and sample-length conditions could one sufficiently long artifact support a result?
F-07 · 125	What causes the marked repetition and incomplete task fulfillment: watermark, prompt formatting, sampling, model behavior or interaction?
F-08 · 128	How should repeated context and repeated context/token windows affect diagnostics, eligible denominator and inferential scoring?
F-09 · 130	What performance/uncertainty claim does the experiment actually have enough independent data to assess?
F-10 · 132	How do normalization, copy/paste, extraction, formatting and other owned deterministic transformations affect the registered signal?
F-11 · 134	Can a passive/stylometric tool help inspect code without confusing generator style with correctness or sloppiness?
F-12 · 136	What evidence can link an artifact to a system/workflow versus to a person, organization, or threat actor?
F-13 · 138	Do actual native validators correctly classify signed trusted/untrusted, tampered, unsigned, derivative and malformed artifacts?
F-14 · 140	Can the actual current Python claim policy be aligned, built and bound to a small Lean certificate and cross-language vectors?

ID	Question / probe
F-15 · 142	Which broader statistical, temporal, adversarial and operational properties remain outside the first claim-policy kernel?
F-16 · 144	Can the field rows be reproduced from a clean exporter/content-addressed snapshot while retaining true acquisition time?
F-17 · 147	Should fresh qwen-stdin-prompt.v1 receipts be supported by a versioned minimal adapter or explicitly deferred?
F-18 · 149	Can candidate radio mappings be validated against controlled differences and independent operator authority?
F-19 · 151	Which native and external runtimes are actually supported on constrained or second-architecture devices, and under which evidence, input and thermal conditions?
F-20 · 154	Can the runner reliably stop, resume, preserve failures and release held-out work without losing attempts or exceeding authority?
F-21 · 156	Can the application show nominal scores, uncertainty, absence and conflict without converting them into a misleading accuracy percentage?
F-22 · 158	What remains untested before real dropped-file, LAN/public or cross-origin operation?
F-23 · 161	How could owned memory/state, bias-evaluation methods or future local-model architectures extend this assurance work?
F-24 · 163	Could omitted archives, inaccessible peer receipts or stale summaries conceal a relevant result or contradiction?
A-01 · 165	Do prospective unique-window hit counts/calibration agree between host and independently run observer?
A-02 · 167	Can a keyed tokenizer-generation coupling change decoded outputs in a detectable, calibrated and quality-preserving way?
A-03 · 168	Does apparent same-model agreement survive independent role, context and provenance controls?
A-04 · 169	Does a single predeclared prompt-format change improve suitability without changing other generation factors?
A-05 · 170	Are invalid scalar values reachable in real callers of the watermark module?
A-06 · 171	Which watermark-removal or disclosure obligations apply to a specific actor, artifact and jurisdiction?
U-01 · 172	What variable is absent from the current protocol/profile fingerprint but can change token or export behavior?
U-02 · 173	What negative control should have produced a signal if the verifier/acquisition path were functioning?

ID	Question / probe
U-03 · 174	Where might the same failure be selected out in both producer and verifier paths?
U-04 · 175	Could source timestamps, export paths, dependency upgrades or metadata change explain an apparent mechanism change?
U-05 · 176	What unsupported conclusion could a downstream reader still infer from the current UI or certificate wording?
U-06 · 177	What genuinely new observation fails to fit every registered explanation?

F-01 questions

Does Anthropic use green-list biasing, a Gumbel-family method, another sampling construction, or a different/combined carrier on each surface?

Research: R02 · R04 · R06 · R08 · R09 · R10 · R17 · R35 · Sources: S-QUESTIONS

Current effective disposition

Layer

freeze_updates

Value

Primary source rechecked: SynthID-Text family publicly declared; exact private profile, historic applicability and access remain unresolved. Contrary statement in new research report rejected.

Preserved historical records

Historical stage: `baseline_record`

Field	Historical value
Id	F-01
Category	Provider mechanism
Question	Does Anthropic use green-list biasing, a Gumbel-family method, another sampling construction, or a different/combined carrier on each surface?
Epistemic state	unresolved
Current boundary	provider_method_unidentified
Next discriminating test or decision	A supported official verifier/specification or controlled positive acquisition with documented applicability.
Prohibited inference	Do not name a private vendor algorithm from local-lab behavior or chatbot self-description.
Reopen trigger	Current exact surface documentation or an official verifier becomes usable.

Historical stage: `analytics_update`

Field	Historical value
Id	F-01
Question	Does Anthropic use green-list biasing, a Gumbel-family method, another sampling construction, or a different/combined carrier on each surface?
Prior epistemic state	unresolved
Analytics state	provider_documentation_refreshed_mechanism_open
New observation	Opus5 current support and detector private-preview documented; August applicability and private scheme not established.
Next discriminating test or decision	A supported official verifier/specification or controlled positive acquisition with documented applicability.
Reopen trigger	Current exact surface documentation or an official verifier becomes usable.

Additive status history

Layer
local_reconciliation

Value

State
partially_resolved_provider_declaration

Observation
Anthropic publicly identifies the SynthID-Text family. Proprietary details, historic specimen applicability and operator detector access remain unresolved.

Next step
Correct the paper and retain an exact-surface/historical verification trigger.

Next step executed by this review
No

Layer
freeze_updates

Value
Primary source rechecked: SynthID-Text family publicly declared; exact private profile, historic applicability and access remain unresolved. Contrary statement in new research report rejected.

F-02 questions

What do the original negative byte scans actually exclude, and what ordinary-format or exporter-induced signals were not tested?

Research: R01 · R03 · R04 · R05 · R06 · Sources: S-QUESTIONS

Current effective disposition

Layer
local_reconciliation

Value

State
bounded_absence_preserved

Observation
Negative byte scans constrain the checked files and features only; ordinary formatting and other carriers were not universally excluded.

Next step

Retain original scan/control scope; reopen for a specified new feature or export comparison.

Next step executed by this review

No

Publication custody update

Six retained summary copies and early audit outputs/scripts are recovered and hash-matched where a historical ledger exists. The 17 original audit inputs remain unavailable in the reviewed material. No carrier audit was rerun by recovery.

Preserved historical records

Historical stage: `baseline_record`

Field	Historical value
Id	F-02
Category	Whitespace and Unicode
Question	What do the original negative byte scans actually exclude, and what ordinary-format or exporter-induced signals were not tested?
Epistemic state	bounded_negative_historical
Current boundary	historical_scan_not_universal_falsification
Next discriminating test or decision	Re-run exact original-byte scans with synthetic positive controls; separate exotic characters, ordinary whitespace choices, parsing and export transforms.
Prohibited inference	Do not claim every whitespace watermark is impossible merely because no exotic whitespace was found.
Reopen trigger	A specified new carrier or raw export comparison is proposed.

Historical stage: `analytics_update`

Field	Historical value
Id	F-02
Question	What do the original negative byte scans actually exclude, and what ordinary-format or exporter-induced signals were not tested?
Prior epistemic state	bounded_negative_historical
Analytics state	preserved_open_no_new_empirical_test
New observation	No new empirical resolution in this pass; original question, test and reopening trigger retained.
Next discriminating test or decision	Re-run exact original-byte scans with synthetic positive controls; separate exotic characters, ordinary whitespace choices, parsing and export transforms.
Reopen trigger	A specified new carrier or raw export comparison is proposed.

Additive status history

Layer

local_reconciliation

Value

State

bounded_absence_preserved

Observation

Negative byte scans constrain the checked files and features only; ordinary formatting and other carriers were not universally excluded.

Next step

Retain original scan/control scope; reopen for a specified new feature or export comparison.

Next step executed by this review

No

F-03 questions

Could keyed/custom segmentation, tokenizer behavior or token-boundary conventions carry or change an observable provenance signal?

Research: R03 · R04 · R11 · R21 · R22 · Sources: S-QUESTIONS

Current effective disposition

Layer

local_reconciliation

Value

State

simple_relabeling_null_confirmed_broader_coupling_open

Observation

All 24 consistent toy relabelings preserve decoded distributions. Output-changing coupling remains untested and novelty unestablished.

Next step

Optional bounded constructive toy study after essential receipts; broader model integration belongs to V2.

Next step executed by this review

No

Preserved historical records

Historical stage: `baseline_record`

Field	Historical value
Id	F-03
Category	Tokenizer hypothesis
Question	Could keyed/custom segmentation, tokenizer behavior or token-boundary conventions carry or change an observable provenance signal?
Epistemic state	untested
Current boundary	hypothesis_only
Next discriminating test or decision	Owned known-key lab with explicit tokenizer behavior/profile identities and fixed raw text/export controls; distinguish segmentation from output-token selection.
Prohibited inference	No inference that vendor tokenizers are keyed or have been recovered.
Reopen trigger	A concrete owned tokenizer experiment is separately approved.

Historical stage: `analytics_update`

Field	Historical value
Id	F-03
Question	Could keyed/custom segmentation, tokenizer behavior or token-boundary conventions carry or change an observable provenance signal?
Prior epistemic state	untested
Analytics state	minimal_subhypothesis_tested_broader_question_open
New observation	All24 consistent ID relabelings preserve decoded distribution; two segmentations collapse to one text. Coupled generation remains untested.
Next discriminating test or decision	Owned known-key lab with explicit tokenizer behavior/profile identities and fixed raw text/export controls; distinguish segmentation from output-token selection.
Reopen trigger	A concrete owned tokenizer experiment is separately approved.

Additive status history

Layer

local_reconciliation

Value

State

simple_relabeling_null_confirmed_broader_coupling_open

Observation

All 24 consistent toy relabelings preserve decoded distributions. Output-changing coupling remains untested and novelty unestablished.

Next step

Optional bounded constructive toy study after essential receipts; broader model integration belongs to V2.

Next step executed by this review

No

F-04 questions

Are keys/contexts scoped per request, account, model, workspace, time or nonce, and how would each affect reproducibility?

Research: R08 · R09 · R11 · R17 · R35 · Sources: S-QUESTIONS

Current effective disposition

Layer

local_reconciliation

Value

State

partially_documented_scope_open

Observation

Provider documentation says the mark is not an embedded user/chat identifier; this does not disclose every key-management scope.

Next step

Retain exact key/nonce scope as an unresolved implementation question.

Next step executed by this review

No

Preserved historical records

Historical stage: `baseline_record`

Field	Historical value
Id	F-04
Category	Key and context scope
Question	Are keys/contexts scoped per request, account, model, workspace, time or nonce, and how would each affect reproducibility?
Epistemic state	untested
Current boundary	candidate_explanations_not_vendor_facts
Next discriminating test or decision	A consented matched acquisition protocol with fixed prompts, exact identities and independently logged context; no secret-key search.
Prohibited inference	Same/different wording across two accounts does not identify key scope.
Reopen trigger	Verifier scope or a controlled account comparison provides a discriminating test.

Historical stage: `analytics_update`

Field	Historical value
Id	F-04
Question	Are keys/contexts scoped per request, account, model, workspace, time or nonce, and how would each affect reproducibility?
Prior epistemic state	untested
Analytics state	preserved_open_no_new_empirical_test
New observation	No new empirical resolution in this pass; original question, test and reopening trigger retained.
Next discriminating test or decision	A consented matched acquisition protocol with fixed prompts, exact identities and independently logged context; no secret-key search.
Reopen trigger	Verifier scope or a controlled account comparison provides a discriminating test.

Additive status history

Layer

local_reconciliation

Value

State

partially_documented_scope_open

Observation

Provider documentation says the mark is not an embedded user/chat identifier; this does not disclose every key-management scope.

Next step

Retain exact key/nonce scope as an unresolved implementation question.

Next step executed by this review

No

F-05 questions

Which observed bytes originate in generation, browser/API export, PDF construction, copying, or credential attachment?

Research: R05 · R12 · R13 · R15 · R35 · R36 · Sources: S-QUESTIONS

Current effective disposition

Layer

local_reconciliation

Value

State

export_stage_distinction_documented

Observation

Current documented media signing can occur at download, separate from generation/container bytes. Historic artifacts remain scoped to their actual route.

Next step

Record original/download/derivative hashes if a new acquisition is admitted.

Next step executed by this review

No

Preserved historical records

Historical stage: `baseline_record`

Field	Historical value
Id	F-05
Category	Export path and container
Question	Which observed bytes originate in generation, browser/API export, PDF construction, copying, or credential attachment?
Epistemic state	unresolved
Current boundary	export_path_specific
Next discriminating test or decision	Preserve paired actual downloads from one completion; hash container-source and exported bytes separately; validate credentials only under applicable scope.
Prohibited inference	A clean Markdown scan does not explain PDF/media metadata or every export path.
Reopen trigger	One supported paired-export/control route is selected.

Historical stage: `analytics_update`

Field	Historical value
Id	F-05
Question	Which observed bytes originate in generation, browser/API export, PDF construction, copying, or credential attachment?
Prior epistemic state	unresolved
Analytics state	current_export_path_documented_not_acquired
New observation	Current FilesAPI signing is on downloaded supported media; PDF/text exclusions matter for selected controls. Historical artifacts were not recreated.
Next discriminating test or decision	Preserve paired actual downloads from one completion; hash container-source and exported bytes separately; validate credentials only under applicable scope.
Reopen trigger	One supported paired-export/control route is selected.

Additive status history

Layer

local_reconciliation

Value

State

export_stage_distinction_documented

Observation

Current documented media signing can occur at download, separate from generation/container bytes. Historic artifacts remain scoped to their actual route.

Next step

Record original/download/derivative hashes if a new acquisition is admitted.

Next step executed by this review

No

F-06 questions

Under which known detector/profile/key and sample-length conditions could one sufficiently long artifact support a result?

Research: R01 · R06 · R09 · R33 · Sources: S-QUESTIONS

Current effective disposition

Layer

local_reconciliation

Value

State

claim_specific_unresolved

Observation

One sufficiently long specimen can carry a profile-specific signal; available local evidence does not give a calibrated universal authorship verdict.

Next step

Define scheme, detector, key, eligibility and calibration before any single-artifact inference.

Next step executed by this review

No

Preserved historical records

Historical stage: `baseline_record`

Field	Historical value
Id	F-06
Category	Single-file detection
Question	Under which known detector/profile/key and sample-length conditions could one sufficiently long artifact support a result?
Epistemic state	unresolved
Current boundary	no_validated_unknown_provider_detector
Next discriminating test or decision	Known-positive/negative samples with applicable detector and calibration; report length/domain/abstention and specificity.
Prohibited inference	Do not repeat the blanket assertion that a single Markdown file is inherently undetectable.
Reopen trigger	An applicable detector and declared decision scope exist.

Historical stage: `analytics_update`

Field	Historical value
Id	F-06
Question	Under which known detector/profile/key and sample-length conditions could one sufficiently long artifact support a result?
Prior epistemic state	unresolved
Analytics state	preserved_open_no_new_empirical_test
New observation	No new empirical resolution in this pass; original question, test and reopening trigger retained.
Next discriminating test or decision	Known-positive/negative samples with applicable detector and calibration; report length/domain/abstention and specificity.
Reopen trigger	An applicable detector and declared decision scope exist.

Additive status history

Layer

local_reconciliation

Value

State

claim_specific_unresolved

Observation

One sufficiently long specimen can carry a profile-specific signal; available local evidence does not give a calibrated universal authorship verdict.

Next step

Define scheme, detector, key, eligibility and calibration before any single-artifact inference.

Next step executed by this review

No

F-07 questions

What causes the marked repetition and incomplete task fulfillment: watermark, prompt formatting, sampling, model behavior or interaction?

Research: R19 · Sources: S-QUESTIONS

Current effective disposition

Layer

freeze_updates

Value

Optional at-most-12-generation single-factor diagnostic deferred; no pre-freeze execution needed.

Preserved historical records

Historical stage: `baseline_record`

Field	Historical value
Id	F-07
Category	Sampling/quality interaction
Question	What causes the marked repetition and incomplete task fulfillment: watermark, prompt formatting, sampling, model behavior or interaction?
Epistemic state	observed_adverse_cause_untested
Current boundary	one_pair_not_causal
Next discriminating test or decision	Methods chooses retain/change/stop; at most one approved bounded new development comparison with other factors fixed.
Prohibited inference	Do not infer causation from one paired seed or delete the adverse sample.
Reopen trigger	Immediate methods decision; new data only after a frozen approved plan.

Historical stage: `analytics_update`

Field	Historical value
Id	F-07
Question	What causes the marked repetition and incomplete task fulfillment: watermark, prompt formatting, sampling, model behavior or interaction?
Prior epistemic state	observed_adverse_cause_untested
Analytics state	adverse_pair_characterized_cause_open
New observation	Both outputs cap-reached withoutEOS; marked repeated paragraph; no causal attribution from one pair.
Next discriminating test or decision	Methods chooses retain/change/stop; at most one approved bounded new development comparison with other factors fixed.
Reopen trigger	Immediate methods decision; new data only after a frozen approved plan.

Additive status history

Layer

local_reconciliation

Value

State

quality_problem_confirmed_cause_open

Observation

Both native outputs have task-quality issues; marked repetition is substantial. A single pair does not identify cause.

Next step

Existing-trace rescore first; then at most one 12-attempt formatting diagnostic if useful.

Next step executed by this review

No

Layer

continuation_updates

Value

Quality and causal question remain open; no new native generations were needed for this rescore.

Layer

freeze_updates

Value

Optional at-most-12-generation single-factor diagnostic deferred; no pre-freeze execution needed.

F-08 questions

How should repeated context and repeated context/token windows affect diagnostics, eligible denominator and inferential scoring?

Research: R13 · R18 · R20 · Sources: S-QUESTIONS

Current effective disposition

Layer

freeze_updates

Value

Measured membership and repeated-unit diagnostics retained; no calibration promotion.

Preserved historical records

Historical stage: `baseline_record`

Field	Historical value
Id	F-08
Category	Dependence and scoring
Question	How should repeated context and repeated context/token windows affect diagnostics, eligible denominator and inferential scoring?
Epistemic state	observed_dependence_unqualified_inference
Current boundary	counts_replayed_not_calibrated
Next discriminating test or decision	Define exact deduplication unit; preserve raw endpoint; new inferential profile requires new calibration and independent replay.
Prohibited inference	Deduplication is not an automatic proof of independence or calibrated probability.
Reopen trigger	Immediate endpoint decision.

Historical stage: `analytics_update`

Field	Historical value
Id	F-08
Question	How should repeated context and repeated context/token windows affect diagnostics, eligible denominator and inferential scoring?
Prior epistemic state	observed_dependence_unqualified_inference
Analytics state	new_partial_identification_and_synthetic_evidence
New observation	Distinct marked windows93; unique hits39..75 compatible with aggregate168. Synthetic duplicate windows inflate nominal z. Actual native unique hits unknown.
Next discriminating test or decision	Define exact deduplication unit; preserve raw endpoint; new inferential profile requires new calibration and independent replay.
Reopen trigger	Immediate endpoint decision.

Additive status history

Layer

local_reconciliation

Value

State

dependence_confirmed_endpoint_decision_required

Observation

Context and context-plus-target units differ. Exact repeated-window counts and 39-75 feasible hits independently checked; actual keyed distinct hits still missing.

Next step

Highest-priority local existing-key rescore, reporting raw and both distinct units without retroactive calibration.

Next step executed by this review

No

Layer

continuation_updates

Value

Actual membership measured: marked primary 60/93 distinct contexts/windows, clean primary 53/249 contexts and 53/251 windows. All six raw results and independent scalar implementation agree. Calibration remains open.

Layer
freeze_updates

Value
Measured membership and repeated-unit diagnostics retained; no calibration promotion.

F-09 questions

What performance/uncertainty claim does the experiment actually have enough independent data to assess?

Research: R09 · R11 · R18 · R20 · R33 · Sources: S-QUESTIONS

Current effective disposition

Layer
local_reconciliation

Value

State
descriptive_scope_selected_for_recommendation

Observation
The existing pair and 30/30 draft cannot support broad low-FPR performance claims. Weak evidence remains useful when labeled.

Next step
Keep V1 descriptive; enlarge a study only under a new claim-sized prospective decision.

Next step executed by this review
No

Preserved historical records

Historical stage: `baseline_record`

Field	Historical value
Id	F-09
Category	Calibration and rare false positives
Question	What performance/uncertainty claim does the experiment actually have enough independent data to assess?
Epistemic state	unexecuted_study
Current boundary	pilot_not_1_percent_guarantee
Next discriminating test or decision	Freeze claim before sample size; prompt/pair unit, eligibility, threshold and multiplicity; report empirical uncertainty including zero-error bounds.
Prohibited inference	Two wrong keys and 360 measurements are not 360 independent negatives or familywise control.
Reopen trigger	A stronger performance claim is explicitly requested.

Historical stage: `analytics_update`

Field	Historical value
Id	F-09
Question	What performance/uncertainty claim does the experiment actually have enough independent data to assess?
Prior epistemic state	unexecuted_study
Analytics state	finite_sample_limit_calculated_calibration_unrun
New observation	Zero/30 gives95% upper9.5034% under independent fixed-threshold negatives; no inferred1%FPR.
Next discriminating test or decision	Freeze claim before sample size; prompt/pair unit, eligibility, threshold and multiplicity; report empirical uncertainty including zero-error bounds.
Reopen trigger	A stronger performance claim is explicitly requested.

Additive status history

Layer
local_reconciliation

Value

State
descriptive_scope_selected_for_recommendation

Observation
The existing pair and 30/30 draft cannot support broad low-FPR performance claims. Weak evidence remains useful when labeled.

Next step
Keep V1 descriptive; enlarge a study only under a new claim-sized prospective decision.

Next step executed by this review
No

F-10 questions

How do normalization, copy/paste, extraction, formatting and other owned deterministic transformations affect the registered signal?

Research: R14 · R25 · R34 · Sources: S-QUESTIONS

Current effective disposition

Layer
local_reconciliation

Value

State
bounded_followon_optional

Observation
Synthetic repetition is measured; real owned transform survival remains unmeasured here.

Next step
Optional fixed small matrix with parent-child hashes and identity control; no universal robustness claim.

Next step executed by this review
No

Preserved historical records

Historical stage: `baseline_record`

Field	Historical value
Id	F-10
Category	Transform survival
Question	How do normalization, copy/paste, extraction, formatting and other owned deterministic transformations affect the registered signal?
Epistemic state	tooling_exists_campaign_untested
Current boundary	no_robustness_acceptance
Next discriminating test or decision	Finite frozen transform matrix, parent-child hashes and same applicable detector/profile, matched controls and failures.
Prohibited inference	No intent or deliberate-removal conclusion from signal loss.
Reopen trigger	Transform robustness becomes an included claim.

Historical stage: `analytics_update`

Field	Historical value
Id	F-10
Question	How do normalization, copy/paste, extraction, formatting and other owned deterministic transformations affect the registered signal?
Prior epistemic state	tooling_exists_campaign_untested
Analytics state	preserved_open_no_new_empirical_test
New observation	No new empirical resolution in this pass; original question, test and reopening trigger retained.
Next discriminating test or decision	Finite frozen transform matrix, parent-child hashes and same applicable detector/profile, matched controls and failures.
Reopen trigger	Transform robustness becomes an included claim.

Additive status history

Layer
local_reconciliation

Value

State
bounded_followon_optional

Observation
Synthetic repetition is measured; real owned transform survival remains unmeasured here.

Next step
Optional fixed small matrix with parent-child hashes and identity control; no universal robustness claim.

Next step executed by this review
No

F-11 questions

Can a passive/stylometric tool help inspect code without confusing generator style with correctness or sloppiness?

Research: R07 · R10 · R23 · R24 · Sources: S-QUESTIONS

Current effective disposition

Layer
local_reconciliation

Value

State
future_protocol_preserved

Observation
Passive and emergent interaction fingerprints are research ideas, not current attribution capability.

Next step
Keep mirror-agent factorial protocol in V2, with shared-context and cross-model controls.

Next step executed by this review
No

Preserved historical records

Historical stage: `baseline_record`

Field	Historical value
Id	F-11
Category	Passive detection and code quality
Question	Can a passive/stylometric tool help inspect code without confusing generator style with correctness or sloppiness?
Epistemic state	untested
Current boundary	not_a_validated_detector
Next discriminating test or decision	Matched content/format/domain baselines, unseen-generator split, measured operational false positives and independent code-quality ground truth.
Prohibited inference	AI origin is neither a code defect nor attribution to a malicious actor.
Reopen trigger	An explicit passive-detector product claim is proposed.

Historical stage: `analytics_update`

Field	Historical value
Id	F-11
Question	Can a passive/stylometric tool help inspect code without confusing generator style with correctness or sloppiness?
Prior epistemic state	untested
Analytics state	preserved_open_no_new_empirical_test
New observation	No new empirical resolution in this pass; original question, test and reopening trigger retained.
Next discriminating test or decision	Matched content/format/domain baselines, unseen-generator split, measured operational false positives and independent code-quality ground truth.
Reopen trigger	An explicit passive-detector product claim is proposed.

Additive status history

Layer
local_reconciliation

Value

State
future_protocol_preserved

Observation
Passive and emergent interaction fingerprints are research ideas, not current attribution capability.

Next step
Keep mirror-agent factorial protocol in V2, with shared-context and cross-model controls.

Next step executed by this review
No

F-12 questions

What evidence can link an artifact to a system/workflow versus to a person, organization, or threat actor?

Research: R02 · R07 · R14 · R23 · R24 · R25 · R34 · Sources: S-QUESTIONS

Current effective disposition

Layer
local_reconciliation

Value

State
attribution_ceiling_retained

Observation
Known-key compatibility, provider declaration and file provenance do not identify a person, intent, ownership or truth.

Next step
Apply the ceiling to paper tables/cards; reopen only with separate corroborating evidence.

Next step executed by this review
No

Preserved historical records

Historical stage: `baseline_record`

Field	Historical value
Id	F-12
Category	Attribution scope
Question	What evidence can link an artifact to a system/workflow versus to a person, organization, or threat actor?
Epistemic state	unresolved
Current boundary	provenance_not_identity_or_intent
Next discriminating test or decision	Multiple independent applicable evidence sources with explicit alternatives, access records and claim boundaries.
Prohibited inference	No human identity, ownership, intent or APT attribution from a watermark alone.
Reopen trigger	A separately scoped defensive-attribution case exists.

Historical stage: `analytics_update`

Field	Historical value
Id	F-12
Question	What evidence can link an artifact to a system/workflow versus to a person, organization, or threat actor?
Prior epistemic state	unresolved
Analytics state	preserved_open_no_new_empirical_test
New observation	No new empirical resolution in this pass; original question, test and reopening trigger retained.
Next discriminating test or decision	Multiple independent applicable evidence sources with explicit alternatives, access records and claim boundaries.
Reopen trigger	A separately scoped defensive-attribution case exists.

Additive status history

Layer
local_reconciliation

Value

State
attribution_ceiling_retained

Observation
Known-key compatibility, provider declaration and file provenance do not identify a person, intent, ownership or truth.

Next step
Apply the ceiling to paper tables/cards; reopen only with separate corroborating evidence.

Next step executed by this review
No

F-13 questions

Do actual native validators correctly classify signed trusted/untrusted, tampered, unsigned, derivative and malformed artifacts?

Research: R12 · R35 · Sources: S-QUESTIONS

Current effective disposition

Layer
local_reconciliation

Value

State
deferred_unqualified

Observation
Synthetic policy tests are not native trusted/untrusted/tampered C2PA qualification.

Next step
Exclude a native C2PA capability claim from V1 unless all actual controls are acquired and validated.

Next step executed by this review
No

Preserved historical records

Historical stage: `baseline_record`

Field	Historical value
Id	F-13
Category	C2PA trust and files
Question	Do actual native validators correctly classify signed trusted/untrusted, tampered, unsigned, derivative and malformed artifacts?
Epistemic state	qualification_absent_in_selected_record
Current boundary	synthetic_tests_only
Next discriminating test or decision	Pinned official backend and trust snapshot, authorized owned/official controls, isolated worker and raw validation receipts.
Prohibited inference	A synthetic pass or valid signature does not prove semantic truth or ownership.
Reopen trigger	C2PA capability is included in a release or a control campaign is approved.

Historical stage: `analytics_update`

Field	Historical value
Id	F-13
Question	Do actual native validators correctly classify signed trusted/untrusted, tampered, unsigned, derivative and malformed artifacts?
Prior epistemic state	qualification_absent_in_selected_record
Analytics state	preserved_open_no_new_empirical_test
New observation	No new empirical resolution in this pass; original question, test and reopening trigger retained.
Next discriminating test or decision	Pinned official backend and trust snapshot, authorized owned/official controls, isolated worker and raw validation receipts.
Reopen trigger	C2PA capability is included in a release or a control campaign is approved.

Additive status history

Layer
local_reconciliation

Value

State
deferred_unqualified

Observation
Synthetic policy tests are not native trusted/untrusted/tampered C2PA qualification.

Next step
Exclude a native C2PA capability claim from V1 unless all actual controls are acquired and validated.

Next step executed by this review
No

F-14 questions

Can the actual current Python claim policy be aligned, built and bound to a small Lean certificate and cross-language vectors?

Research: R16 · R26 · R42 · Sources: S-QUESTIONS

Current effective disposition

Layer
local_reconciliation

Value

State
source_mismatch_supported_formal_claim_deferred

Observation
One source-level policy mismatch is documented. The transcript claim of seven cases lacks an inspected final case inventory; no current Lean build was performed.

Next step
Use one bounded finding; request cases only if the seven-case claim is retained. Defer certification.

Next step executed by this review
No

Preserved historical records

Historical stage: `baseline_record`

Field	Historical value
Id	F-14
Category	Formal policy
Question	Can the actual current Python claim policy be aligned, built and bound to a small Lean certificate and cross-language vectors?
Epistemic state	source_mismatch_observed
Current boundary	not_formally_certified
Next discriminating test or decision	Resolve insufficient-length counterexample, pin toolchain/policy/evidence, build and independently replay.
Prohibited inference	A proof seed or successful policy theorem does not prove empirical detector accuracy.
Reopen trigger	Formal capability is claimed or its work package is approved.

Historical stage: `analytics_update`

Field	Historical value
Id	F-14
Question	Can the actual current Python claim policy be aligned, built and bound to a small Lean certificate and cross-language vectors?
Prior epistemic state	source_mismatch_observed
Analytics state	preserved_open_no_new_empirical_test
New observation	No new empirical resolution in this pass; original question, test and reopening trigger retained.
Next discriminating test or decision	Resolve insufficient-length counterexample, pin toolchain/policy/evidence, build and independently replay.
Reopen trigger	Formal capability is claimed or its work package is approved.

Additive status history

Layer
local_reconciliation

Value

State
source_mismatch_supported_formal_claim_deferred

Observation
One source-level policy mismatch is documented. The transcript claim of seven cases lacks an inspected final case inventory; no current Lean build was performed.

Next step
Use one bounded finding; request cases only if the seven-case claim is retained. Defer certification.

Next step executed by this review
No

F-15 questions

Which broader statistical, temporal, adversarial and operational properties remain outside the first claim-policy kernel?

Research: R26 · R42 · Sources: S-QUESTIONS

Current effective disposition

Layer
local_reconciliation

Value

State
future_scope_preserved

Observation
Broader formal/statistical/temporal/adversarial assurance is outside this case study.

Next step
Keep explicit future scope and reopen on a defined property and bounded proof/test plan.

Next step executed by this review
No

Preserved historical records

Historical stage: `baseline_record`

Field	Historical value
Id	F-15
Category	Advanced verification
Question	Which broader statistical, temporal, adversarial and operational properties remain outside the first claim-policy kernel?
Epistemic state	untested
Current boundary	future_certification_scope
Next discriminating test or decision	Choose one property/assumption boundary at a time; pair formal derivations with empirical evidence where required.
Prohibited inference	Do not use Lean as a generic accuracy or security certificate.
Reopen trigger	A concrete property beyond the first kernel is scoped.

Historical stage: `analytics_update`

Field	Historical value
Id	F-15
Question	Which broader statistical, temporal, adversarial and operational properties remain outside the first claim-policy kernel?
Prior epistemic state	untested
Analytics state	preserved_open_no_new_empirical_test
New observation	No new empirical resolution in this pass; original question, test and reopening trigger retained.
Next discriminating test or decision	Choose one property/assumption boundary at a time; pair formal derivations with empirical evidence where required.
Reopen trigger	A concrete property beyond the first kernel is scoped.

Additive status history

> **Layer**
> local_reconciliation
>
> **Value**
>
> **State**
> future_scope_preserved
>
> **Observation**
> Broader formal/statistical/temporal/adversarial assurance is outside this case study.
>
> **Next step**
> Keep explicit future scope and reopen on a defined property and bounded proof/test plan.
>
> **Next step executed by this review**
> No

F-16 questions

Can the field rows be reproduced from a clean exporter/content-addressed snapshot while retaining true acquisition time?

Research: R27 · R28 · R31 · R32 · Sources: S-QUESTIONS

> **Current effective disposition**
>
> **Layer**
> freeze_updates
>
> **Value**
> Unavailable exporter disposition closes response obligation; stronger clean-producer claim excluded, future question open.

Preserved historical records

Historical stage: `baseline_record`

Field	Historical value
Id	F-16
Category	field node clean lineage
Question	Can the field rows be reproduced from a clean exporter/content-addressed snapshot while retaining true acquisition time?
Epistemic state	missing_required_receipt
Current boundary	dirty_manifests_present
Next discriminating test or decision	One clean re-export per included family with IDs, source/input hashes, validation and independent acceptance; otherwise explicit exclusion.
Prohibited inference	Flipping source_dirty to false or reusing old timestamps is not a repair.
Reopen trigger	Immediate if the field families remain in the finite milestone.

Historical stage: `analytics_update`

Field	Historical value
Id	F-16
Question	Can the field rows be reproduced from a clean exporter/content-addressed snapshot while retaining true acquisition time?
Prior epistemic state	missing_required_receipt
Analytics state	preserved_open_no_new_empirical_test
New observation	No new empirical resolution in this pass; original question, test and reopening trigger retained.
Next discriminating test or decision	One clean re-export per included family with IDs, source/input hashes, validation and independent acceptance; otherwise explicit exclusion.
Reopen trigger	Immediate if the field families remain in the finite milestone.

Additive status history

Layer
local_reconciliation

Value

State
conditional_v1_field_gate

Observation
Clean/content-addressed exporter evidence remains distinct from installed-wheel import acceptance.

Next step
field node supplies honest source-bound re-exports for included families or the paper narrows field-reproducibility claims.

Next step executed by this review
No

Layer
continuation_updates

Value
Exporter source provenance is the next field request; stronger field claims remain pending.

Layer
field_return_updates

Value

State
unavailable_excluded_from_stronger_v1_claims

Observation
One historical model row and one radio row verify at file/canonical-row level, with source_dirty=true and no new export or complete exporter snapshot. Task B response accepted as unavailable.

Reopen trigger
Select these families for a stronger reproducible field-performance/clean-exporter claim, or receive a new complete source-bound export.

Future execution authorized
No

Layer

freeze_updates

Value

Unavailable exporter disposition closes response obligation; stronger clean-producer claim excluded, future question open.

F-17 questions

Should fresh qwen-stdin-prompt.v1 receipts be supported by a versioned minimal adapter or explicitly deferred?

Research: R30 · Sources: S-QUESTIONS

Current effective disposition

Layer

field_return_updates

Value

State

deferred_not_needed_for_selected_case_study

Observation

No new qwen-stdin-prompt.v1 adapter or fresh compact receipt selected.

Preserved historical records

Historical stage: `baseline_record`

Field	Historical value
Id	F-17
Category	Compact runtime schema
Question	Should fresh qwen-stdin-prompt.v1 receipts be supported by a versioned minimal adapter or explicitly deferred?
Epistemic state	unsupported_current_dispatch
Current boundary	decision_needed
Next discriminating test or decision	Real sanitized positive, malformed and hash-mismatch fixtures; declare producer versus verifier claims.
Prohibited inference	Do not rename schema to an older supported value.
Reopen trigger	Immediate support-matrix decision; implementation only when in scope.

Historical stage: `analytics_update`

Field	Historical value
Id	F-17
Question	Should fresh qwen-stdin-prompt.v1 receipts be supported by a versioned minimal adapter or explicitly deferred?
Prior epistemic state	unsupported_current_dispatch
Analytics state	preserved_open_no_new_empirical_test
New observation	No new empirical resolution in this pass; original question, test and reopening trigger retained.
Next discriminating test or decision	Real sanitized positive, malformed and hash-mismatch fixtures; declare producer versus verifier claims.
Reopen trigger	Immediate support-matrix decision; implementation only when in scope.

Additive status history

Layer
local_reconciliation

Value

State
defer_by_default

Observation
Fresh qwen-stdin-prompt.v1 support is not implied by accepted qwen-fast-contract import.

Next step
Add a versioned adapter only if that fresh compact receipt is selected for accepted ingestion.

Next step executed by this review
No

Layer
field_return_updates

Value

State
deferred_not_needed_for_selected_case_study

Observation
No new qwen-stdin-prompt.v1 adapter or fresh compact receipt selected.

F-18 questions

Can candidate radio mappings be validated against controlled differences and independent operator authority?

Research: R29 · Sources: S-QUESTIONS

Current effective disposition

Layer
local_reconciliation

Value

State
independent_authority_open

Observation

Producer self-declared operator verification checks consistency, not independent radio semantics.

Next step

Defer stronger mappings until a separate operator/control receipt supports them.

Next step executed by this review

No

Preserved historical records

Historical stage: `baseline_record`

Field	Historical value
Id	F-18
Category	Radio authority and semantics
Question	Can candidate radio mappings be validated against controlled differences and independent operator authority?
Epistemic state	candidate_only
Current boundary	not_semantic_acceptance
Next discriminating test or decision	Separate operator authority receipt; known controlled-difference experiment only under the radio lane scope.
Prohibited inference	Repeated identical images do not validate channel semantics; models cannot self-promote them.
Reopen trigger	Separate radio experiment or verified-semantics claim is approved.

Historical stage: `analytics_update`

Field	Historical value
Id	F-18
Question	Can candidate radio mappings be validated against controlled differences and independent operator authority?
Prior epistemic state	candidate_only
Analytics state	preserved_open_no_new_empirical_test
New observation	No new empirical resolution in this pass; original question, test and reopening trigger retained.
Next discriminating test or decision	Separate operator authority receipt; known controlled-difference experiment only under the radio lane scope.
Reopen trigger	Separate radio experiment or verified-semantics claim is approved.

Additive status history

Layer

local_reconciliation

Value

State

independent_authority_open

Observation

Producer self-declared operator verification checks consistency, not independent radio semantics.

Next step

Defer stronger mappings until a separate operator/control receipt supports them.

Next step executed by this review

No

F-19 questions

Which native and external runtimes are actually supported on constrained or second-architecture devices, and under which evidence, input and thermal conditions?

Research: R27 · R28 · R31 · R40 · Sources: S-QUESTIONS

Current effective disposition

Layer

freeze_updates

Value

Bounded field node observer return accepted as peer-reported execution with exact nonsecret counts/arithmetic; broader native/accelerator claims remain open.

Preserved historical records

Historical stage: `baseline_record`

Field	Historical value
Id	F-19
Category	Hardware and portability
Question	What actually runs on field device, mobile device, particular accelerators, and under which evidence/prompt/thermal budget?
Epistemic state	profile_limited_observations
Current boundary	no_general_accelerator_or_pixel_proof
Next discriminating test or decision	Exact-device/model/backend input-budget controls and execution-path receipts. Preserve local host native and field node external-runtime results separately.
Prohibited inference	Hardware availability or a device node does not establish accelerator use.
Reopen trigger	A new supported device/backend is expressly in scope.

Historical stage: `analytics_update`

Field	Historical value
Id	F-19
Question	What actually runs on field device, mobile device, particular accelerators, and under which evidence/prompt/thermal budget?
Prior epistemic state	profile_limited_observations
Analytics state	preserved_open_no_new_empirical_test
New observation	No new empirical resolution in this pass; original question, test and reopening trigger retained.
Next discriminating test or decision	Exact-device/model/backend input-budget controls and execution-path receipts. Preserve local host native and field node external-runtime results separately.
Reopen trigger	A new supported device/backend is expressly in scope.

Additive status history

Layer

local_reconciliation

Value

State

bounded_device_replay_recommended

Observation

Current evidence does not establish same-native-profile equivalence between local host and field device.

Next step

Obtain a standard-library observer portability receipt; keep native backend/model porting as a separate proposal.

Next step executed by this review

No

Layer

continuation_updates

Value

Self-contained stdlib observer packet rehearsed on local host; actual field node execution pending.

Layer
field_return_updates

Value

State
bounded_observer_portability_accepted_broader_hardware_question_open

Observation
field node/second-architecture standard-library observer output matches expected bytes; 251 clean and 93 marked distinct windows. No new native model/backend, mobile device or accelerator result.

Future execution authorized
No

Layer
freeze_updates

Value
Bounded field node observer return accepted as peer-reported execution with exact nonsecret counts/arithmetic; broader native/accelerator claims remain open.

F-20 questions

Can the runner reliably stop, resume, preserve failures and release held-out work without losing attempts or exceeding authority?

Research: R10 · R13 · R14 · R33 · R36 · R38 · R39 · R41 · R42 · Sources: S-QUESTIONS

Current effective disposition

Layer
freeze_updates

Value
Final bounded analytics reproduction passes; test-layout repair verified; historical full runner explicitly needs original workspace.

Preserved historical records

Historical stage: `baseline_record`

Field	Historical value
Id	F-20
Category	Experiment lifecycle
Question	Can the runner reliably stop, resume, preserve failures and release held-out work without losing attempts or exceeding authority?
Epistemic state	one_pair_harness_not_campaign_runner
Current boundary	implementation_pending_method
Next discriminating test or decision	Finite ledger/admission/cleanup regression matrix; explicit restart/attempt semantics; no uncontrolled loop.
Prohibited inference	Looping a one-pair harness is not a qualified campaign.
Reopen trigger	Pilot continuation is selected after methods reconciliation.

Historical stage: `analytics_update`

Field	Historical value
Id	F-20
Question	Can the runner reliably stop, resume, preserve failures and release held-out work without losing attempts or exceeding authority?
Prior epistemic state	one_pair_harness_not_campaign_runner
Analytics state	lifecycle_contract_specified_not_implemented
New observation	Finite transition/journal/threshold/resume tests specified; no new campaign runner or native execution.
Next discriminating test or decision	Finite ledger/admission/cleanup regression matrix; explicit restart/attempt semantics; no uncontrolled loop.
Reopen trigger	Pilot continuation is selected after methods reconciliation.

Additive status history

Layer
local_reconciliation

Value

State
conditional_runner_work

Observation
The one-pair harness is not a full campaign runner. Browser offline replay also has timeout and exact-float portability issues.

Next step
Repair offline replay now; implement attempted/failure/resource controls for any admitted native diagnostic, full phase machinery only for a real campaign.

Next step executed by this review
No

Layer
continuation_updates

Value
Amended offline replay passes with a declared deadline and narrow relative float tolerance. No larger native campaign admitted.

Layer
freeze_updates

Value
Final bounded analytics reproduction passes; test-layout repair verified; historical full runner explicitly needs original workspace.

F-21 questions

Can the application show nominal scores, uncertainty, absence and conflict without converting them into a misleading accuracy percentage?

Research: R01 · R15 · R17 · R24 · R30 · R38 · Sources: S-QUESTIONS

Current effective disposition

Layer
local_reconciliation

Value

State

paper_representation_now_product_ui_deferred

Observation

Nominal and weak evidence need clear fields, abstention and limits; full browser UI acceptance remains separate.

Next step

Use receipt-bound tables/figures/illustrative cards now; no percentage-of-AI conversion.

Next step executed by this review

No

Preserved historical records

Historical stage: `baseline_record`

Field	Historical value
Id	F-21
Category	Evidence-card semantics
Question	Can the application show nominal scores, uncertainty, absence and conflict without converting them into a misleading accuracy percentage?
Epistemic state	acceptance_not_complete
Current boundary	presentation_only
Next discriminating test or decision	Six canonical fixture states, visible claim ceiling and prohibitions, accessibility/errors, local target verification.
Prohibited inference	A confidence slider cannot manufacture calibration or origin certainty.
Reopen trigger	Canonical fixtures accepted; UI implementation then reviewed.

Historical stage: `analytics_update`

Field	Historical value
Id	F-21
Question	Can the application show nominal scores, uncertainty, absence and conflict without converting them into a misleading accuracy percentage?
Prior epistemic state	acceptance_not_complete
Analytics state	presentation_contract_amended
New observation	Raw score must be separated from unique counts, completion flags and calibration state; no authorship slider fromz.
Next discriminating test or decision	Six canonical fixture states, visible claim ceiling and prohibitions, accessibility/errors, local target verification.
Reopen trigger	Canonical fixtures accepted; UI implementation then reviewed.

Additive status history

Layer

local_reconciliation

Value

State

paper_representation_now_product_ui_deferred

Observation

Nominal and weak evidence need clear fields, abstention and limits; full browser UI acceptance remains separate.

Next step

Use receipt-bound tables/figures/illustrative cards now; no percentage-of-AI conversion.

Next step executed by this review

No

F-22 questions

What remains untested before real dropped-file, LAN/public or cross-origin operation?

Research: R15 · R16 · R30 · Sources: S-QUESTIONS

Current effective disposition

Layer

freeze_updates

Value

Implemented branch diagnostics remain unchanged; no further merge or deployment required.

Preserved historical records

Historical stage: `baseline_record`

Field	Historical value
Id	F-22
Category	Service and parser boundary
Question	What remains untested before real dropped-file, LAN/public or cross-origin operation?
Epistemic state	outside_current_scope
Current boundary	not_service_ready
Next discriminating test or decision	Separate authenticated/origin/CSRF/file-boundary/resource and target-host qualification with explicit release decision.
Prohibited inference	Static cards or loopback smoke tests do not establish safe public ingestion.
Reopen trigger	A network or live-ingestion release is actually requested.

Historical stage: `analytics_update`

Field	Historical value
Id	F-22
Question	What remains untested before real dropped-file, LAN/public or cross-origin operation?
Prior epistemic state	outside_current_scope
Analytics state	preserved_open_no_new_empirical_test
New observation	No new empirical resolution in this pass; original question, test and reopening trigger retained.
Next discriminating test or decision	Separate authenticated/origin/CSRF/file-boundary/resource and target-host qualification with explicit release decision.
Reopen trigger	A network or live-ingestion release is actually requested.

Additive status history

Layer

local_reconciliation

Value

State

direct_api_patch_qualified_in_copy_public_surface_open

Observation

Scalar-validation patch passes the complete canonical suite in an isolated copy; no canonical merge or public-ingress exploitation established.

Next step

Adopt bounded validation with regressions in the next source-writing task; public/LAN/dropped-file qualification remains deferred.

Next step executed by this review

No

Layer

continuation_updates

Value

Strict scalar validation and additive diagnostics implemented on local branch; 597 core tests pass, 2 skip. No release, public endpoint or deployment claim.

Layer
freeze_updates

Value
Implemented branch diagnostics remain unchanged; no further merge or deployment required.

F-23 questions

How could owned memory/state, bias-evaluation methods or future local-model architectures extend this assurance work?

Research: R40 · Sources: S-QUESTIONS

Current effective disposition

Layer
local_reconciliation

Value

State
future_project_preserved

Observation
Owned memory/state, owned bias-evaluation tests and architectures remain distinct projects.

Next step
Preserve hypotheses and require a scoped new proposal to reopen.

Next step executed by this review
No

Preserved historical records

Historical stage: `baseline_record`

Field	Historical value
Id	F-23
Category	Local sovereignty / learning extensions
Question	How could owned memory/state, owned bias-evaluation bias tests or future local-model architectures extend this assurance work?
Epistemic state	unscoped_future_work
Current boundary	not_current_acceptance_requirement
Next discriminating test or decision	Open a separately scoped property or local-model experiment with preserved source and falsifiable evaluation.
Prohibited inference	Do not turn memristor speculation, cognitive-bias ontology or broad agent autonomy into this release blocker.
Reopen trigger	A specific follow-on question is deliberately selected.

Historical stage: `analytics_update`

Field	Historical value
Id	F-23
Question	How could owned memory/state, owned bias-evaluation bias tests or future local-model architectures extend this assurance work?
Prior epistemic state	unscoped_future_work
Analytics state	preserved_open_no_new_empirical_test
New observation	No new empirical resolution in this pass; original question, test and reopening trigger retained.
Next discriminating test or decision	Open a separately scoped property or local-model experiment with preserved source and falsifiable evaluation.
Reopen trigger	A specific follow-on question is deliberately selected.

Additive status history

Layer
local_reconciliation

Value

State
future_project_preserved

Observation
Owned memory/state, owned bias-evaluation tests and architectures remain distinct projects.

Next step
Preserve hypotheses and require a scoped new proposal to reopen.

Next step executed by this review
No

F-24 questions

Could omitted archives, inaccessible peer receipts or stale summaries conceal a relevant result or contradiction?

Research: R01 · R04 · R05 · R06 · R10 · R28 · R32 · R36 · R37 · R39 · R41 · Sources: S-QUESTIONS

Current effective disposition

Layer
local_reconciliation

Value

State
research_artifact_received_reconciliation_complete

Observation
The completed research report is now local and reconciled with Analytics; missing sandbox-linked artifacts and opaque citations remain publication repairs.

Next step
Repair only references/artifacts used in the final paper; retain an intake trigger for genuinely new contradictory receipts.

Next step executed by this review
No

Publication custody update

Six retained summary copies and early audit outputs/scripts are recovered and hash-matched where a historical ledger exists. The 17 original audit inputs remain unavailable in the reviewed material. No carrier audit was rerun by recovery.

Preserved historical records

Historical stage: `baseline_record`

Field	Historical value
Id	F-24
Category	Missing or inaccessible evidence
Question	Could omitted archives, inaccessible peer receipts or stale summaries conceal a relevant result or contradiction?
Epistemic state	coverage_limited
Current boundary	absence_in_selection_not_global_absence
Next discriminating test or decision	Keep searched/packaged scope, missing inputs and trusted acquisition times explicit; reconcile a newly supplied result by identity.
Prohibited inference	Not present in this packet does not mean never existed or never tested anywhere.
Reopen trigger	A concrete new artifact, contradiction or support-matrix change arrives.

Historical stage: `analytics_update`

Field	Historical value
Id	F-24
Question	Could omitted archives, inaccessible peer receipts or stale summaries conceal a relevant result or contradiction?
Prior epistemic state	coverage_limited
Analytics state	completed_research_artifact_not_accessible
New observation	Do not infer launch request is finished report or substitute prior intake; attach exact report and reconcile when available.
Next discriminating test or decision	Keep searched/packaged scope, missing inputs and trusted acquisition times explicit; reconcile a newly supplied result by identity.
Reopen trigger	A concrete new artifact, contradiction or support-matrix change arrives.

Additive status history

Layer
local_reconciliation

Value

State
research_artifact_received_reconciliation_complete

Observation
The completed research report is now local and reconciled with Analytics; missing sandbox-linked artifacts and opaque citations remain publication repairs.

Next step
Repair only references/artifacts used in the final paper; retain an intake trigger for genuinely new contradictory receipts.

Next step executed by this review
No

A-01 analytics subquestions

Do prospective unique-window hit counts/calibration agree between host and independently run observer?

Research: R18 · R31 · Sources: S-QUESTIONS

Current effective disposition

Layer
field_return_updates

Value

State
trace_arithmetic_device_replay_accepted_keyed_and_calibration_limits_preserved

Observation
Actual keyed distinct hits remain the October 7 local host rescore. October 8 field node confirms trace counts and reported-score arithmetic without keys; it does not independently measure those hits or calibrate them.

Computation date
2026-10-07T23:26:27.485292Z

Return review date
2026-10-08

Preserved historical records

Historical stage: `analytics_record`

Field	Historical value
Id	A-01
Parents	• F-08
Question	Do prospective unique-window hit counts/calibration agree between host and independently run observer?
State	native membership missing
Test	existing-key local rescore; separate independent receipt
Claim limit	aggregate bounds are not actual dedup scores

Additive status history

Layer
local_reconciliation

Value
Actual keyed distinct hits remain the next local measurement; aggregate bounds are reproduced.

Layer
continuation_updates

Value
Native membership gap resolved locally; separate field node observer portability return pending.

Layer
field_return_updates

Value

State
trace_arithmetic_device_replay_accepted_keyed_and_calibration_limits_preserved

Observation
Actual keyed distinct hits remain the October 7 local host rescore. October 8 field node confirms trace counts and reported-score arithmetic without keys; it does not independently measure those hits or calibrate them.

A-02 analytics subquestions

Can a keyed tokenizer-generation coupling change decoded outputs in a detectable, calibrated and quality-preserving way?

Research: R21 · R22 · Sources: S-QUESTIONS

Current effective disposition

Layer

local_reconciliation

Value

Optional bounded constructive toy extension; identity-only control already reproduced; native coupling remains future work.

Preserved historical records

Historical stage: `analytics_record`

Field	Historical value
Id	A-02
Parents	• F-03
Question	Can a keyed tokenizer-generation coupling change decoded outputs in a detectable, calibrated and quality-preserving way?
State	untested
Test	future tiny owned controlled generator with identity-only negative and explicitly output-changing positive; frozen correct/wrong controls
Claim limit	no novelty/provider-mechanism conclusion

Additive status history

Layer

local_reconciliation

Value

Optional bounded constructive toy extension; identity-only control already reproduced; native coupling remains future work.

A-03 analytics subquestions

Does apparent same-model agreement survive independent role, context and provenance controls?

Research: R23 · R38 · Sources: S-QUESTIONS

Current effective disposition

Layer

local_reconciliation

Value

Preserve controlled V2 protocol; no agent experiment launched.

Preserved historical records

Historical stage: `analytics_record`

Field	Historical value
Id	A-03
Parents	• F-11 • F-12
Question	Does apparent same-model agreement survive independent role, context and provenance controls?
State	conversation hypothesis, not tested
Test	future blind factorial same/different-model and shared/separate-context comparison
Claim limit	shared style/agreement is not subconscious recognition, intent or intentional watermark

Additive status history

Layer

local_reconciliation

Value

Preserve controlled V2 protocol; no agent experiment launched.

A-04 analytics subquestions

Does a single predeclared prompt-format change improve suitability without changing other generation factors?

Research: R19 · Sources: S-QUESTIONS

Current effective disposition

Layer

local_reconciliation

Value

One conditional formatting diagnostic, at most 12 attempts; no new native run in this review.

Preserved historical records

Historical stage: `analytics_record`

Field	Historical value
Id	A-04
Parents	• F-07
Question	Does a single predeclared prompt-format change improve suitability without changing other generation factors?
State	prospective bounded diagnostic
Test	at most12 separately authorized generations once; all attempts retained
Claim limit	three prompts are a development diagnostic, not general efficacy

Additive status history

Layer

local_reconciliation

Value

One conditional formatting diagnostic, at most 12 attempts; no new native run in this review.

A-05 analytics subquestions

Are invalid scalar values reachable in real callers of the watermark module?

Research: R16 · Sources: S-QUESTIONS

Current effective disposition

Layer

continuation_updates

Value

Bounded API validation change implemented and tested; broader public-ingress reachability remains outside scope.

Preserved historical records

Historical stage: `analytics_record`

Field	Historical value
Id	A-05
Parents	• F-22
Question	Are invalid scalar values reachable in real callers of the watermark module?
State	direct API accepts shown cases; canonical reachability untested
Test	local caller/full-suite review and optional hash-bound validation patch
Claim limit	not evidence of a current public exploitable endpoint

Additive status history

Layer

local_reconciliation

Value

Patch source identity and full-suite compatibility now locally checked in an isolated copy; public-ingress reachability remains open.

Layer

continuation_updates

Value

Bounded API validation change implemented and tested; broader public-ingress reachability remains outside scope.

A-06 analytics subquestions

Which watermark-removal or disclosure obligations apply to a specific actor, artifact and jurisdiction?

Research: R34 · Sources: S-QUESTIONS

Current effective disposition

Layer

local_reconciliation

Value

Primary legal background now checked; application to specific facts/jurisdiction remains unresolved.

Publication scope note

Historical source review is background only. This design packet supplies no fact-specific legal advice or clearance; final legal assertions would need their own appropriate source check.

Preserved historical records

Historical stage: `analytics_record`

Field	Historical value
Id	A-06
Parents	• F-10 • F-12
Question	Which watermark-removal or disclosure obligations apply to a specific actor, artifact and jurisdiction?
State	legal analysis not performed
Test	primary-law/terms review by actual research return with qualified counsel as needed
Claim limit	technical possibility/public method is not legal clearance

Additive status history

Layer

local_reconciliation

Value

Primary legal background now checked; application to specific facts/jurisdiction remains unresolved.

U-01 residual unknown unknown probes

What variable is absent from the current protocol/profile fingerprint but can change token or export behavior?

Research: R37 · Sources: S-QUESTIONS

Current effective disposition

Layer

baseline/analytics

Value

Preserved unchanged; probe is not a finding.

Preserved historical records

Historical stage: `baseline_record`

Field	Historical value
Id	U-01
Probe	What variable is absent from the current protocol/profile fingerprint but can change token or export behavior?
Trigger	Equal declared profiles produce irreconcilable behavior.
Not a finding	Yes

Historical stage: `analytics_record`

Field	Historical value
Id	U-01
Probe	What variable is absent from the current protocol/profile fingerprint but can change token or export behavior?
Trigger	Equal declared profiles produce irreconcilable behavior.
Not a finding	Yes

U-02 residual unknown unknown probes

What negative control should have produced a signal if the verifier/acquisition path were functioning?

Research: R37 · Sources: S-QUESTIONS

Current effective disposition

Layer

local_editorial_clarification

Value

The baseline says negative control should have produced a signal. Preserve that historical wording, but use positive control expected to signal / negative control expected not to signal when implementing the probe.

Preserved historical records

Historical stage: `baseline_record`

Field	Historical value
Id	U-02
Probe	What negative control should have produced a signal if the verifier/ acquisition path were functioning?
Trigger	Expected positive control fails or negative control unexpectedly matches.
Not a finding	Yes

Historical stage: `analytics_record`

Field	Historical value
Id	U-02
Probe	What negative control should have produced a signal if the verifier/ acquisition path were functioning?
Trigger	Expected positive control fails or negative control unexpectedly matches.
Not a finding	Yes

Additive status history

Layer

local_editorial_clarification

Value

The baseline says negative control should have produced a signal. Preserve that historical wording, but use positive control expected to signal / negative control expected not to signal when implementing the probe.

U-03 residual unknown unknown probes

Where might the same failure be selected out in both producer and verifier paths?

Research: R23 · R37 · R38 · Sources: S-QUESTIONS

Current effective disposition

Layer

baseline/analytics

Value

Preserved unchanged; probe is not a finding.

Preserved historical records

Historical stage: `baseline_record`

Field	Historical value
Id	U-03
Probe	Where might the same failure be selected out in both producer and verifier paths?
Trigger	Attempt/denominator mismatch, missing records or shared implementation assumptions.
Not a finding	Yes

Historical stage: `analytics_record`

Field	Historical value
Id	U-03
Probe	Where might the same failure be selected out in both producer and verifier paths?
Trigger	Attempt/denominator mismatch, missing records or shared implementation assumptions.
Not a finding	Yes

U-04 residual unknown unknown probes

Could source timestamps, export paths, dependency upgrades or metadata change explain an apparent mechanism change?

Research: R37 · Sources: S-QUESTIONS

Current effective disposition

Layer

baseline/analytics

Value

Preserved unchanged; probe is not a finding.

Preserved historical records

Historical stage: `baseline_record`

Field	Historical value
Id	U-04
Probe	Could source timestamps, export paths, dependency upgrades or metadata change explain an apparent mechanism change?
Trigger	New version/surface or provenance mismatch.
Not a finding	Yes

Historical stage: `analytics_record`

Field	Historical value
Id	U-04
Probe	Could source timestamps, export paths, dependency upgrades or metadata change explain an apparent mechanism change?
Trigger	New version/surface or provenance mismatch.
Not a finding	Yes

U-05 residual unknown unknown probes

What unsupported conclusion could a downstream reader still infer from the current UI or certificate wording?

Research: R37 · Sources: S-QUESTIONS

Current effective disposition

Layer

baseline/analytics

Value

Preserved unchanged; probe is not a finding.

Preserved historical records

Historical stage: `baseline_record`

Field	Historical value
Id	U-05
Probe	What unsupported conclusion could a downstream reader still infer from the current UI or certificate wording?
Trigger	A reviewed card or report implies identity, intent, accuracy or safety beyond its receipts.
Not a finding	Yes

Historical stage: `analytics_record`

Field	Historical value
Id	U-05
Probe	What unsupported conclusion could a downstream reader still infer from the current UI or certificate wording?
Trigger	A reviewed card or report implies identity, intent, accuracy or safety beyond its receipts.
Not a finding	Yes

U-06 residual unknown unknown probes

What genuinely new observation fails to fit every registered explanation?

Research: R37 · Sources: S-QUESTIONS

Current effective disposition

Layer

baseline/analytics

Value

Preserved unchanged; probe is not a finding.

Preserved historical records

Historical stage: `baseline_record`

Field	Historical value
Id	U-06
Probe	What genuinely new observation fails to fit every registered explanation?
Trigger	A concrete anomaly with retained bytes/trace and failed reproduction of expected behavior.
Not a finding	Yes

Historical stage: `analytics_record`

Field	Historical value
Id	U-06
Probe	What genuinely new observation fails to fit every registered explanation?
Trigger	A concrete anomaly with retained bytes/trace and failed reproduction of expected behavior.
Not a finding	Yes

A03 Original 64 gate obligations

Original obligations and reported historical status preserved with generic operational roles. This is an index, not 64 newly passed tests.

Original obligation retained as history. The selected descriptive engineering milestone is closed; broader capabilities are not qualified by this publication. Read effective question dispositions before interpreting historical required actions.

Each row is a historical requirement recorded during development, with its original wording, the evidence reported at the time and the requested disposition. Printing it here does not re-test it; the checks that apply to this package are described in docs/REPRODUCE.md and scripts/VERIFICATION_SCOPE.md.

INT · 4 original obligations

INT-001 · Artifact existence and identity

Historical field	Preserved statement
Evidence reported	Verified for reviewed artifacts and packets; current hashes match.
Required action / disposition reported	Repeat for the final return; do not generalize to unseen inputs.
Source	S-ENGINEERING · S-QUESTIONS

INT-002 · Duplicate/ghost upload guard

Historical field	Preserved statement
Evidence reported	Exact named archive and member identities checked in this pass.
Required action / disposition reported	Use immutable selection/hash in final intake, not filename alone.
Source	S-ENGINEERING · S-QUESTIONS

INT-003 · Archive safety

Historical field	Preserved statement
Evidence reported	Bounded capture/archive regressions exist; full aggregate parser/resource qualification is not established.
Required action / disposition reported	Keep trusted-local scope; complete limits/containment before general untrusted intake.
Source	S-ENGINEERING · S-QUESTIONS

INT-004 · Immutable original

Historical field	Preserved statement
Evidence reported	Original qualification preserved; transform tooling creates derivatives.
Required action / disposition reported	Keep originals immutable and bind every future child to a parent.
Source	S-ENGINEERING · S-QUESTIONS

REC · 4 original obligations

REC-001 · Typed receipt

Historical field	Preserved statement
Evidence reported	Multiple versioned native/field node/statistical contracts exist; universal receipt convergence incomplete.
Required action / disposition reported	Declare authoritative schema per family and normalize without dropping authority distinctions.
Source	S-ENGINEERING · S-QUESTIONS

REC-002 · Tool identity

Historical field	Preserved statement
Evidence reported	Release and qualification identities are recorded; field fixtures retain dirty producer state.
Required action / disposition reported	Freeze current executor and obtain clean/content-addressed field node source evidence.
Source	S-ENGINEERING · S-QUESTIONS

REC-003 · Output integrity

Historical field	Preserved statement
Evidence reported	Release, browser, field node and 25 native artifact hashes verified.
Required action / disposition reported	Carry raw/normalized relationships and hashes into the final packet.
Source	S-ENGINEERING · S-QUESTIONS

REC-004 · Secrets exclusion

Historical field	Preserved statement
Evidence reported	Reviewed browser archive excludes key files/weights; no private keys read this pass.
Required action / disposition reported	Perform a final allowlist/content preflight; hashes alone do not prove absence of secrets.
Source	S-ENGINEERING · S-QUESTIONS

POL · 8 original obligations

POL-001 · No signal does not imply human

Historical field	Preserved statement
Evidence reported	Python policy/evidence-state regressions prohibit human authorship from no signal.
Required action / disposition reported	Preserve in final golden vectors and UI; no formal proof is claimed.
Source	S-ENGINEERING · S-QUESTIONS

POL-002 · Passive is not cryptographic

Historical field	Preserved statement
Evidence reported	Evidence families are separated in Python; passive signals are not credential validation.
Required action / disposition reported	Exercise explicit passive-only negative conformance vector for claimed implementations.
Source	S-ENGINEERING · S-QUESTIONS

POL-003 · C2PA scope

Historical field	Preserved statement
Evidence reported	Synthetic trusted-credential mapping and prohibited claims are tested.
Required action / disposition reported	Retain ceiling; real native C2PA fixture acceptance remains separate.
Source	S-ENGINEERING · S-QUESTIONS

POL-004 · Lineage-loss preconditions

Historical field	Preserved statement
Evidence reported	Python policy requires linked ancestor, same profile and positive/negative evidence.
Required action / disposition reported	Supply actual linked campaign receipts before claiming measured signal loss.
Source	S-ENGINEERING · S-QUESTIONS

POL-005 · No intent inference

Historical field	Preserved statement
Evidence reported	Malicious-removal intent is globally prohibited by current Python policy.
Required action / disposition reported	Preserve prohibition in every card and formal/conformance target.
Source	S-ENGINEERING · S-QUESTIONS

POL-006 · Conflict handling

Historical field	Preserved statement
Evidence reported	Four-valued conflict handling and grouped-source checks are tested.
Required action / disposition reported	Independently reconcile actual conflicting returned receipts; preserve disagreement.
Source	S-ENGINEERING · S-QUESTIONS

POL-007 · Downward closure

Historical field	Preserved statement
Evidence reported	Python downward closure and implication order have local tests.
Required action / disposition reported	Bind registry/policy identity and compare all claimed language implementations.
Source	S-ENGINEERING · S-QUESTIONS

POL-008 · Compiler idempotence

Historical field	Preserved statement
Evidence reported	Deterministic compiler implementation/test exists; this review did not separately certify all repeated invocations.
Required action / disposition reported	Include identical-input canonical-output equality in final acceptance.
Source	S-ENGINEERING · S-QUESTIONS

STAT · 10 original obligations

STAT-001 · Score type declaration

Historical field	Preserved statement
Evidence reported	Typed statistical results and raw-score/probability-language rejection are tested.
Required action / disposition reported	Use those semantics in final analysis and cards; nominal z is not probability.
Source	S-ENGINEERING · S-QUESTIONS

STAT-002 · Hypotheses

Historical field	Preserved statement
Evidence reported	Profile-bound statistical receipt design exists; main experiment is not frozen.
Required action / disposition reported	Freeze the null, alternative, scope and endpoint with the new protocol.
Source	S-ENGINEERING · S-QUESTIONS

STAT-003 · Unit of independence

Historical field	Preserved statement
Evidence reported	Exact-content split binding exists; prompt dependence is not removed by hashing.
Required action / disposition reported	Declare prompt/pair as experimental unit; keep development/calibration/ held-out boundaries.
Source	S-ENGINEERING · S-QUESTIONS

STAT-004 · Sample length

Historical field	Preserved statement
Evidence reported	Minimum scorable-token and context allowances are tested.
Required action / disposition reported	Preserve count/abstention logic; version any change to eligible unique windows.
Source	S-ENGINEERING · S-QUESTIONS

STAT-005 · Exact known-key null

Historical field	Preserved statement
Evidence reported	Current native scorer reports nominal normal-tail statistics, not an exact validated null.
Required action / disposition reported	Keep descriptive scope or qualify a registered empirical/null procedure before inference.
Source	S-ENGINEERING · S-QUESTIONS

STAT-006 · Repeated-context dependence

Historical field	Preserved statement
Evidence reported	Repetition is measured; marked qualification has 63.10% repeated contexts.
Required action / disposition reported	Required browser methods decision; prospective dependence handling or explicit inferential abstention.
Source	S-ENGINEERING · S-QUESTIONS

STAT-007 · Multiplicity

Historical field	Preserved statement
Evidence reported	Correction/alpha/threshold consistency checks exist; no executed family-control study.
Required action / disposition reported	Freeze all actual inferential tests; two wrong keys remain diagnostic in the pilot.
Source	S-ENGINEERING · S-QUESTIONS

STAT-008 · Exploratory search

Historical field	Preserved statement
Evidence reported	Freeze/multiplicity gates exist; no validated exploratory-search pipeline is claimed.
Required action / disposition reported	Record development choices; suppress significance for post-outcome searches.
Source	S-ENGINEERING · S-QUESTIONS

STAT-009 · Power planning

Historical field	Preserved statement
Evidence reported	60-prompt draft is observation-only; no public-performance power acceptance.
Required action / disposition reported	Do not expand by default; size a separate acceptance study for any stronger requested claim.
Source	S-ENGINEERING · S-QUESTIONS

STAT-010 · Confidence intervals

Historical field	Preserved statement
Evidence reported	Core score intervals exist; no completed experiment-level uncertainty report.
Required action / disposition reported	Report appropriate sample/pair-level uncertainty; do not reuse token independence assumptions.
Source	S-ENGINEERING · S-QUESTIONS

CAL · 6 original obligations

CAL-001 · Held-out calibration

Historical field	Preserved statement
Evidence reported	30/30 split and content binding verified in draft; no completed calibration/ held-out outputs.
Required action / disposition reported	Calibration first; freeze threshold receipt before any held-out generation or inspection.
Source	S-ENGINEERING · S-QUESTIONS

CAL-002 · Probability semantics

Historical field	Preserved statement
Evidence reported	Raw score versus calibrated-probability gate has regressions.
Required action / disposition reported	Keep probability language disabled unless its own receipt requirements are met.
Source	S-ENGINEERING · S-QUESTIONS

CAL-003 · Reliability

Historical field	Preserved statement
Evidence reported	No calibrated probability model or reliability campaign is present.
Required action / disposition reported	Exclude probability claims from the laboratory milestone; required if later offered.
Source	S-ENGINEERING · S-QUESTIONS

CAL-004 · Base-rate scenarios

Historical field	Preserved statement
Evidence reported	Research reference scenarios exist; no deployed posterior claim is justified.
Required action / disposition reported	No posterior-style percentage; require declared prevalence/scenarios for future probability output.
Source	S-ENGINEERING · S-QUESTIONS

CAL-005 · Abstention

Historical field	Preserved statement
Evidence reported	Scorer/receipt abstentions exist; risk-coverage and prospective quality policy incomplete.
Required action / disposition reported	Freeze eligibility and report failed/short/repetitive outcomes plus both attempted/eligible denominators.
Source	S-ENGINEERING · S-QUESTIONS

CAL-006 · OOD applicability

Historical field	Preserved statement
Evidence reported	One native model/pair qualified; no OOD/generalization study.
Required action / disposition reported	Limit to exact supported profile; broader provider/domain/codec claims need separate tests.
Source	S-ENGINEERING · S-QUESTIONS

WM · 6 original obligations

WM-001 · Frozen profile

Historical field	Preserved statement
Evidence reported	Qualification identities agree; full study execution freeze is incomplete.
Required action / disposition reported	Create a new real-path protocol ID after the methods decision; pin every effective setting.
Source	S-ENGINEERING · S-QUESTIONS

WM-002 · Correct-key control

Historical field	Preserved statement
Evidence reported	One correct-key marked signal is observed with dependence caveat.
Required action / disposition reported	Complete accepted pilot for descriptive results; claim-sized study required for performance acceptance.
Source	S-ENGINEERING · S-QUESTIONS

WM-003 · Wrong-key controls

Historical field	Preserved statement
Evidence reported	Two wrong-key diagnostics exist; original 128-key/familywise requirement is unmet.
Required action / disposition reported	Keep pilot ceiling explicit; full directive needs frozen family controls and error accounting.
Source	S-ENGINEERING · S-QUESTIONS

WM-004 · Clean paired controls

Historical field	Preserved statement
Evidence reported	One real paired control exists for the selected current profile and a 60-prompt paired matrix is drafted; older field node prototype work has separate scope.
Required action / disposition reported	Run accepted calibration/held-out pairs or close unsuitable profile with retained failures.
Source	S-ENGINEERING · S-QUESTIONS

WM-005 · Quality cost

Historical field	Preserved statement
Evidence reported	Marked repetition is adverse quality evidence; clean also fails to give a simple completed answer.
Required action / disposition reported	Predeclare quality rubric; reject/downgrade unsuitable profile rather than tune indefinitely.
Source	S-ENGINEERING · S-QUESTIONS

WM-006 · Transformation survival

Historical field	Preserved statement
Evidence reported	Deterministic transformation tools exist; real watermark survival campaign absent.
Required action / disposition reported	No robustness claim in pilot; full directive requires frozen transform matrix and intervals.
Source	S-ENGINEERING · S-QUESTIONS

PAS · 6 original obligations

PAS-001 · AUROC plus operational metrics

Historical field	Preserved statement
Evidence reported	No accepted passive-detector study is present.
Required action / disposition reported	Explicitly exclude passive classification from this milestone; later adapter needs operational metrics.
Source	S-ENGINEERING · S-QUESTIONS

PAS-002 · Future/unseen generator split

Historical field	Preserved statement
Evidence reported	No held-out unseen-generator passive validation is present.
Required action / disposition reported	Keep future/unseen-generator claims disabled; separate preregistered future study.
Source	S-ENGINEERING · S-QUESTIONS

PAS-003 · Matched data

Historical field	Preserved statement
Evidence reported	Historical comparison corpora do not establish a passive detector's matched-data validity.
Required action / disposition reported	Any future passive study must audit content/format/codec confounds.
Source	S-ENGINEERING · S-QUESTIONS

PAS-004 · Model candidate set

Historical field	Preserved statement
Evidence reported	No accepted open-set provider/model attribution tool is present.
Required action / disposition reported	Keep provider labels unresolved absent applicable verifier; do not force candidate choice.
Source	S-ENGINEERING · S-QUESTIONS

PAS-005 · Feature sensitivity

Historical field	Preserved statement
Evidence reported	Artifact/topology comparisons exist, not validated passive feature attribution.
Required action / disposition reported	Defer feature ablations with the passive detector rather than blocking the known-key pilot.
Source	S-ENGINEERING · S-QUESTIONS

PAS-006 · Human subgroup audit

Historical field	Preserved statement
Evidence reported	No passive human-subgroup audit is present.
Required action / disposition reported	No deployment of passive authorship decisions; require subgroup FPR evidence if later proposed.
Source	S-ENGINEERING · S-QUESTIONS

C2PA · 5 original obligations

C2PA-001 · Official validator

Historical field	Preserved statement
Evidence reported	Official-compatible wrapper design exists; pinned native backend qualification absent.
Required action / disposition reported	Select and hash one official tool/version; satisfy current worker controls before native acceptance.
Source	S-ENGINEERING · S-QUESTIONS

C2PA-002 · Known-positive credential

Historical field	Preserved statement
Evidence reported	Synthetic normalization tests only; trusted native fixture test is skipped.
Required action / disposition reported	Run trusted and signed-untrusted controls under an explicit trust snapshot.
Source	S-ENGINEERING · S-QUESTIONS

C2PA-003 · Tampered hard binding

Historical field	Preserved statement
Evidence reported	Invalid-binding mappings are tested synthetically.
Required action / disposition reported	Return actual tampered-asset native validation and raw output.
Source	S-ENGINEERING · S-QUESTIONS

C2PA-004 · Soft-binding semantics

Historical field	Preserved statement
Evidence reported	No accepted native soft-binding recovery evidence.
Required action / disposition reported	Test if supported; otherwise mark unavailable and omit recovery claims.
Source	S-ENGINEERING · S-QUESTIONS

C2PA-005 · Semantic ceiling

Historical field	Preserved statement
Evidence reported	Python policy forbids semantic truth/ownership from credential validity.
Required action / disposition reported	Keep the ceiling in real-control and UI tests; policy tests do not prove native crypto.
Source	S-ENGINEERING · S-QUESTIONS

LIN · 3 original obligations

LIN-001 · Parent-child hashes

Historical field	Preserved statement
Evidence reported	Transform receipts/hash relationships exist in source and tests.
Required action / disposition reported	Produce the actual experiment's parent/child graph and validate all links.
Source	S-ENGINEERING · S-QUESTIONS

LIN-002 · Same-profile comparison

Historical field	Preserved statement
Evidence reported	Same-profile precondition exists in Python policy.
Required action / disposition reported	Verify actual detector/profile hashes on ancestor and derivative; no loss inference from unmatched reports.
Source	S-ENGINEERING · S-QUESTIONS

LIN-003 · Transformation family

Historical field	Preserved statement
Evidence reported	Bounded deterministic transform tooling exists.
Required action / disposition reported	Freeze exact selected tool/version/parameters; do not silently include paraphrase or external processing.
Source	S-ENGINEERING · S-QUESTIONS

LEAN · 5 original obligations

LEAN-001 · Policy source hash

Historical field	Preserved statement
Evidence reported	Archived seed is not bound to current Python policy or empirical receipts.
Required action / disposition reported	Synchronize current semantics and bind policy/evidence/theorem hashes before certification.
Source	S-ENGINEERING · S-QUESTIONS

LEAN-002 · No sorry/admit

Historical field	Preserved statement
Evidence reported	Seed claims no placeholders; no accepted pinned build log is present.
Required action / disposition reported	Inspect final sources and independently replay a successful build; static absence alone is insufficient.
Source	S-ENGINEERING · S-QUESTIONS

LEAN-003 · Pinned toolchain

Historical field	Preserved statement
Evidence reported	Seed pins Lean 4.32.0 and uses Std; no current execution receipt.
Required action / disposition reported	Record actual toolchain/dependency identity; add a Mathlib revision only if used.
Source	S-ENGINEERING · S-QUESTIONS

LEAN-004 · Certificate semantics

Historical field	Preserved statement
Evidence reported	Bounded certificate wording exists in design; no live certificate feature qualifies.
Required action / disposition reported	Display policy-derivation scope only when a real bound certificate exists.
Source	S-ENGINEERING · S-QUESTIONS

LEAN-005 · Cross-language conformance

Historical field	Preserved statement
Evidence reported	No active Go implementation or Python-Go-Lean golden acceptance receipt.
Required action / disposition reported	Full original directive requires conformance; lab-only closeout explicitly defers it.
Source	S-ENGINEERING · S-QUESTIONS

PUB · 4 original obligations

PUB-001 · No magic percentage

Historical field	Preserved statement
Evidence reported	Generic dashboard exists, not the planned evidence cards.
Required action / disposition reported	Render nominal/descriptive values accurately; no unsupported magic percentage.
Source	S-ENGINEERING · S-QUESTIONS

PUB-002 · What this does not prove

Historical field	Preserved statement
Evidence reported	Reports carry claim/prohibition fields; product card acceptance not complete.
Required action / disposition reported	Make prohibited inferences visible in each canonical fixture state.
Source	S-ENGINEERING · S-QUESTIONS

PUB-003 · Local-first privacy

Historical field	Preserved statement
Evidence reported	Existing proof uses local same-origin service; full public/privacy acceptance absent.
Required action / disposition reported	Keep static/local mode; separately qualify any file upload, provider request or public companion.
Source	S-ENGINEERING · S-QUESTIONS

PUB-004 · Abstention

Historical field	Preserved statement
Evidence reported	Backend supports insufficient/unknown/blocked states; card behavior not accepted.
Required action / disposition reported	Render unresolved, conflict, rejection and unavailable capabilities explicitly.
Source	S-ENGINEERING · S-QUESTIONS

AUD · 3 original obligations

AUD-001 · Independent rerun

Historical field	Preserved statement
Evidence reported	This pass independently checks retained integrity/arithmetic and reruns core tests, not keyed native detection.
Required action / disposition reported	Final spine independently recomputes detection/analysis under controlled local access; retain disagreement.
Source	S-ENGINEERING · S-QUESTIONS

AUD-002 · Negative result retention

Historical field	Preserved statement
Evidence reported	Unfavorable native text and original draft remain preserved.
Required action / disposition reported	Keep every future attempt, failure and abstention; no selective regeneration or hidden exclusions.
Source	S-ENGINEERING · S-QUESTIONS

AUD-003 · Reproducibility

Historical field	Preserved statement
Evidence reported	Source/package identities and retained receipts agree; clean field producer/ campaign replay incomplete.
Required action / disposition reported	Final packet must reproduce deterministic checks; declare runtime nondeterminism and field-source limits.
Source	S-ENGINEERING · S-QUESTIONS

A04 Claim ledger and explicit exclusions

The ledger keeps supported bounded statements and explicitly excluded stronger statements together. An excluded statement is included so that the boundary is inspectable; it is not a result. Source IDs here do not imply direct claim-to-question or claim-to-gate links.

Shared status and reopening rule

Status scope

Status of the preserved engineering result. This publication revision adds public evidence and calculation routes; it performs no new native or field experiment.

Reopening condition

A concrete contradictory receipt, integrity/method error, or separately selected stronger claim with a fixed protocol.

L-01 · locally_recomputed

Claim

The retained marked specimen has 60 primary-key hits in 93 distinct context/target windows.

Limit

One retained specimen; not calibrated performance.

Public reproduction

Exact completion arrays support 93 distinct-window counts; supplied aggregate hits support nominal arithmetic. Native membership remains unavailable.

Shared status and reopening rule

Sources: S-RESULTS

L-02 · locally_recomputed

Claim

All six historical raw results match and a separate membership implementation agrees.

Limit

Shared sample and construction; not independent scientific replication.

Public reproduction

All 18 public supplied-hit score calculations are checkable. The separate historical keyed implementation receipt remains a private-source report.

Shared status and reopening rule

Sources: S-RESULTS

L-03 · bounded_interpretation

Claim

Marked repetition increases the raw statistic; positive keyed excess remains after exact-repeat removal.

Limit

No causality, independence or authorship probability.

Public reproduction

Public recurrence rows and nominal score tables make the unit change inspectable; native membership and causal repetition mechanisms are not independently reproduced.

Shared status and reopening rule

Sources: S-RESULTS

L-04 · locally_reproduced

Claim

The fixed synthetic JSON reproduces, including 1502/4096 raw crossings for the 64-repeat motif.

Limit

Fixture/key-ensemble behavior, not natural-text FPR.

Public reproduction

Exact public fixtures, public-key derivation, source and frozen expected output are included; run scripts/reproduce_controls.py.

Shared status and reopening rule

Sources: S-ENGINEERING

L-05 · locally_reproduced

Claim

Identity-only token-ID relabeling preserves decoded probability mass for all 24 toy permutations.

Limit

Output-changing coupling and novelty remain open.

Public reproduction

All 24 toy relabelings and the segmentation example are reproducible with scripts/reproduce_controls.py.

Shared status and reopening rule

Sources: S-ENGINEERING

L-06 · locally_tested

Claim

The changed core passes 597 tests, with 2 expected skips.

Limit

Optional Torch and native trusted-C2PA capability not qualified.

Public reproduction

Historical core-test receipt summary; the full original core test environment is not distributed in this atlas.

Shared status and reopening rule

Sources: S-ENGINEERING

L-07 · locally_tested

Claim

The amended browser replay passes in the current local environment.

Limit

101 top-level plus 9 subtests; one Torch skip, two toy instances omitted.

Public reproduction

Historical analytical replay receipt summary; distinct from the local browser checks of this publication site.

Shared status and reopening rule

Sources: S-ENGINEERING

L-08 · locally_tested

Claim

Wheel/sdist build and six wheel-based smoke commands pass.

Limit

Unreleased branch test packages; accepted 0.1.4 release unchanged.

Public reproduction

Historical watermark toolkit wheel/sdist and smoke-test receipts; the original wheel/runtime is not redistributed in this atlas.

Shared status and reopening rule

Sources: S-ENGINEERING

L-09 · accepted_bounded_field_return

Claim

A field contributor reported execution of the nonsecret observer on a second architecture; returned output exactly matched the reference supplied with the request.

Limit

Shared code and fixed fixture; device identity and execution are peer-reported. No independent key-membership result, new sample, calibrated efficacy, native model equivalence, accelerator proof or independent event authentication.

Public reproduction

Public observer receipt, expected counts and contribution chronology are included; supplied-output equality does not authenticate the field event.

Shared status and reopening rule

Sources: S-OBSERVER

L-10 · excluded_unavailable

Claim

Included model/radio field rows have reproducible source-bound exporter provenance.

Limit

Historical model/radio row bytes and descriptors verify but source_dirty=true; no clean or complete exporter snapshot, fresh acquisition or re-export. Excluded from reproducible field-performance/clean-producer claims. Current clean watermark toolkit source does not repair historical exporter state.

Public reproduction

Seven public numerical rows and summaries are included. The stronger complete clean historical exporter-source claim remains unavailable.

Shared status and reopening rule

Sources: S-FIELD-HISTORY

L-11 · provider_declaration

Claim

Provider mechanism-family and access background has changed.

Limit

Not local verification of historic provider specimens.

Public reproduction

Dated primary-source bibliography and preserved review history are included; declarations are background, not activation evidence for earlier specimens.

Shared status and reopening rule

Sources: S-ENGINEERING

L-12 · excluded_unqualified

Claim

Current Python policy is formally certified by compiled Lean.

Limit

One source-level counterexample; no current compiled certificate.

Public reproduction

The discrepancy and missing current compiled certificate remain documented; no public proof artifact establishes the excluded claim.

Shared status and reopening rule

Sources: S-QUESTIONS

L-13 · excluded_unsupported

Claim

The milestone supplies calibrated generic AI authorship or person attribution.

Limit

No such conclusion follows from these observations.

Public reproduction

Toy controls and finite-sample planning calculations are public; neither supplies generic authorship calibration or person attribution.

Shared status and reopening rule

Sources: S-ENGINEERING

L-14 · locally_verified_with_documented_exception

Claim

The external archive hash and 23 payload entries match; the original return manifest has one incorrect self-hash.

Limit

Original retained unchanged; receiver manifest adds byte binding, not source authentication. Do not report blanket original-manifest success.

Public reproduction

The observer projection preserves the original self-hash defect and separate receiver correction; it does not recreate the private original return archive.

Shared status and reopening rule

Sources: S-ENGINEERING

A05 Sources and dated bibliography

A public derivative route is not an independent original. Private and unavailable originals remain explicitly unavailable. No public reader is expected to reconstruct a local path or request undisclosed credentials. All included file routes below are relative to this package.

S-HISTORY · Full investigation history and operator methodological requests

Origin

Retained historical source represented by identified public derivatives

Availability

Underlying original retained privately; not an open raw-data route in this publication payload.

Verification ceiling

A summary route is not independent evidence or source authentication.

Included derivative files:

- `data/research-register.public.json`

S-EARLY-DIALOGUE · Early exploratory model dialogue

Origin

Retained historical source represented by identified public derivatives

Availability

Underlying original retained privately; not an open raw-data route in this publication payload.

Verification ceiling

A summary route is not independent evidence or source authentication.

Included derivative files:

- `data/research-register.public.json`

S-INFORMAL-LEADS · Informal third-party mechanism hypotheses

Origin

Retained historical source represented by identified public derivatives

Availability

Underlying original retained privately; not an open raw-data route in this publication payload.

Verification ceiling

A summary route is not independent evidence or source authentication.

Included derivative files:

- `data/research-register.public.json`

S-COMPARATIVE · Historical paired-corpus, arbitrary-key and container analyses

Origin

Retained historical source represented by identified public derivatives

Availability

Numerical corpus/null and container projections are public; original corpora remain privately retained.

Verification ceiling

Selected aggregates can be recomputed from public rows; original-input and pooled-ngram analyses cannot be rerun from this projection.

Included derivative files:

- `data/research-register.public.json`
- `data/comparative-observations.public.json`
- `data/comparative-observations.public.csv`

S-CONTRACTS · Historical cross-format and contract analyses

Origin

Retained historical source represented by identified public derivatives

Availability

Cross-format counts, code/schema discrepancies and extractor findings are public projections; original fixture bundles remain private.

Verification ceiling

Passing implementation tests did not establish schema equivalence. Raw PDF byte matches do not establish parsed actions.

Included derivative files:

- `data/research-register.public.json`
- `data/comparative-observations.public.json`

S-ENGINEERING · Frozen bounded engineering case study and accepted receipts

Origin

Retained historical source represented by identified public derivatives

Availability

Public methods and exact toy fixtures/source/expected results are included. Native runtime and secret keys remain private.

Verification ceiling

Toy controls and planning arithmetic are reproducible. This does not reproduce native membership or perform calibration.

Included derivative files:

- `data/research-register.public.json`
- `data/native-methods.public.json`
- `data/synthetic-controls.expected.json`
- `fixtures/synthetic-controls.public.json`
- `scripts/reproduce_controls.py`
- `sources/controls/membership.py`

S-QUESTIONS · Original questions, probes and additive current dispositions

Origin

Retained historical source represented by identified public derivatives

Availability

Underlying original retained privately; not an open raw-data route in this publication payload.

Verification ceiling

A summary route is not independent evidence or source authentication.

Included derivative files:

- `data/future-work.public.json`
- `data/original-gates.public.json`

S-FIELD-HISTORY · Historical field-runtime and adjacent radio reports

Origin

Retained historical source represented by identified public derivatives

Availability

Seven numerical field rows and dated contribution receipts are projected publicly. Complete clean historical exporter source is unavailable; raw operational logs remain private.

Verification ceiling

Row summaries are reproducible. Historical execution, utility, accelerator use and radio semantics are not established by this projection.

Included derivative files:

- `data/research-register.public.json`
- `data/field-observations.public.json`
- `data/field-observations.public.csv`
- `data/field-contributions.public.json`

S-EARLY-AUDITS · Recovered early audit material and six retained summaries

Origin

Retained historical source represented by identified public derivatives

Availability

Public measurements include 17 audit rows and six summary records. Six source copies were recovered privately; the 17 original audit inputs remain unavailable.

Verification ceiling

Totals and pairwise reported values are inspectable; this is not a fresh original-input byte audit.

Included derivative files:

- `data/research-register.public.json`
- `data/historical-audits.public.json`

S-RESULTS · Retained-pair key-holder aggregate results and disclosed trace projection

Origin

Retained historical source represented by identified public derivatives

Availability

Exact owned prompt/completion text, completion arrays, scientific settings and aggregate results are public. Original receipts are held privately.

Verification ceiling

Included arrays support exact recurrence counts; aggregate hits support arithmetic. Native secret-key membership cannot be recomputed publicly.

Included derivative files:

- `data/results.public.json`
- `fixtures/profile.public.json`
- `fixtures/clean-completion.public.json`
- `fixtures/marked-completion.public.json`
- `data/observation-tracks.public.json`
- `data/native-methods.public.json`
- `fixtures/native-prompt.public.txt`
- `fixtures/native-clean.public.txt`
- `fixtures/native-marked.public.txt`
- `data/results.public.csv`

S-OBSERVER · Accepted peer-reported keyless execution and exact output agreement

Origin

Retained historical source represented by identified public derivatives

Availability

The public observer receipt and expected counts are included; the original operational return is retained privately.

Verification ceiling

Peer-reported execution with shared inputs and supplied expected output; the public projection does not authenticate the device event.

Included derivative files:

- `data/observer-receipt.public.json`
- `data/keyless-check.expected.json`
- `data/field-contributions.public.json`

Dated external bibliography

The experimental record closes on 9 October 2026. The provider pages, the MarkSec abstract, the misuse reports and the education guidance below were checked on 11 October 2026. These sources explain the problem that motivated the investigation. They do not add measurements on the Vitreos specimens.

Source	Background role	Limit
Anthropic: How Claude's text watermarking works	Published 14 August 2026, updated 1 September; identifies the method family	Does not establish historical specimen activation, exact private profile or local detector access
Anthropic: Detecting and countering misuse of AI, September 2026	Published 10 September, as dated by the publisher's report index; the Malaysia case illustrates rewritten reporting and obscured source attribution	Provider-reported investigation covering disruptions from December 2025 through August 2026; not a watermark-detection experiment or a Vitreos case
MarkSec: Capability-Aware Evaluation of Adversarial Attacks Against LLM Watermarks	Version 1 submitted 15 September 2026; abstract identifies attack capabilities and quality constraints as evaluation dimensions	Abstract and bibliographic record checked; methods not reproduced or fully adjudicated here
OpenAI: Our approach to EU text provenance rules	Published 5 October 2026; provider announcement and stated detection limits	Provider declaration is separate from local experimental evidence; no claim that all earlier outputs were marked
OpenAI: Disrupting AI-enabled false-front operations	Published 8 October 2026; the Bogus Bylines case illustrates deceptive journalist personas and published articles	Provider-reported investigation of earlier and ongoing activity, not incidents all occurring on publication day; not independently authenticated here

Source	Background role	Limit
Turnitin: Using the AI Writing Report	Current guidance accessed 11 October 2026; education use, possible misclassification and the need for further scrutiny	Vendor guidance about an AI-writing classifier, not evidence that Vitreos detects student misconduct or evaluates that product
C2PA specification 2.4	Credential and soft-binding background	No native validator or trust qualification follows from citing the specification
DHMark research entry	Future research reference identified in the supplied review	Source/topic checked; methods not reproduced or fully adjudicated here

Technique background used for the map in S01, checked on 12 October 2026:

Source	Role
Kirchenbauer et al., A Watermark for Large Language Models (ICML 2023)	Green-list token watermark; the family of the owned construction
Dathathri et al., Scalable watermarking for identifying large language model outputs (Nature 634, 818-823, 2024)	SynthID-Text tournament sampling
Kuditipudi et al., Robust Distortion-free Watermarks for Language Models (TMLR)	Key-driven sampling that preserves the output distribution
Krishna et al., Paraphrasing evades detectors of AI-generated text, but retrieval is an effective defense (NeurIPS 2023)	Paraphrase attacks on watermarks and classifiers
Jovanović, Staab and Vechev, Watermark Stealing in Large Language Models (ICML 2024)	Learning a watermark rule from outputs to scrub or spoof it
Zhang et al., Character-Level Perturbations Disrupt LLM Watermarks (arXiv, September 2025)	Typos, swaps and other character-level attacks
Liang et al., GPT detectors are biased against non-native English writers (Patterns, 2023)	Uneven false positives in AI-writing classifiers
Li et al., textGrain: Entropy-Calibrated Watermarking for Language Model Text (OpenAI technical report, 5 October 2026)	Keyed vocabulary blocks coupled by optimal transport under an entropy budget; first-occurrence scoring of repeated context windows
Google DeepMind, SynthID Text reference implementation	Research reference code with mean, weighted-mean and Bayesian detectors; not for production use

These papers supply terminology; their methods were not reproduced here.

The C2PA and DHMark entries come from the 9 October review and were not re-examined on 11 October. MarkSec and DHMark remain pointers for deliberately selected future work. Their inclusion does not establish a reproduced method, detailed novelty comparison, legal conclusion or robustness result. The related research questions remain open under their own conditions.

The connection between these developments and the follow-up questions in S01 is the atlas's research rationale. It is not a measured forecast of future misuse, a prevalence estimate, or a claim that a watermark alone would have resolved the reported incidents. The examples are selective rather than a comprehensive history of synthetic media or anti-watermarking tools.

The original investigation transcripts and operational records are not distributed. `data/source-map.public.json` says which public derivatives are included and which raw routes remain unavailable. This is an explicit verification limitation, not a broken local hyperlink to be guessed or replaced. Source summaries do not become independent corroboration by being duplicated across HTML, PDF and JSON.

Reported keyless observer

Schema

vitreos.public-product-observer-receipt.v1

Status

Public wording of the preserved observer report; not hardware attestation.

Reported operation

Second-architecture keyless count and reported-score arithmetic check

Reported outcome

Historical returned output exactly matched the expected output supplied with the request.

Counts

Clean

Eligible positions

252

Distinct contexts

249

Distinct windows

251

Marked

Eligible positions

252

Distinct contexts

93

Distinct windows

93

Historical score arithmetic checks

6

Models or secret keys used by reported observer

No

Scientific array note

The disclosed completion arrays are equal to the retained source arrays. Public wrappers and identities are new.

Reproduction command

python3 scripts/verify_public_package.py

Expected output

data/keyless-check.expected.json

Source ids

- S-OBSERVER
- S-RESULTS

Limitations

- The expected output was supplied with the historical request.
- Local replay of this derivative is not a new device run or event authentication.
- Keyless counting does not measure supplied membership hits.
- No model generation, hardware performance or calibrated detector claim follows.

Derivative note

Scientific counts and limits preserved from the design-input derivative; the package-relative reproduction command now addresses the actual output verifier. This is not a new device run.

A06 Contribution and design decisions

Reviewers and implementers worked from shared artifacts. Local implementation/key-holder work, field execution reports, research/context review, independent arithmetic review and editorial integration are distinct contributions. Multiple reviews do not create independent generations, keys, observations or institutional provenance.

The original planning disposition and reason below are preserved as history. Candidate implementation evidence is a separate statement about this edition. In particular, historical descriptions of then-unbuilt pages do not describe the current rendering.

LC01 · incorporated

Retain research, operator and collaborator reading routes, including negative and unresolved evidence.

Planning reason

Audience routes must reach the same evidence without changing its claim strength.

Candidate disposition

implemented

Implementation evidence

Contents and S01 give three routes into the same complete evidence; S03, S06, A01 and A02 retain negative and unresolved material.

LC02 · incorporated

Evidence-atlas architecture and existing 20 candidate families.

Planning reason

The seven-page study already balances observation, explanation, custody and open constructions. No new chart engine is needed.

Candidate disposition

implemented

Implementation evidence

S01-S07 narrative, ten vector evidence figures, M01 methods and complete A01-A07 registers/indexes; all 20 format decisions are recorded separately.

LC03 · amended

42 research titles, 36 question/probe IDs and 64 gate IDs.

Planning reason

Exact authoritative bodies now resolve locally; the newer future snapshot adds freeze_updates only. R16 now carries A-05 plus its distinct October episode; map regenerated from the planning source. Final pages remain unbuilt.

Candidate disposition

implemented

Implementation evidence

A01 prints all 42 substantive entries; A02 all 36 current/history records; A03 all 64 original obligations. R16 has the separate A-05 episode. coverage.json binds actual anchors and rendered pages.

LC04 · incorporated

Show the transition from proprietary-output investigation to owned known-key experiment to evidence tooling.

Planning reason

These are related questions, not one mechanism identification result.

Candidate disposition

implemented

Implementation evidence

S01-S02 and S05 distinguish proprietary-output questions, local registered-key experiment and evidence tooling; F06 and F05 show separate layers/operations.

LC05 · incorporated

Correct the overbroad historical whitespace and single-artifact conclusions.

Planning reason

Finite feature checks do not exclude all possible carriers; a supported detector can sometimes score one sufficiently informative artifact.

Candidate disposition

implemented

Implementation evidence

S02-S03, R03 and R09 retain ordinary-separator and single-artifact observability distinctions without excluding all carriers.

LC06 · amended

Early six-summary bytes and audit scripts now recovered.

Planning reason

Custody updated: six retained copies and ZIP match historical hashes; complete generation state and activation remain unresolved. R05 separately retains missing 17-input custody.

Candidate disposition

implemented

Implementation evidence

S03, F01, R05/R06 and current question dispositions distinguish recovered six copies/audit outputs from missing 17 audit inputs and unestablished activation.

LC07 · incorporated

Primary raw 168/252 and distinct-window 60/93 marked results.

Planning reason

Retain all three units and key-holder membership ceiling. Receipt created October 7 at 23:26 UTC; October 8 is return/review, not a second rescore.

Candidate disposition

implemented

Implementation evidence

S05, F07 and the searchable 18-row table preserve raw/context/window units, keyed hit origins, and October 7 computation versus October 8 review.

LC08 · incorporated

Earlier 39-75 feasible-hit bound and later 60/93 result.

Planning reason

Preserve how uncertainty was reduced, without treating the bound as an alternative measurement or another sample.

Candidate disposition

implemented

Implementation evidence

S05 and R18 show the earlier 39-75 feasible bound followed by 60/93 measurement on the same specimen.

LC09 · incorporated

Earlier field inference work, local native qualification, later nonsecret observer.

Planning reason

Retain historical field inference/runtime, local native pair, local keyed rescore and October keyless field observer as separate operations. Do not assign the keyed result to the keyless field node.

Candidate disposition

implemented

Implementation evidence

S05-S06, F05, R27 and R31 distinguish historical field inference, local native/keyed work and reported keyless observer.

LC10 · amended

Output match and peer execution receipt.

Planning reason

Bounded wording already present in the freeze/design. Amend the public receipt draft before use; exact output agreement and a peer receipt do not authenticate execution.

Candidate disposition

implemented

Implementation evidence

S05, M01, R31 and the public observer derivative preserve expected-output supply and peer-reported execution, with no device-event authentication claim.

LC11 · incorporated

Allowlist-first public/private split, source-map and metadata redaction.

Planning reason

Fresh public allowlist required. Private originals remain unchanged. Draft observer notes still contain internal names and will be rewritten in the public derivative.

Candidate disposition

implemented

Implementation evidence

Public files are an explicit derivative selection with generic roles and package-relative routes. R39/R40 disclose abstraction. Private originals and integration receipts are separate.

LC12 · incorporated

Broaden inspection from byte strings to parsed PDF content and archive headers.

Planning reason

Local parsed-page and TAR inspection confirms the channels. Adopt screening plus explicit decoded-format/container/pixel coverage; the helper alone is insufficient.

Candidate disposition

implemented_with_scope

Implementation evidence

Candidate verification covers decoded HTML/JSON/SVG, PDF text/metadata/links, rendered pages and archive members/headers. Results are bound to actual candidate artifacts in the accompanying review record. A scanner pass does not prove complete confidentiality or authorize publication.

LC13 · incorporated

Attribution scope as a separate fifth analytical dimension.

Planning reason

A scheme signal, provider claim, workflow, person, ownership and truth are not interchangeable identification targets.

Candidate disposition

implemented

Implementation evidence

S01, F06, R17 and R35 separate scheme, provider/workflow, person, ownership, intent and truth claims.

LC14 · amended

Five-state evidence vocabulary.

Planning reason

Use measured/receipt-bound/provider-declared as origin tags; excluded/open as claim or task states. Preserve not measured, inspected absence, missing, conflicting and superseded separately.

Candidate disposition

implemented

Implementation evidence

A01 preserves literal independent fields, A02 current/history dimensions, and F01 explicit missingness categories; no flat keyword classifier assigns strength.

LC15 · incorporated

Feynman-inspired and shared-input diagram grammars.

Planning reason

Borrow visual syntax only. Existing C06/C07 clearly mark schematic events and dependencies; do not invent telemetry or silently imply independence.

Candidate disposition

implemented

Implementation evidence

F05 adopts the useful artifact/dependency grammar with named shared inputs, no measured timing, amplitudes, weighted bonds or independent-generation claim.

LC16 · incorporated

Searchable numerical table and redundant recurrence encoding.

Planning reason

Raster plots contain the exact score labels but PDF text does not; readers need units, hits, denominators and limits in accessible text.

Candidate disposition

implemented

Implementation evidence

F04 uses words/patterns and exact indexed positions. F07 and the full 18-row searchable table expose hits/n/unit and nominal z; figure text remains vector text.

LC17 · amended

Provider-family update.

Planning reason

Dated provider declarations checked; SynthID-Text correction was already accepted October 8. Preserve historic activation uncertainty and avoid treating discovery by a reviewer as publication date.

Candidate disposition

implemented

Implementation evidence

S02/S06, R35, current F-01 and dated bibliography preserve the earlier accepted provider-family correction and historical applicability/access uncertainty.

LC18 · amended

Keyed tokenizer/segmentation coupled to generation.

Planning reason

Retain the original promising construction. ID-only renaming null is narrower. Use novelty unestablished; no novelty search or working coupling experiment was completed.

Candidate disposition

implemented

Implementation evidence

S06 and R22 retain untested generation-coupled construction separately from R21 identity-only permutation null; novelty remains unestablished.

LC19 · deferred

Eight-generation repetition micro-factorial.

Planning reason

Deferred optional protocol. Preserve the older maximum-12 formatting option separately; no generation is assigned by planning.

Candidate disposition

deferred

Implementation evidence

S06-S07 and A02 retain eight-generation factorial and older maximum-12 formatting option as separate optional protocols. No new generation or execution assignment.

Reopening trigger

Owner selects a repetition-causality claim and approves a fixed protocol/budget.

LC20 · deferred

Transformation, spoofing, C2PA and passive interaction-fingerprint campaigns.

Planning reason

These extend scope and need their own controls. Literature or plausibility is not a completed local test.

Candidate disposition

deferred

Implementation evidence

S06-S07, R23-R25/R34 and A02 retain transformation, false-implication, credential and interaction studies as separate unperformed programs requiring controls.

Reopening trigger

A separately selected robustness, credential-validation or attribution claim requires that exact experiment.

LC21 · amended

Public legal and compliance implications.

Planning reason

Keep technical/legal distinctions as scoped background; detailed legal claims require exact primary-source review if retained. No fact-specific legal determination is made.

Candidate disposition

implemented_with_scope

Implementation evidence

The report distinguishes technical evidence from legal/compliance outcomes and makes no fact-specific legal determination or new detailed legal claim.

LC22 · incorporated

Historical manifest defect and source-dirty exporter disposition.

Planning reason

Original producer-manifest defect and source_dirty history remain unchanged, with receiver binding and explicit exclusion. No new clean-exporter demand.

Candidate disposition

implemented

Implementation evidence

S06, R32 and A02 preserve source_dirty history, producer-manifest defect/receiver binding and unavailable clean exporter; stronger claim L-10 remains excluded.

LC23 · excluded

Restarting settled P0, private-key or field work for this design.

Planning reason

Exclude a default restart: current source and October 8 final disposition resolve the previously unseen merge/test-layout state. Reopen only an affected claim on a concrete contradiction.

Candidate disposition

excluded

Implementation evidence

No default restart of P0, key or field work. S07 and current statuses allow only a concrete contradictory receipt/method defect or a deliberately selected stronger claim.

LC24 · incorporated

Worked source-to-claim example, shared-input labels and continuation entry.

Planning reason

New collaborators should be able to challenge or extend an item without reconstructing hidden history.

Candidate disposition

implemented

Implementation evidence

M01 provides units, formulas and bounded command; F10 source-to-claim examples, F05 dependencies and per-record next discriminators support challenge and extension.

LC25 · amended

Generic public roles for reviewers.

Planning reason

Generic names must preserve shared-source/operator relationships. Separate alternate implementation, separate execution and institutional independence.

Candidate disposition

implemented

Implementation evidence

S02/S05, M01, F05 and A06 distinguish local implementation/key-holder, alternate arithmetic, reported execution and editorial review without fabricated organizational independence.

LC26 · incorporated

Unknown unknowns and weak/absent signals.

Planning reason

Retain the six discovery probes, anomaly capture and specific discriminators; do not manufacture a list of all unimagined risks or collapse missing evidence into a negative result.

Candidate disposition

implemented

Implementation evidence

S03/S06/S07, F01/F09, full R01-R42 and U-01-U-06 preserve weak/absent/unavailable/unmeasured states, overlooked knowledge and observable discovery triggers.

Analytical-format selection

The original decks supplied candidate explanatory formats, not measured data. These twenty candidate identities remain distinct from the figures selected for this report. Format selection changes presentation, not source claims.

C01 · Evidence availability and observation map

Field	Design disposition
Original candidate status	recommend
Source formats	Missingness Matrix
Reader question	What is absent, unmeasured, unavailable or actually observed?
Selected disposition	selected
Reason / evidence	Literal availability, observed-result and sensitivity columns preserve unavailable inputs, inspected absence and unmeasured quantities without a strength heatmap.
Original semantic limit	A blank is forbidden without a defined missingness state. Availability is not truth and a tested absence is not a missing record.

C02 · Hypothesis episodes

Field	Design disposition
Original candidate status	recommend
Source formats	Swimmer Plot, Persistence Barcode, Timeline Layout
Reader question	How did each explanation change through the investigation?
Selected disposition	adapted
Reason / evidence	Hypothesis lanes distinguish events and open/excluded endpoints; arcs show corrections and bound resolution. This adaptation replaces the original duration guidance: positions encode reading order, not duration or confidence.
Original semantic limit	Length means duration in the record, not credibility, statistical survival or topological significance. No borrowed long-bar-is-truth rule.

C03 · Claim, warrant and rebuttal

Field	Design disposition
Original candidate status	recommend
Source formats	Toulmin Argument Map
Reader question	What exactly follows from this observation?
Selected disposition	selected
Reason / evidence	Toulmin diagrams connect observations through warrants to bounded claims, with separate custody and rebuttal links. R05 and R18 retain their different evidence and limits.
Original semantic limit	A drawn arrow is a documented inference step, not proof that a premise is correct. Separate custody from inference.

C04 · Aligned observation tracks

Field	Design disposition
Original candidate status	recommend
Source formats	Genome Browser Track, Annotation Tiers, Sequence Alignment
Reader question	Where does evidence repeat, and which interpretation shares that coordinate?
Selected disposition	selected
Reason / evidence	All 512 exact position rows are drawn as separate window/context annotation tracks. Counts are derived from the public source; recurrence does not represent key membership.
Original semantic limit	Position is not time; repeat class is not secret-key membership. Do not invent token-level labels or equate deduplication with independence.

C05 · Source-to-claim translation

Field	Design disposition
Original candidate status	recommend
Source formats	Interlinear Gloss, Clause Matrix
Reader question	How does a source field become a reported statement?
Selected disposition	selected
Reason / evidence	Three explicitly distinct source routes align a field, reading and ceiling. This is interpretation, not literal translation or pooled evidence.
Original semantic limit	Interpretation is not a literal translation. Preserve alternative readings and do not align unrelated evidence as if identical.

C06 · Artifact exchange diagram

Field	Design disposition
Original candidate status	prototype
Source formats	Feynman Diagram, Swimlane Flowchart, Sequence Diagram
Reader question	Which event produced new evidence, and which event reviewed the same artifact?
Selected disposition	adapted
Reason / evidence	Artifact nodes and event vertices carry labeled exchanges. One pair feeds keyed, analytical and keyless operations; dates identify receipts. Interaction geometry implies no physical amplitudes or measured time scale.
Original semantic limit	Schematic order, no physical quantities or amplitudes. Review count is not independent sample count; execution remains peer-reported.

C07 · Shared-input map

Field	Design disposition
Original candidate status	prototype
Source formats	Tensor Network, Bipartite Graph, Dependency Graph
Reader question	What do apparently separate checks have in common?
Selected disposition	adapted
Reason / evidence	Named shared IDs/profile connections expose input reuse. Observer code and expected output are supplied separately; categorical bonds imply neither measured covariance nor independent samples.
Original semantic limit	This is a graph inspired by the layout, not an entanglement/tensor result or measured covariance. Label what is shared.

C08 · Provenance cutaway

Field	Design disposition
Original candidate status	recommend
Source formats	Exploded Assembly, Building Section, C4 / Architecture Diagram
Reader question	At which layer was a property observed or left untested?
Selected disposition	adapted
Reason / evidence	Separated conceptual planes, alignment guides and source-linked callouts distinguish bytes, export, profile, verification and claim. Depth and spacing encode no confidence or universal causal pipeline.
Original semantic limit	Layer order is conceptual, not physical assembly order or a universal causal pipeline. Height/area must not mean confidence.

C09 · Measured comparisons with conditions

Field	Design disposition
Original candidate status	recommend
Source formats	Small Multiples, Dumbbell Chart, Cleveland Dot Plot
Reader question	What changed numerically, and what does that comparison permit?
Selected disposition	selected
Reason / evidence	Three paired nominal-z rows include exact primary hit counts and denominators. The full 18-row searchable table belongs beside this figure in the report; no invented intervals. Historical seven-row costs remain narrative/table content because they need separate units and lineage limits.
Original semantic limit	No invented interval or causal before/after story. Different count units and lack of calibration remain visible.

C10 · Claim-condition crosswalk

Field	Design disposition
Original candidate status	recommend
Source formats	Clause Matrix, Decision Table
Reader question	Which evidence supports a statement, and which requirement remains unmet?
Selected disposition	adapted
Reason / evidence	Selected ledger claims are shown with their explicit evidence and missing conditions. L-10/L-12/L-13 remain visibly excluded. No claim-to-question/gate edges are invented; full identities are separate report registers.
Original semantic limit	A matrix cell summarizes one condition; do not reduce nuance to an overall pass percentage.

C11 · Static detail and research index

Field	Design disposition
Original candidate status	recommend
Source formats	Archival Index Table, Drill-Through Drawer, Accordion Trace Log
Reader question	Can the reader inspect a weak result without opening another application?
Selected disposition	adapted_to_document
Reason / evidence	Use the full visible 42-record appendix, stable IDs, indexes and page references in the report. A separate figure would duplicate the navigation and could hide essential limits.
Original semantic limit	Essential caveats stay visible on the figure page; appendix links supplement rather than conceal them.

C12 · Correction and dependency links

Field	Design disposition
Original candidate status	prototype
Source formats	Arc Diagram, Citation Network
Reader question	Where did an earlier statement get revised or reuse an older source?
Selected disposition	adapted
Reason / evidence	Sparse labeled arcs show documented corrections and bound resolution in F02. F10 retains source-interpretation routes. Connections do not measure authority or independent evidence.
Original semantic limit	No line density as authority; same source cited twice does not add evidence. Keep one story per diagram.

C13 · Overlapping research themes

Field	Design disposition
Original candidate status	conditional
Source formats	UpSet Plot
Reader question	Which questions share mechanisms or evidence gaps?
Selected disposition	deferred
Reason / evidence	No explicit multi-label taxonomy and reviewed membership matrix support an UpSet count. Record relationships are not assumed to be set membership.
Original semantic limit	Research themes overlap; do not force a partition or treat record counts as signal strength.

C14 · Experiment lifecycle and stop states

Field	Design disposition
Original candidate status	conditional
Source formats	Petri Net, Finite State Machine
Reader question	What can legitimately happen next, including abstention and unavailable evidence?
Selected disposition	adapted
Reason / evidence	A continuation-state map retains closure, unavailable evidence, known unknowns, overlooked existing knowledge and discovery probes. It is conceptual, not a formal state machine or mandatory queue.
Original semantic limit	A process illustration is not a formal reachability proof. Avoid turning every open research item into pending mandatory work.

C15 · Future experiment selection

Field	Design disposition
Original candidate status	defer_quantitative
Source formats	Abatement Cost (MACC), 2×2 Priority Matrix
Reader question	Which optional question is worth investigating next?
Selected disposition	deferred_quantitative
Reason / evidence	Optional continuation rules are retained categorically. Comparable measured cost, utility and information-gain inputs do not exist; no quantitative priority ranking is drawn.
Original semantic limit	Do not invent bar area, dollar estimates or expected information gains; interactions would invalidate a simple additive ranking.

C16 · Research trajectory concept

Field	Design disposition
Original candidate status	defer_quantitative
Source formats	Phase Portrait, Connected Scatter Plot
Reader question	Is there a defined state trajectory rather than a persuasive visual metaphor?
Selected disposition	deferred_quantitative
Reason / evidence	Categorical history supplies the supported substitute. No sampled numeric state trajectory, confidence coordinates or attractor is available.
Original semantic limit	No invented attractor, convergence curve or confidence coordinates. If a conceptual sketch is retained, call it schematic and remove quantitative axes.

C17 · Multidimensional profile

Field	Design disposition
Original candidate status	defer
Source formats	RadViz, Parallel Coordinates, Radar / Spider Chart
Reader question	Would projection clarify evidence without compressing incompatible axes?
Selected disposition	deferred
Reason / evidence	Custody, observation and novelty are not commensurate numeric axes; no radar, RadViz or parallel-coordinate synthesis is justified.
Original semantic limit	No radial overall-confidence score or similarity-based identity conclusion; projection and axis ordering can create apparent clusters.

C18 · Monitoring and calibrated behavior

Field	Design disposition
Original candidate status	defer
Source formats	Horizon Chart, Live Rolling Strip Chart, Control Chart (SPC), Calibration Plot
Reader question	Do we have a time series, baseline or calibration set to support this?
Selected disposition	deferred
Reason / evidence	No continuous telemetry, control baseline or qualified calibration population is available. No live badge, noise envelope, gauge or confidence percentage is drawn.
Original semantic limit	No arbitrary noise envelope, probability gauge or LIVE status in the PDF.

C19 · Branching and exchange

Field	Design disposition
Original candidate status	defer_quantitative
Source formats	Collision Event, Sankey Diagram, Chord Diagram
Reader question	Are flows actually counted and conserved, or merely related?
Selected disposition	deferred_quantitative
Reason / evidence	Named exchanges and shared-input bonds supply a categorical schematic. No conserved count supports weighted ribbons or a 100-percent partition.
Original semantic limit	No decorative ribbon widths or implied conservation; branching hypotheses may overlap and do not sum to 100%.

C20 · Design before pagination

Field	Design disposition
Original candidate status	recommend
Source formats	Style Tile, Wireframe
Reader question	Can we compare visual directions before laying out the complete report?
Selected disposition	amended
Reason / evidence	The original three-direction comparison remains amended. This revision compares four earlier plates with their analytical-card redesigns on unchanged evidence, then reviews the integrated pages. It is not completion of the original three-direction exercise.
Original semantic limit	A layout proof is not the final PDF, a complete reading order or a shipping decision.

A07 Glossary and subject index

Definitions here fix the meaning used by this report. The subject index points to substantive records; it does not imply that every listed concept was measured or qualified.

Absence and expected observability

A feature not observed under a named method is informative only relative to opportunity and method sensitivity. It is not proof of human origin or proof of hidden marking.

R01 · R05 · R06 · R09

Artifact versus generation event

A new file or conversion may preserve the same linguistic material. Artifact count must not be treated as independent generation count.

R13 · R14

Attribution scope

Compatibility with a registered construction, a provider declaration, an artifact chain and a claim about a person or intent are distinct targets.

R17 · R25 · R35

Calibration

Empirical characterization of a fixed decision rule on a defined population. The nominal scores in this report are not calibrated performance estimates.

R18 · R33

Claim ceiling

The strongest inference justified by the particular observation, custody, method and assumptions; printed separately from the observation.

R01 · R18 · R35

Collection mistake

The retained six-summary collection includes an initial effort-setting mistake. Its labels and comparative implications remain part of the record.

R06

Context and window

A context is the exact four preceding completion-token IDs. A window adds the target ID. First-occurrence selection preserves original eligible-position order.

R18 · R20

Contract equivalence

Agreement on ordinary tests does not prove that implementations accept the same input language or reject the same invalid values.

R16

Custody

Which exact bytes, reports, scripts or derivatives are actually available, private or missing. Custody is separate from an observation being positive or negative.

R05 · R06 · R32

Dependence and repetition

Repeated represented material can inflate raw nominal scores without providing additional independent content. Exact-repeat removal does not establish independence of the remainder.

R18 · R19 · R20

Discovery probes

Observable triggers and anomaly capture for revisiting assumptions. They preserve discovery capacity without enumerating or quantifying unknown hidden causes.

R37 · R42

Excluded stronger claim

A statement retained in the ledger specifically to show it is not supported or included as a result. Its exclusion and reason must accompany its text.

R26 · R32

False implication and transplantation

A compatible scheme signal does not by itself identify original authorship; transfer or construction alternatives require their own controlled tests.

R25 · R34

Field resource observations

Historical small-batch context, throughput and memory observations under their recorded source/quality limits; not a qualified general deployment benchmark.

R27 · R28

Formal assurance

A proof must bind the actual current policy and implementation. An earlier sketch or satisfiable configuration is not a compiled current-policy certificate.

R26 · R42

Identity-only tokenizer null

Consistent bijective renaming of token identities while preserving probabilities and decoding is a bounded invariance control. It does not test generation-changing tokenizer coupling.

R21 · R22

Instrument artifact

A pattern created by extraction, parsing, reporting or an implementation defect rather than a demonstrated property of the original material.

R15 · R30

Keyless reproduction

Public tuple counting and arithmetic on supplied aggregate hits. It does not recover secret-key membership, regenerate text or authenticate a device event.

R18 · R31

Known unknowns

Named unresolved questions with specified evidence gaps or next discriminators. Their existence does not establish a positive hidden mechanism.

R03 · R22 · R33

Noise floor

A physical noise-floor claim needs a defined channel, statistic, background and sensitivity. The project did not measure a universal physical floor.

R01

Nominal z

The reported standardized arithmetic (h - n gamma) / sqrt(n gamma (1 - gamma)). Its numerical value is not a calibrated tail probability or a person-level attribution score.

R18 · R20 · R33

Novelty status

A distinction among established methods, local observations and project proposals. Global novelty remains unassessed where no qualifying search or comparison was completed.

R22 · R23 · R42

Ordinary separator positions

A question about arrangements of ordinary visible separators, distinct from unusual Unicode or metadata carriers. Finite byte-feature scans do not settle it.

R03 · R05

Output suitability

Repetition, truncation and usefulness are outcomes separate from a watermark-compatible statistic. The native pair does not establish useful-text performance.

R19 · R28

Passive interaction and style

Untested same-model interaction and code/style triage proposals. A plausible association is not a qualified identification mechanism or quality metric.

R23 · R24

Provider declarations

Dated background statements about a provider or family. They do not establish historical activation on an earlier specimen, exact private implementation or local verifier access.

R02 · R04 · R35

Radio repeatability

A repeatable observation under an earlier collection setup does not identify its semantics or establish a watermarking capability.

R29

Shared-input review

Multiple implementations, recounts and reviews can reuse the same trace, profile, operator or supplied expectation. Such agreement is not another independent generation.

R02 · R31 · R38

Stopping and reopening

Close the declared bounded milestone; reopen an affected item only for a concrete contradiction, method error or deliberately selected stronger claim with a fixed protocol.

R33 · R41

Unknown knowns

Relevant information already recorded but overlooked, unretrieved or not integrated. This is a retrieval/workflow problem distinct from unmeasured signal.

R36 · R38

Unknown unknowns

Unanticipated causes or observations cannot be comprehensively listed in advance. Preserve discovery capacity and revision triggers instead of assigning invented probabilities.

R37

Unmeasured, unavailable and unqualified

An unmeasured outcome, missing log/artifact and result without established sensitivity or semantics are different states. None should be silently encoded as zero.

R28 · R29 · R32

Visible abstraction

A disclosed public reformulation that preserves the research lesson while omitting private personal or operational context.

R39 · R40

Known knowns within declared scope

Retained observations, calculations and bounded controls whose evidence and assumptions are specified here. Their acceptance is local to that scope; it does not turn an instrument result into unrestricted certainty or establish an untested stronger claim.

R18 · R20 · R21 · R30

Subject index

Subject	Record destinations
Absence and expected observability	R01 · 68 · R05 · 72 · R06 · 73 · R09 · 76
Artifact versus generation event	R13 · 80 · R14 · 81
Attribution scope	R17 · 85 · R25 · 93 · R35 · 103
Calibration	R18 · 86 · R33 · 101
Claim ceiling	R01 · 68 · R18 · 86 · R35 · 103
Collection mistake	R06 · 73
Context and window	R18 · 86 · R20 · 88
Contract equivalence	R16 · 83
Custody	R05 · 72 · R06 · 73 · R32 · 100
Dependence and repetition	R18 · 86 · R19 · 87 · R20 · 88
Discovery probes	R37 · 105 · R42 · 110
Excluded stronger claim	R26 · 94 · R32 · 100
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Field resource observations	R27 · 95 · R28 · 96
Formal assurance	R26 · 94 · R42 · 110
Identity-only tokenizer null	R21 · 89 · R22 · 90
Instrument artifact	R15 · 82 · R30 · 98
Keyless reproduction	R18 · 86 · R31 · 99
Known knowns within declared scope	R18 · 86 · R20 · 88 · R21 · 89 · R30 · 98
Known unknowns	R03 · 70 · R22 · 90 · R33 · 101
Noise floor	R01 · 68
Nominal z	R18 · 86 · R20 · 88 · R33 · 101
Novelty status	R22 · 90 · R23 · 91 · R42 · 110
Ordinary separator positions	R03 · 70 · R05 · 72
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Passive interaction and style	R23 · 91 · R24 · 92
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Subject	Record destinations
Shared-input review	R02 · 69 · R31 · 99 · R38 · 106
Stopping and reopening	R33 · 101 · R41 · 109
Unknown knowns	R36 · 104 · R38 · 106
Unknown unknowns	R37 · 105
Unmeasured, unavailable and unqualified	R28 · 96 · R29 · 97 · R32 · 100
Visible abstraction	R39 · 107 · R40 · 108

A08 Evidence tables and downloadable artifacts

The full research record includes measurements below the level of a headline finding: negative inspections, weak overlap, missing fields, failed acceptance and corrected interpretations. This supplement exposes the available measurements and methods. The distinction between a historical report, public projection and reproducible calculation travels with each artifact.

Seven historical field rows

Row	Prompt tokens	TTFT (s)	Post-first tokens/s	Peak RSS (KiB)	Wall (s)
COMPACT-01	437	12.433	8.331	645664	49.838
COMPACT-02	437	12.475	8.637	645580	49.573
COMPACT-03	437	12.431	8.089	645252	50.108
COMPACT-04	437	12.469	8.340	651700	49.871
BROAD-01	1749	62.995	3.927	1467988	109.046
BROAD-02	1749	62.935	3.826	1490020	109.313
BROAD-03	1749	63.073	3.859	1489736	109.295

The four compact rows ran Qwen2 INT4 0.5B (vendor ONNX format) under KY-ORT 1.2.2; their model-file and receipt hashes are in the hardware source projection. The model variant for the three broad rows was not recovered. All seven rows report 64 generated tokens and null usefulness. Four compact log deltas were captured and reported only periodic Wi-Fi messages; the three broad deltas were absent. Device presence is not acceleration evidence. JSON retains full precision, start/end thermal measurements, load/generation times and device flags. JSON · CSV.

Metric	Compact mean [min, max]	Broad mean [min, max]
prompt tokens	437.000 [437.000, 437.000]	1749.000 [1749.000, 1749.000]
generated tokens	64.000 [64.000, 64.000]	64.000 [64.000, 64.000]
first token seconds	12.452 [12.431, 12.475]	63.001 [62.935, 63.073]
post first tokens per second	8.349 [8.089, 8.637]	3.871 [3.826, 3.927]
peak rss kib	647049.000 [645252.000, 651700.000]	1482581.333 [1467988.000, 1490020.000]

Metric	Compact mean [min, max]	Broad mean [min, max]
run wall seconds	49.847 [49.573, 50.108]	109.218 [109.046, 109.313]

These are descriptive nonrandom batches, with different context sizes and incomplete control of other conditions. The historical exporter was dirty and its complete clean snapshot is unavailable. Means/ranges can be recomputed from public rows; stronger causal or device-authentication claims remain unsupported. R27, R28, R32.

What the Orange Pi RV2 field study contributed

2026-08-14 - Cross-format implementation return

Orange Pi RV2 field study. Commit 3c2bb98 accepted for service comparison and external ONNX Runtime executor import. Fourteen ordered segments were checked. A raw /AA byte hit did not establish a PDF action; ten U+007F extraction artifacts were ordinary bullet glyphs under another extractor and visual inspection.

Implementation and extraction findings; no provider watermark or hardware-use claim.

2026-08-15 - Acceptance failure retained

Orange Pi RV2 field study. Direct ONNX Runtime checks passed at 99ac647. The full smoke check failed because the synthetic receipt lacked model_authority/runtime fields.

A passing direct runtime check did not imply full contract acceptance.

2026-08-16 - Fixture repair and revalidation

Orange Pi RV2 field study. Repaired fixture 13062417 passed full acceptance and the direct ONNX Runtime regression. The imported external receipt declared watermark.enabled=false.

The import preserves a producer's declaration; it does not independently prove how output was generated.

2026-08-22 - Historical field measurements and data import

Orange Pi RV2 field study. Four compact and three broad context model rows were returned alongside separate radio observations. A subsequent strict import (format 1.1) of the model and radio records reported validity and priority-zero checks passing.

Utility remained unscored, acceleration unqualified and radio semantics unvalidated. Dirty/incomplete exporter lineage remains visible.

2026-09-09 - Installed AxisPL 0.1.4 qualification

Orange Pi RV2 field study. A fresh offline wheel installation, core self-check and external ONNX Runtime executor smoke checks passed. The device clock showed August 25 during this September 9 collection; collector UTC governs this chronology.

This did not qualify native Hugging Face generation or C2PA verification on Orange Pi RV2.

2026-09-11 - Native executor preparation

Field contributor. The field native Hugging Face path was unavailable. A separate CPU executor was prepared and configuration/tokenizer imports were checked before native execution.

No weights were loaded and no native text was generated by this preparation event. The owned pair was generated separately on September 12.

2026-10-08 - Second-architecture keyless observer

Orange Pi RV2 field study. The board reported riscv64 / Python 3.12.3 execution agreeing on clean 251 and marked 93 distinct windows, plus the bounded supplied-hit arithmetic. No model, tokenizer or secret key was used. The original return manifest had a self-hash defect, and a corrected inventory was computed on receipt; the original is preserved.

The observer used shared retained inputs and supplied expected output. Agreement is not independent generation, historical device authentication or clean exporter qualification; source_dirty remained true.

Chronology JSON with original source-document hashes.

The negative byte audit remains a measured result

All 17 original inputs are unavailable in the reviewed collection. These are retained audit measurements, not a new audit of original bytes. The report totals 237,506 bytes, 234,326 characters and 2,644 lines. The inspected invisible-codepoint count is zero; two trailing-space lines contain 35 ordinary spaces. This does not establish sensitivity to an undisclosed token-level mechanism.

Specimen	Bytes	Characters	Lines	Invisible hits	Trailing-space lines
AUDIT-01	4646	4603	54	0	1
AUDIT-02	25943	25259	289	0	1
AUDIT-03	22808	22493	273	0	0
AUDIT-04	12406	12276	151	0	0
AUDIT-05	14499	14311	150	0	0
AUDIT-06	11148	11036	133	0	0
AUDIT-07	10813	10709	131	0	0
AUDIT-08	10134	9972	118	0	0
AUDIT-09	12126	11987	151	0	0

Specimen	Bytes	Characters	Lines	Invisible hits	Trailing-space lines
AUDIT-10	18861	18675	209	0	0
AUDIT-11	12576	12445	115	0	0
AUDIT-12	19491	19250	191	0	0
AUDIT-13	12238	12101	127	0	0
AUDIT-14	10225	10101	124	0	0
AUDIT-15	13260	13074	139	0	0
AUDIT-16	13640	13474	155	0	0
AUDIT-17	12692	12560	134	0	0

Six summaries: overlap and collection limits

Repeated short and long summaries of shared radio-operating instructions; shared topic and prompt constraints can produce overlap. Six retained copies recovered privately in October; public measurements below do not distribute the underlying conversations. The failed collection assumption and possible hidden context remain in R06; descriptive condition labels do not identify which user produced a pair.

Specimen / condition	Bytes	Body words	Unique words
SUMMARY-01 / HIGH_B_long	1151	184	124
SUMMARY-02 / HIGH_A_long	1071	170	112
SUMMARY-03 / MAX_long	1305	204	137
SUMMARY-04 / HIGH_B_short	497	76	58
SUMMARY-05 / HIGH_A_short	525	79	63
SUMMARY-06 / MAX_short	572	85	67

Focused pair	Character ratio	Word ratio	Word cosine	Longest shared run
SUMMARY-02 / SUMMARY-01	0.7043	0.6384	0.9118	13
SUMMARY-02 / SUMMARY-03	0.6227	0.5508	0.8928	15
SUMMARY-05 / SUMMARY-04	0.8031	0.7355	0.8691	11
SUMMARY-05 / SUMMARY-06	0.6167	0.5732	0.8272	18
SUMMARY-01 / SUMMARY-03	0.5552	0.4742	0.8682	7
SUMMARY-04 / SUMMARY-06	0.5573	0.4969	0.7833	10

All 15 pairs, Unicode/whitespace counts, shared-ngram counts and ZIP measurements remain in the audit projection. A six-pair focused view adds no new samples. Shared task constraints and style remain alternatives to a hidden carrier. R05, R06, R07.

Paired corpora and the arbitrary-key null

Nine prose pairs and 32 constrained continuations share byte-identical prompt inputs. None of the nine prose or 32 completion pairs is exactly identical. Twenty of 32 continuations share the first word, twelve share the first two, and eight share the first four. That is a weak conditional observation with prompt/context confounds, not a person-level fingerprint.

Pair	Words A / B	Sequence ratio	TF-IDF cosine	Shared 8-grams
PROSE-01 long	1888 / 1908	0.2329	0.7099	21
PROSE-02 long	1934 / 1838	0.1803	0.7027	7
PROSE-03 long	1854 / 1855	0.1833	0.6414	7
PROSE-04 medium	1084 / 1044	0.1118	0.5190	0
PROSE-05 medium	1065 / 992	0.1371	0.5605	1
PROSE-06 medium	1011 / 934	0.2180	0.5138	7
PROSE-07 short	525 / 528	0.2374	0.4017	16
PROSE-08 short	551 / 522	0.2535	0.5955	11
PROSE-09 short	547 / 539	0.1657	0.4305	2

Corpus	Surrogate tokens	Mean nominal z	Population SD	Range	z ≥ 4 / ≥ 6
A	12292	0.128255	0.882910	-1.979167 to 2.187500	0 / 0
B	12219	0.008489	0.925328	-2.063430 to 2.241042	0 / 0

This scan used 64 deterministic arbitrary keys, a 3,902-entry word-plus-punctuation vocabulary, width 4 and gamma 0.25. It checks the local laboratory profile under those keys; no provider key, tokenizer or detector was supplied. It is separate from the later 4,096-key motif control. Full comparison JSON · All 41 pair rows as CSV.

Format and instrument findings

The cross-format report retained 300 structured rows: 288 valid and 12 invalid. Clean JSONL and CSV agreed on values/order. Five HTML and nine PDF source-segment word sequences matched; two code segments also matched after removing whitespace. Conversion did not create independent linguistic samples.

- Raw /AA byte hit occurred inside a compressed PDF stream, not an action dictionary.
- Pypdf mapped ten bullet glyphs to U+007F; pdftotext and visual rendering show normal bullets.
- Line-ending byte counts are emitted for binary ZIP data.
- Whole-bundle compare needs section- or manifest-aware containment semantics.

The code/schema discrepancies remain concrete, rather than being hidden by passing tests:

- Schema rejects additional properties; both implementations accept them.
- Schema rejects surrounding whitespace in event_id/source; both implementations accept it.
- Python accepts several noncanonical timestamp forms rejected by schema and Go.
- Go aborts on a line over 1 MiB despite its comment saying it becomes a record rejection; Python has no equivalent bound.

Five related images had no observed embedded C2PA/signed-provenance structure in the inspected container channels. Public fields retain export methods and structural scan results. Undisclosed steganography, unknown sensitivity and generation provenance remain outside that observation. R12, R13, R15, R16.

Reproducible toy controls and planning arithmetic

Fixture	Raw positions	Distinct windows	Raw z ≥ 4	Distinct z ≥ 4
unique_1028	1024	1024	0/4096	0/4096
cycle16_68	64	16	116/4096	1/4096
cycle16_260	256	16	795/4096	1/4096
cycle16_1028	1024	16	1502/4096	1/4096

The same 16-window motif is repeated 4, 16 and 64 times. The fixed public key trials and full expected receipt are supplied; there was no native-key search. Raw threshold crossings are toy behavior, not natural-text FPR. All 24 consistent four-symbol ID permutations

preserve decoded probability mass. The distinct sequences [1,2] and [3] both decode to “watermark”; segmentation alone need not survive decoding. Generation-coupled alternatives remain open.

Independent held-out negatives, zero false positives	One-sided 95% upper bound
30	0.095034
60	0.048703
100	0.029513
299	0.009969
300	0.009936

The bound assumes a fixed rule and independent held-out negatives. It takes 299 such zero-error negatives to put the upper bound at or below 1%. With 30 continuous exchangeable calibration scores the smallest split-conformal p value is 1/31, while an empirical error-rate increment is 1/30. A separate illustrative 90% sensitivity / 99% specificity / 1% prevalence calculation gives PPV 0.47619. These are planning calculations, not attained operating performance.

```
python3 -I -B scripts/reproduce_controls.py
```

Reproduction guide · Data dictionary.

Publication correction record

CHANGE-R39 - Restore nonpersonal forecast topics and the failed causal inference obscured by the prior abstraction.

Earlier wording: Peripheral forecasts and informal observations motivated discussion, but lacked prospective resolution criteria and sufficient evidence for a causal interpretation. Personal circumstances are omitted from this public abstraction.

Current wording: Historical conjectures concerned access/cost stratification, resource constraints, information disorder, and marking/guardrails. They lacked prospective resolution criteria and a scored baseline. A separate personal-contact anecdote did not substantiate the assistant-suggested NDA or gated-lab explanation; neither silence nor speculation established hidden intent. Personal circumstances remain abstracted.

No predictive success or causal claim added; historical unscored status preserved.

CHANGE-R38-TITLE - Describe the research-process observation without internal workflow vocabulary.

Earlier wording: Agreement quality and the human acceptance firewall

Current wording: Agreement quality and human acceptance of AI-assisted findings

Wording only; the observation, its limits and its links are unchanged.

CHANGE-R38-OBSERVATION - Describe the research-process observation without internal workflow vocabulary.

Earlier wording: The user asks for conflict reporting and human scrutiny of agreeable outputs; partial cross-lane visibility and assistant confidence inflation recur.

Current wording: Conflict reporting and human scrutiny of agreeable AI outputs were requested throughout; parallel AI-assisted analyses had only partial visibility of one another, and confidence inflation in assistant responses recurred.

Wording only; the observation, its limits and its links are unchanged.

CHANGE-R38-ASSUMPTIONS - Describe the research-process observation without internal workflow vocabulary.

Earlier wording: Multiple agents sharing context/models/fixtures are not independent samples merely because they have separate role names.

Current wording: Multiple AI assistants sharing context, models or fixtures are not independent samples merely because they have separate role names.

Wording only; the observation, its limits and its links are unchanged.

CHANGE-R38-NOVELTY-STATUS - Describe the research-process observation without internal workflow vocabulary.

Earlier wording: Project workflow doctrine; established epistemic concern.

Current wording: Project methodology; established epistemic concern.

Wording only; the observation, its limits and its links are unchanged.

CHANGE-R41-TITLE - Describe the research-process observation without internal workflow vocabulary.

Earlier wording: Workflow stop gates and file proliferation

Current wording: Research-process losses: unnecessary stops and file proliferation

Wording only; the observation, its limits and its links are unchanged.

CHANGE-R41-OBSERVATION - Describe the research-process observation without internal workflow vocabulary.

Earlier wording: Needless pause gates, continuity/export losses and repeated directories were reported; collection and synthesis were explicitly distinguished.

Current wording: Unnecessary pauses, losses of continuity between sessions and exports, and duplicated working directories were reported in the AI-assisted research process; collection and synthesis were explicitly distinguished.

Wording only; the observation, its limits and its links are unchanged.

CHANGE-R41-CLAIM-CEILING - Describe the research-process observation without internal workflow vocabulary.

Earlier wording: Preserve a stable scoped handoff and finite stopping rule; do not turn unknowns into endless mandatory gates.

Current wording: Keep a stable scope and a finite stopping rule; open questions should not become endless mandatory checkpoints.

Wording only; the observation, its limits and its links are unchanged.

CHANGE-R41-NEXT-DISCRIMINATOR - Describe the research-process observation without internal workflow vocabulary.

Earlier wording: Use one companion, preserve immutable incoming bytes and link current status; reopen only for a concrete error or selected new question.

Current wording: Use one working copy, preserve incoming files unchanged and link their current status; reopen only for a concrete error or a selected new question.

Wording only; the observation, its limits and its links are unchanged.

CHANGE-R41-REOPENING-CONDITION - Describe the research-process observation without internal workflow vocabulary.

Earlier wording: Use one companion, preserve immutable incoming bytes and link current status; reopen only for a concrete error or selected new question.

Current wording: Use one working copy, preserve incoming files unchanged and link their current status; reopen only for a concrete error or a selected new question.

Wording only; the observation, its limits and its links are unchanged.

CHANGE-R41-RETENTION-REQUIREMENT - Describe the research-process observation without internal workflow vocabulary.

Earlier wording: Do not turn every uncertainty into a new mandatory lane. Respect intake, research, analytics, synthesis and publication stages.

Current wording: Do not turn every uncertainty into a new mandatory task. Respect intake, research, analysis, synthesis and publication stages.

Wording only; the observation, its limits and its links are unchanged.

CHANGE-NATIVE-METHODS-IMPLEMENTATION - Identify the experimental software by name.

Earlier wording: Implementation identified only as watermark toolkit version 0.1.4, commit 7f09d34ca6fef346312907f5647bcab7f28a8d45.

Current wording: Implementation named Axis Provenance Lab (AxisPL), distribution axis-provenance-lab, version 0.1.4, commit 7f09d34ca6fef346312907f5647bcab7f28a8d45, scheme hmac-sha256-splitmix64-greenlist-v1.

Identity only; version, commit, settings and results are unchanged.

CHANGE-C09-EVIDENCE - List the new figures that apply the measured-comparison format.

Earlier wording: A06/C09 records this selection and its limit. Actual figures: fig-F07.

Current wording: A06/C09 records this selection and its limit. Actual figures: fig-F07, fig-F11, fig-F12.

Presentation only; no measurement or format decision changed.

Artifact catalog

Hashes below identify the included artifact bytes. Original receipt hashes within evidence files identify separate source artifacts. Neither is a signature or independent authentication. The PDF/HTML and their destinations are bound later by the package manifest and coverage record.

E01 · E02 · E03 · E04 · E05 · E06 · E07 · E08 · E09 · E10 · E11 · E12

E01 - Owned native specimens and methods

Inspect the exact owned prompt and output quality, match text hashes and inspect scientific conditions.

Original generation receipts are retained privately; scientific settings and text are public. Native keys, model assets and private runtime are not included.

- data/native-methods.public.json
- fixtures/native-prompt.public.txt
- fixtures/native-clean.public.txt
- fixtures/native-marked.public.txt

E02 - Native counts and supplied-hit results

Recompute exact contexts/windows and all 18 nominal scores from supplied hit counts.

Public wrappers preserve the completion arrays. Aggregate membership hits remain key-holder reported; no public native membership labels are supplied.

- fixtures/profile.public.json
- fixtures/clean-completion.public.json
- fixtures/marked-completion.public.json
- data/results.public.json
- data/results.public.csv
- data/observation-tracks.public.json
- data/keyless-check.expected.json

E03 - Seven historical field observations

Recompute means/ranges; inspect missing utility, log availability and thermal/device observations.

Numerical original report fields are projected publicly with neutral row IDs. Complete clean historical exporter source is unavailable; raw operational logs remain private.

- data/field-observations.public.json
- data/field-observations.public.csv
- data/hardware.public.json
- data/hardware-source.public.json

E04 - Orange Pi RV2 field-study chronology

Distinguish implementation failures/repairs, installed-package qualification, executor preparation and the later observer.

Original handoffs are retained privately and identified by document hashes. No event authentication or new sample independence follows from receipt agreement.

- data/field-contributions.public.json
- data/observer-receipt.public.json

E05 - Early negative audit and six summaries

Inspect 17 audit rows, six summary records and all 15 summary pairs; recompute retained aggregate counts.

The 17 original audit inputs are unavailable. Six summary copies were recovered privately. Their measured fields are public; raw personal context is not redistributed.

- data/historical-audits.public.json

E06 - Corpus, null, format and image observations

Inspect nine prose pairs, 32 constrained continuations, arbitrary-key nulls and parser/format findings.

Original corpora and converted assets are retained privately. The selected aggregates are reproducible from rows; raw-corpus and pooled-ngram recomputation remain unavailable here.

- data/comparative-observations.public.json
- data/comparative-observations.public.csv

E07 - Frozen toy controls

Repeat all frozen public-key motif trials, all 24 relabelings and finite-sample calculations.

Exact fixtures, the retained 4096 public key values, focused source and full expected receipt are included. This is a toy replay, not native membership verification or field calibration.

- fixtures/synthetic-controls.public.json
- fixtures/public-null-keys.json
- data/synthetic-controls.expected.json
- scripts/reproduce_controls.py
- sources/controls/membership.py
- sources/controls/diagnostics.py
- sources/controls/experiments.py
- sources/controls/README.md
- sources/controls/LICENSE.controls.txt

E08 - Full research register

Inspect all 42 records, weak/absent signals, assumptions, corrections and open discovery probes.

The original conversation is retained privately. Scientific record fields and publication corrections are public; private locators/personal context are abstracted.

- data/research-register.public.json
- data/glossary.public.json

- data/publication-changes.public.json
- data/casebooks.public.json

E09 - Questions and original gates

Inspect 36 questions/probes and their histories, plus all 64 original gate obligations.

Historical pending wording is preserved alongside later dispositions. Printing a gate does not recertify it.

- data/future-work.public.json
- data/original-gates.public.json
- data/content-map.public.json

E10 - Claims and contribution decisions

Inspect 14 bounded/excluded claims, 26 contribution dispositions and 20 format decisions.

Shared inputs are explicit. Multiple reviews are not independent generations or replicated scientific experiments.

- data/claim-ledger.public.json
- data/contribution-decisions.public.json
- data/contribution-implementation.public.json
- data/format-implementation.public.json
- design/format-candidates.public.json
- design/VISUAL_GRAMMAR.md

E11 - Twelve analytical figures

Download SVG/PNG plates with captions, units, links and limits. The F01-F10 identities are retained; F11 and F12 were added in this edition.

Figures organize the same evidence. Hypothesis lanes, Toulmin maps, the shared-input network and provenance cutaway are explanatory structures, not new measurements.

- figures/figures.json
- figures/f01-availability,-observation-and-sensitivity.svg
- figures/f02-hypothesis-episodes-and-revisions.svg
- figures/f03-two-arguments-with-visible-limits.svg
- figures/f04-recurrence-at-every-completion-position.svg
- figures/f05-shared-inputs,-different-operations.svg
- figures/f06-provenance-layers-and-the-claim-boundary.svg

- figures/f07-primary-key-numerical-comparisons.svg
- figures/f08-claim-conditions-and-explicit-exclusions.svg
- figures/f09-continuation-states-and-finite-stopping.svg
- figures/f10-from-source-field-to-bounded-interpretation.svg
- figures/f11-repetition-and-nominal-threshold-crossings.svg
- figures/f12-orange-pi-rv2-field-runs-by-context-size.svg
- figures/f01-availability,-observation-and-sensitivity.png
- figures/f02-hypothesis-episodes-and-revisions.png
- figures/f03-two-arguments-with-visible-limits.png
- figures/f04-recurrence-at-every-completion-position.png
- figures/f05-shared-inputs,-different-operations.png
- figures/f06-provenance-layers-and-the-claim-boundary.png
- figures/f07-primary-key-numerical-comparisons.png
- figures/f08-claim-conditions-and-explicit-exclusions.png
- figures/f09-continuation-states-and-finite-stopping.png
- figures/f10-from-source-field-to-bounded-interpretation.png
- figures/f11-repetition-and-nominal-threshold-crossings.png
- figures/f12-orange-pi-rv2-field-runs-by-context-size.png

E12 - Methods, manuscripts and build sources

Read method definitions, check payload integrity and rebuild the static reader/report locally.

Build sources are delivered as source files. They are not a deployed backend; rebuilding changes output identities and requires fresh visual review.

- manuscript/overview.md
- manuscript/hardware.md
- docs/EDITION_NOTES.md
- docs/AXISPL_METHODOLOGY.md
- manuscript/narrative.md
- manuscript/methods.md
- manuscript/background-sources.md
- docs/REPRODUCE.md
- docs/DATA_DICTIONARY.md
- data/source-map.public.json
- sources/build_atlas.py

- sources/build_supplement.py
- sources/build_figures.py
- sources/format_cards_history.py
- sources/format_cards_network.py
- sources/format_cards_cutaway.py
- sources/finalize_package.py
- sources/requirements.txt
- scripts/verify_public_package.py
- scripts/verify_supplement.py
- scripts/VERIFICATION_SCOPE.md
- scripts/expected-content.public.json
- sources/make_archive.py

Full byte lengths and SHA-256 values are in the machine-readable catalog. The catalog distinguishes included artifacts from privately held or unavailable originals.